

Disease of Field, Horticulture Crops and their Management

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The Pustakalaya

Publisher:

The Pustakalaya

Jail Road, Bangla Bazar,
Near Indian Bank, Lucknow - 226002
Email - thepustakalaya.in@gmail.com

First Published by The Pustakalaya in 2025

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First Edition 2025

ISBN: 978-93-49921-90-0

Editing - Dr. Gajendra Pratap, Dr. Ratul Moni Ram, Mr. Sachin Sharma,
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Printer - Repro Books Limited

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Preface

Agriculture is still the livelihood of the Indian economy, and field and horticultural crops are crucial for providing food, nutritional, and economic security. The productivity and quality of these crops are seriously jeopardized by numerous biotic stresses, primarily plant diseases caused by fungi, bacteria, viruses, nematodes, and other pathogens. These plant diseases not only reduce yields but also affect farm revenues, food supply chains, as well as trade opportunities.

This book, "Diseases of Field and Horticultural Crops and Their Management", is a sincere effort to present a complete, updated, and useful book for students, researchers, extension staff, and farmers. It deals with important diseases of cereals, pulses, oilseeds, commercial crops, vegetables, fruits, spices, and plantation crops, their symptoms, causal pathogens, epidemiology, and integrated disease management. The material is reader-friendly, complemented by obvious illustrations, diagnostic characteristics, and field-control recommendations at the level of recent developments in plant pathology and sustainable agriculture.

Particular focus is placed on Integrated Disease Management (IDM) that integrates cultural, mechanical, biological, and chemical means of control in order to reduce crop loss while maintaining environmental safety. With newly emerging climate change issues and changing pathogen dynamics, the book also illuminates the necessity for disease forecasting systems, resistant varieties, and environmentally friendly practices for sustainable crop protection.

We are sure that this book will be a useful guide and teaching tool for undergraduate and postgraduate students of agriculture and horticulture, and for practitioners and policy-makers involved in plant protection and crop health management.

We wish to thank all authors, reviewers, and experts who contributed to making this work better. Comments for further improvement are always appreciated.

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Chapter-1

Mango Disease, Causal Organism, Symptoms, Etiology and Their Management

Anthracnose of mango

Common name: Anthracnose of mango

Host: Mango (*Mangifera indica* L.)

Disease category: Fungal disease

Causal organism

The fungus *Colletotrichum gloeosporioides* (Penz.) Penz. & Sacc. (or *Glomerella cingulata* in its teleomorphic form) causes the disease.

Symptoms

On leaves

Small, irregular, dark brown to black spots develop, which can coalesce to give big necrotic patches. In severe infections, leaf blight or premature defoliation can take place.

On flowers and panicles

Black necrotic spots form on flower stalks, flowers, and immature fruits. This can result in flower blight, causing poor fruit set or total loss of flowers.

On immature fruits

Small, rounded, black lesions are seen, which become larger and can cause drop of the fruit.

Black, sunken, round spots with dark edges are seen on the fruit surface during storage or ripening. Such spots can become confluent, causing large areas of rot. In humid environments, a pinkish spore mass can be formed.

Etiology

The fungus forms acervuli (fruiting structures) bearing conidia (asexual spores).

Conidia are hyaline (colorless), single-celled, oblong to cylindrical.

The pathogen persists as dormant mycelium or conidia on infected plant debris, twigs, and mummified fruits.

Disease is spread predominantly by wind, rain splash, and insects.

High humidity (>90%), frequent rains, and moderate temperatures (24-32°C) support disease establishment, particularly at flowering and fruiting stages.

Management

Cultural practices

Remove and destroy the infected twigs and leaves to minimize the inoculum load.

Ensure good orchard sanitation.

Enhance aeration by spacing plants properly and managing the canopy to lower the humidity levels around plants.

Chemical control

Spray 1% Bordeaux mixture or 0.2% Carbendazim or 0.25% copper oxychloride at flowering and fruiting stage, and again at 15-day intervals, particularly during rainy weather.

Post-harvest immersion of fruits in hot water (50-55°C) for 15 minutes or in fungicide solution (Carbendazim 500 ppm) can reduce storage rots.

- ✓ Spray *P. fluorescens* (FP 7) at 3 weeks interval commencing from October at 5g/like on flower branches.
- ✓ 5-7 sprays one to be given on flowers and bunches.
- ✓ Before storage, treat with hot water, (50-55°C) for 15 minutes or dip in Benomyl solution (500ppm) or Thiobendazole (1000ppm) for 5 minutes

Resistant varieties

Some cultivars show tolerance (though no cultivar is totally immune). Growing less susceptible cultivars may prove to be beneficial.

Integrated disease management

Integrate sanitation, pruning, fungicidal sprays at the right time in-season, and post-harvest treatments for effective disease control.



Anthracnose of Mango

Malformation of mango disease

Malformation of Mango

Common name: Malformation of mango

Host: Mango (*Mangifera indica* L.)

Disease category: Fungal disease (linked with physiological disorder)

Causal organism:

The disease is mainly linked with the fungus *Fusarium mangiferae* (formerly *Fusarium moniliforme* var. *subglutinans*).

Symptoms

1 Vegetative malformation

Majorly seen in nursery and young plants.

Characterized by crowded, bunched, and swollen shoots (witches' broom).

Hyper-branched vegetative buds with short internodes.

The deformed shoots tend to dry up with time.

2. Floral malformation (more severe type)

Panicles (flower clusters) are abnormally large, rigid, and bushy.

The flowers are tiny, sterile, and do not produce fruits.

Panicles persist for a long time on the tree without fruiting (persistent panicles). Abnormal panicles can be short internoded with a bushy growth.

Etiology

The disease is caused by *Fusarium mangiferae*, which persists in the malformed tissue.

Spores (conidia) produced by the fungus are wind-borne, rain-borne, and insect-borne.

Nymphs of mango bud mite (*Aceria mangiferae*) are responsible for transmitting the pathogen and infecting meristematic tissues.

Pre-favorable environmental conditions are cool temperatures (10-15°C) during flowering initiation and development.

Management

Cultural practices

Prune and kill (burn) irregularly growing shoots and panicles regularly.

Practice orchard sanitation and remove infected nursery plants.

Chemical control

Spray NAA (naphthalene acetic acid) at 200 ppm in October to inhibit floral malformation.

Application of Carbendazim (0.1%) or Copper oxychloride (0.3%) to the pruning wounds suppresses fungal infection.

Biological and integrated methods

Control of mango bud mite (*Aceria mangiferae*) by acaricides may assist in minimizing the spread of malformation.

Utilization of healthy, disease-free planting material.

Disease-resistant varieties:

Certain mango varieties are tolerant (e.g., Dashehari is susceptible; Langra and Chausa can exhibit relative tolerance).



Malformation of Mango

Bacterial blight of mango disease

Common Name: Bacterial Blight / Black Spot / Canker of Mango

Host: Mango (*Mangifera indica* L.)

Disease Category: Bacterial disease

Causal Organism: The bacterium *Xanthomonas campestris* pv. *mangiferaeindicae* causes the disease.

Symptoms

On leaves

Small, angular, water-soaked spots occur on the veins or margins of leaves.

The spots increase in size, turn brown or black, and may be surrounded by a yellow halo.

In acute cases, the spots join together, resulting in leaf blight and premature defoliation.

On twigs and branches

Development of dark brown to black cankerous spots on twigs, which can crack and ooze bacterial gum.

Severe infection can result in dieback of young twigs.

On fruits

Immature fruits develop raised, black, cankerous spots.

The spots can crack and ooze gummy material.

Infected fruits can fall prematurely or become of poor quality.

Etiology

Characteristics of the pathogen:

The bacterium is rod-shaped, single polar flagellate, and develops yellow colonies on culture media.

Survival

The pathogen overwinters in cankerous lesions on twigs, leaf scars, and infected plant trash.

Spread

Spread is through wind-blown rain, water used for irrigation, and inoculum-laden tools.

Insects and mites could assist transmission.

Favorable conditions

Warm and humid weather (optimum temperature ~25-30°C).

Frequent rains and storms facilitate disease spread.

Management

Cultural practices

Remove and destroy infected twigs and cankered branches.

Avoid twining and proper orchard sanitation.

Sterilize pruning tools in 0.1% mercuric chloride or 10% bleach solution.

Chemical control:

Spray Streptomycin sulphate (200 ppm) + Copper oxychloride (0.3%) at intervals during periods of active growth.

Bordeaux mixture (1%) can also be sprayed to lower the disease incidence.

Resistant varieties:

Some mango varieties such as Langra and Chausa exhibit relative tolerance against bacterial blight.



Bacterial Blight of Mango

Powdery Mildew of mango disease

Alternate name: Powdery mildew

Host: Mango (*Mangifera indica* L.)

Disease type: Fungal disease

Causal organism: *Oidium mangiferae* (also referred to as *Acrosporium mangiferae*)

It is an obligate parasitic fungus in the class Ascomycetes and order Erysiphales.

Symptoms

1 On inflorescence (panicles)

There is white or greyish powdery growth of fungus on panicles, floral buds, and flowers.

Infected flowers tend to wither, dry, and fall off prematurely.

Fruit setting is greatly impaired.

2 On young fruits

Young fruits exhibit powdery coating, become deformed, and usually drop off. Occasionally, survived fruits exhibit blemishes or corky tissue on the surface.

3 On leaves (rare)

Infected young leaves exhibit white powdery patches.

Severe infection leads to leaf curling or distortion.

Etiology

The fungus forms superficial mycelium on the host surface.

Conidia are hyaline, single-celled, and chained.

Conidia are wind-dispersed and start new infections.

The pathogen persists on infected plant tissues or in the form of dormant mycelium on shoots and leaves.

Ideal conditions for disease development:

Cool, dry weather followed by moist conditions.

Temperature: 10°C to 25°C is extremely favourable.

High humidity (approximately 70-90%) increases disease severity.

Characteristic during flowering season (December to February in India).

Management

Cultural practices

Destroy and remove infected plant parts following harvest to minimize inoculum load.

Supply suitable aeration in the orchard by wider spacing and tree training.

Avoid excessive nitrogenous fertilizers that favor succulent growth.

Chemical control

Sulfur dusting (250 mesh) at 20-25 kg/ha or spraying wettable sulfur (0.2%) at 10-15 days interval during flowering.

Systemic fungicides: Spray Carbendazim 0.1% or Hexaconazole 0.1% or Tridemorph 0.05% at the beginning of disease and repeat at 10-15 days interval if necessary.

Dinocap 0.05% or wettable sulfur 0.2% also controls.

Integrated management

Integrate orchard sanitation, pruning, and fungicidal spray at vulnerable growth stages (flowering and fruit set).



Powdery Mildew of Mango

Chapter-2

Citrus Disease, Causal Organism, Symptoms, Etiology and Their Management

Canker of citrus disease

Synonym: Citrus canker

Host: Citrus species (lime, lemon, orange, grapefruit, etc.)

Type of disease: Bacterial disease

Causal organism: *Xanthomonas citri subsp. citri* (previously called *Xanthomonas axonopodis* pv. *citri*)

It is a Gram-negative rod-shaped bacterium that is motile with a single polar flagellum.

Symptoms

1 On leaves

Small, water-soaked, translucent spots form at first.

Spots grow to become round, elevated, and corky with a yellow halo (chlorotic margin).

Older lesions can coalesce, resulting in leaf distortion, cracking, or early defoliation.

2 On twigs

The same corky lesions appear, and often this results in twig dieback.

3 On fruits

Fruit lesions are elevated, rough, and corky with a yellow halo.

The rind cracks, and the fruit becomes blemished and deformed.

Severe infection lowers market value and fruit quality.

Etiology

The bacterium grows in the epidermal tissue and induces cell hyperplasia, resulting in the development of corky canker.

The pathogen infects through stomata or wounds (induced by wind, insects, or mechanical injuries).

Wind-borne rain, insects (particularly leaf miners), infected tools, and nursery stock are the vectors.

Favourable conditions:

Warm (20°C - 30°C), wet weather.

Heavy rains with rigorous winds facilitate quick spread.

Insect vectors such as citrus leaf miner (*Phyllocnistis citrella*) favour increased severity of disease.

Management

Cultural practices

Plant new plantations using disease-free nursery stock.

Remove and kill infected twigs, leaves, and fruits.

Implement windbreaks to minimize wind-swept rain dissemination.

Regulate citrus leaf miner numbers to limit entry points.

Chemical control

Spray copper bactericides (e.g., copper oxychloride 0.3% or copper hydroxide) at 15-20 day intervals with new flush emergence.

Use copper sprays in combination with systemic antibiotics (e.g., streptomycin 100-200 ppm) for enhanced control.

Regulatory measures

Quarantine and rigorous inspection of nursery plants.

Prohibition against the movement of infected plant material from the endemic region.

Integrated management

Merge cultural practices, control of leaf miners (using insecticides or biocontrol insects), and bactericidal sprays.



Canker of Citrus Disease

Gummosis of Citrus disease

Local name: Gummosis or Citrus foot rot

Host: Citrus species (orange, lemon, lime, mandarin, etc.)

Disease category: Fungal disease

Causal organism

Fungus: *Phytophthora* spp. (chiefly *Phytophthora palmivora*, *P. parasitica*, *P. citrophthora*)

Symptoms

The disease is mostly seen in the trunk at ground level and the main roots.

Dark brown to black patches are visible on the bark at the collar area.

Gummy exudation (oozing of gum) from these lesions.

The bark softens, rots, and develops longitudinal cracks.

In late stages, the bark sheds, revealing wood.

Yellowing of leaves, defoliation, dwarfing, and dieback of branches can occur in extreme cases.

Poor fruit set, and tree vigour decreases progressively.

Etiology

The cause is the *Phytophthora* fungi, which are soil-borne and persist as oospores, chlamydospores, or mycelium in infected plant debris and soil.

The pathogen prefers:

Poor drainage or water-logged conditions.

Injuries in the collar area as a result of cultural practices or insect injury.

Excess irrigation or waterlogging in the area of the collar.

Warm humid conditions favor the development of the disease.

Management

Cultural practices:

Grow citrus on ridges or raised beds in well-drained soils.

Prevent waterlogging close to the collar area.

Maintain clean tree base and avoid weeds.

Destroy and remove infected bark and dead tissue.

Chemical control:

Spray Bordeaux paste (copper sulfate-lime mixture) or copper oxychloride paste on the trunk following removal of diseased bark.

Soil drenching around the trunk with fungicides such as metalaxyl (0.1%) or fosetyl-Al according to directions.

Rootstock with resistance

Employ resistant or tolerant rootstocks such as trifoliate orange or Rangpur lime where possible.

Other actions

Adequate spacing between trees to enhance air circulation.

Early detection and treatment of minor infections through regular monitoring.



Gummosis of Citrus

Chapter-3

Grape Vine Disease, Causal Organism, Symptoms, Etiology and Their Management

Powdery mildew of Grape vine disease

Common name: Powdery mildew of grape

Host: Grapevine (*Vitis vinifera* and allied species)

Disease category: Fungal disease

Causal organism

Fungus: *Uncinula necator* (syn. *Oidium tuckeri*)

Symptoms

The disease infects all aerial parts of the vine: leaves, shoots, flowers, and berries.

White to greyish powdery fungal growth on young leaves, shoots, and berries.

Infected leaves can curl, wrinkle, and exhibit distorted growth.

Flower clusters can fall prematurely (poor fruit set).

Infected berries can be small, hard, crack, or shrivel.

Infected fruit skin cracks, making them susceptible to secondary infections.

Etiology

The fungus exists as cleistothecia (sexual fruiting bodies) on plant debris or as latent mycelium in buds.

Spores (conidia) are spread by wind.

Favorable conditions for the disease include:

Warm and dry weather (optimum temperature 20–27°C).

High humidity (but free water is not required for germination of the spores).

Compact canopy that restricts air movement and raises the humidity near the leaves.

Management

Cultural practices:

Train and prune vines correctly to promote adequate air movement and sunlight penetration.

Destroy infested plant material and remove infected shoots.

Chemical control:

Sulfur dusting (fine dust sulfur) at 20–25 kg/ha early morning or late evening when temperature drops below 30°C.

Spray fungicides like wettable sulfur (0.2%), hexaconazole (0.1%), or myclobutanil (0.04%) 10–14 days apart from pre-bloom stage.

Alternate fungicides to prevent resistance build-up.

Resistant varieties:

Apply powdery mildew-resistant varieties if they are available.



Powdery Mildew of Grape Vine

Downy Mildew of Grape Vine

Causal organism) : *Plasmopara viticola* (Berk. & Curt.) Berl. & de Toni

It is an oomycete (water mold) of class Oomycetes, order Peronosporales.

Symptoms

Leaves:

Yellowish oil-spot lesions on upper surface.

Under humid conditions, the spots grow larger and brown.

On the lower side, white downy growth (sporangia) is observed opposite to the spots.

Severe infection causes curling, browning, and early defoliation.

Shoots and young stems:

Water-logged lesions which subsequently become necrotic.

Inflorescences and young berries:

Shriveling, browning and killing of flower clusters and young berries due to infection.

Older berries become leathery and hard ("leather berry").

Etiology

Pathogen nature: Obligate parasite with intercellular mycelium and haustoria.

Sporangia: Lemon-shaped sporangia borne on branched sporangiophores coming out through stomata.

Zoospores: Biflagellate, zoospores released from sporangia in water films.

Sexual stage: Development of oospores (thick-walled resting spores) in infected tissues which assist in overwintering.

Favorable conditions:

High humidity (>85%), rain, and moderate temperature (20-25°C).

Poor air circulation and wet weather facilitate spread of disease.

Disease cycle

Survival of oospores in fallen leaves and soil, germination to form sporangia.

Zoospores are discharged by sporangia, infecting young tissues.

Secondary infection by wind-blown sporangia and rain splash.

Management

Cultural practices

Employ resistant/tolerant varieties (e.g. 'Champanel').

Correct pruning and training for enhancing air circulation.

Remove and destroy fallen leaves and debris to minimize primary inoculum.

Chemical control

Prophylactic sprays with copper-based fungicides (e.g. 0.5% Bordeaux mixture) at 3-4 leaf stage and repeat at every 10-15 days interval during susceptible stages.

Apply systemic fungicides such as metalaxyl + mancozeb or dimethomorph during active growth phase.

Integrated management

Integrate cultural and chemical methods for effective control.

Watch weather and forecast infection periods for timely sprays.



Downy Mildew of Grape Vine

Anthracnose of Grape vine disease

Causal Organism

Grapevine anthracnose is produced by the fungus *Elsinoë ampelina* (de Bary) Shear. The asexual (conidial) phase of this fungus is *Sphaceloma ampelinum*.

This pathogen infects other parts of the vine to cause characteristic lesions, resulting in severe damage if not controlled.

Symptoms

On the leaves, the disease initially manifests as minute, circular spots. These spots eventually become grayish-white centered with a clear dark brown or black border, producing a pattern which is generally called bird's-eye spots. Later, as the disease advances, the lesions might coalesce, and irregular holes in the leaves could be formed, making them ragged or tattered in appearance.

On the tendrils and shoots, infection results in the formation of sunken, dark cankers or lesions. In severe cases of the disease, cankers result in breaking and cracking of shoots, and occasional death of the infected shoots, which makes the vine structure weak.

On the fruits, minute, round, depressed spots with dark edges are observed. As these spots grow, they make the fruits crack, shrivel, or fall prematurely, which results in direct loss in fruit yield and quality.

Etiology

The fungus forms small, cushion-like fruiting bodies known as acervuli on infected plant surfaces. The acervuli develop conidia, which are the asexual spores that cause infection during the growing period.

The conidia are dispersed mainly by rain splash and wind-borne rain, allowing the fungus to infect adjacent healthy tissues, particularly in wet weather.

The disease overwinters as mycelium or stromata in dormancy in the cankers on ancient wood or in infected debris retained in the vineyard.

Pathogen development is strongly favored by hot and humid conditions, with an ideal temperature of around 24°C and high humidity. Wetness for extended periods of time presents a perfect scenario for spore germination and infection.

Disease Cycle

The disease cycle proceeds with survival of the pathogen in the form of dormant mycelium or stromata in cankers on old wood or in debris from the previous season's plant material.

During the spring, when temperature and moisture are conducive, these overwintered structures form conidia, which are disseminated by rain splash to young, susceptible areas, such as shoots, leaves, and developing berries.

When initial infections have become established, cycles of recurring conidial production and new infection can continue under wet weather, enabling the disease to become widely distributed and cause significant damage throughout the season.

Management

Cultural Practices

Growing resistant or tolerant grape varieties, e.g., Isabella, serves to decrease the amount and severity of the disease.

Destruction and pruning of infected shoots and cankers during the dormant period is a critical sanitation practice for decreasing the source of primary inoculum.

Use adequate vine training and spacing to increase aeration in the canopy and reduce humidity, thus minimizing the potential for disease development.

Chemical Control

There should be a dormant season spray of lime sulfur (10%) prior to bud break to kill the overwintering inoculum on cankers and on the ground.

At the growing period, protective fungicide sprays of Bordeaux mixture (0.5%) or copper oxychloride at fortnightly intervals of 10-15 days during the periods of susceptibility between bud break and fruit set should be given.

Systemic fungicides, e.g., carbendazim (0.05%) or thiophanate-methyl, can be applied during the growing stage for the control of well-established infections and to check further development.

Integrated Disease Management

An integrated method consisting of on-time pruning, removal of diseased plant components, and properly planned fungicide spray regimes provides the best management of the disease.

Constant monitoring of the weather ensures forecasting high-risk durations and adjusting the schedule for fungicide sprays to get maximum impact.

Favorable Conditions

The disease flourishes under prolonged wetness resulting from rain or splash irrigation, with warm temperatures of between 20°C and 25°C. A high canopy that traps and holds moisture also plays a critical role in disease development and its quick spread.



Anthrachnose of Grape Vine

Chapter-4

Apple Disease, Causal Organism, Symptoms, Etiology and Their Management

Scab of Apple disease

Causal Organism

Apple scab is caused by the fungus *Venturia inaequalis* (Cooke) Wint. The sexual phase of the pathogen is called *Venturia inaequalis*, and the asexual phase is called *Spilocaea pomi*. It is a fungus that belongs to the phylum Ascomycota and the order **Venturiales**.

Symptoms

On the leaves, the disease starts out as tiny, olive-green to brown, fuzzy spots that usually first occur on the lower surface of the leaves. With further infection, the spots enlarge and turn black and cause leaf curling, distortion, and sometimes early leaf drop, making the tree weak.

On the fruit, the disease causes round, olive-black, slightly sunken lesions with a cracked, rough surface. The lesions frequently result in fruit deformity, rendering the apples unmarketable and unattractive. At severe infection, the fruits are small, might crack, and have poor commercial quality.

On the twigs and flowers, young twigs can develop small, dark lesions. On severely infected flowers, these can wilt and fall early, thereby diminishing fruit set and yield.

Etiology

The pathogen produces pseudo-thecia, which are specialized fruiting structures, on dead leaves in the fall. Within these pseudo-thecia, asci that bear ascospores develop throughout the winter and mature by spring.

Under rainy and cold weather in spring, ascospores are forcibly ejected from pseudo-thecia and wind- or splash-carried to infect young leaves and fruitlets to cause primary infections.

When the primary lesions appear, conidia (asexual spores) are formed on them by the fungus. Conidia are disseminated during wet weather and cause secondary infections throughout the season.

The disease is favored by cool, wet weather conditions, with an optimum temperature range of 12°C to 24°C. The risk of infection is high when leaves remain wet for extended periods (more than 9 hours). Both ascospores and conidia are spread primarily by rain and wind.

Disease Cycle

Overwintering pathogen occurs as pseudo-thecia in the dead leaves that are shed from the base of apple trees. The mature ascospores are shot and dispersed by spring rains, causing primary infections on young, susceptible tissues like new leaves and fruitlets.

Conidia developing on these primary lesions are disseminated by rain splash or wind-borne rain during the season and cause the secondary spread of the disease during wet weather conditions.

Management

Cultural Practices

Falling leave collection and destruction during autumn is a significant sanitation practice that minimizes the inoculum present at the end of the overwintering period.

Introducing resistant or tolerant apple cultivars like Prima or Priscilla minimizes the intensity and transmission of scab.

Good pruning and training of the trees enhance air movement within the canopy, thereby lowering humidity and leaf wetness duration, which in turn makes it less favorable for the disease.

Chemical Control

Spraying with a 5% urea solution on autumn shed leaves accelerates the breakdown of leaves and drastically decreases ascospore production in the spring.

Fungicides need to be applied during main growth stages, such as the green tip, pink bud, petal fall, and fruit development stages. Suitable fungicides are Captan (0.2%), Mancozeb (0.25%), and systemic fungicides such as difenoconazole or myclobutanil, according to recommended schedules and rotation guidelines to prevent resistance.

Integrated Management

A blend of sanitation operations (e.g., removal and destruction of leaf litter) and on-schedule fungicide applications in accordance with disease forecasting models or weather monitoring is the most effective control of apple scab.

Prompt protection of juvenile and differentiating tissues is imperative, as early fruit infection can result in permanent scarring and deformity.

Favorable Conditions

Apple scab is most serious in cool and wet conditions, especially in areas with repeated rain showers in spring. The disease increases when temperatures are between 12°C and 24°C and when leaves remain moist for prolonged periods. Close canopies that hold water also contribute to a greater possibility of disease occurrence.



Scab of Apple

Powdery mildew of Apple

Causal Organism

Apple powdery mildew is produced by the fungus *Podosphaera leucotricha* (Ell. & Ev.) Salm. It is a member of phylum Ascomycota and order **Erysiphales**. It is one of the most important fungal pathogens of apple trees in most of the apple-producing areas of the world.

Symptoms

On the leaves, the disease initially manifests as small white powdery patches of mycelium and conidia of the fungus. These are most evident on young leaves, particularly on their lower surface during early infection. As the infection advances, the leaves become curled, crinkled, and deformed as a result of destruction by the fungal growth. Infected leaves tend to fall off prematurely, which can cut down the leaf photosynthetic area of the tree and impair its vigor.

On the new shoots, a powdery white covering is found on the surface. The covering is made up of fungal mycelium and spores. The shoot growth is usually weakened, and if it is a severe infection, the tips of the shoots may die back, causing loss of new growth.

On flowers, infection can lead to the flowers becoming deformed, and in most instances, the flowers do not open or produce fruit. When fruits become infected, they are apt to develop abnormally and end up small and distorted. Infected fruits exhibit a netted russetting appearance on the surface and are usually of lower market value and attractiveness.

Etiology

The fungus forms an ectophytic mycelium, with the fungal filaments growing on host tissue surfaces. The fungus then forms haustoria through the host epidermal cells to feed.

During the asexual reproduction phase, the fungus produces profuse conidia in chain on the surface of infected tissues. Conidia are loosely attached and can be easily blown by wind, thus causing quick secondary spread of disease during vegetative growth.

The fungus may reproduce sexually, producing cleistothecia (closed fruiting bodies), but in apple trees of India, these are not commonly seen.

The fungus overwinters in the form of dormant mycelium within infected buds. The infected buds are the source of primary inoculum during the following growing season.

Disease is most favored under warm, dry weather conditions between 15°C and 25°C. Though high humidity facilitates conidial germination and infection, free water (rain, dew) is not needed by the fungus for its spores to germinate and infect. This is a special characteristic that makes powdery mildew different from most other fungal pathogens.

Disease Cycle

During the winter, the pathogen overwinters as mycelium in infected, dormant apple tree buds. When buds terminate dormancy during spring, they emit powdery mildew-covered infected shoots. These infected shoots serve as the main inoculum source for the year.

From these initial infections, conidia are formed on the surface of infected tissues. These spores are wind-dispersed over long and short distances and infect healthy shoots, leaves, flowers, and fruits during the growing period.

Management

Cultural Practices

Pruning and removal of infected shoots and buds during the dormant period are essential to remove the overwintering inoculum and decrease the severity of spring primary infections.

Utilization of resistant or tolerant cultivars (where they are available) can assist in minimizing the probability of serious outbreaks.

Effective pruning and training of trees enhance air movement inside the canopy, decreasing humidity and deterring disease development.

Chemical Control

In the winter dormant season, spray lime sulfur before bud break to exterminate overwintering mycelium in infected buds.

During the flowering season, repeated sprays of wettable sulfur (0.3%), Karathane (0.05%), or systemics like hexaconazole (0.05%) and myclobutanil (0.02%) at 10-15 day intervals during the period of active growth assist in maintaining the disease

in check. Use of fungicides should be initiated as soon as the symptoms appear or according to the disease forecasting.

Integrated Management

An integrated strategy incorporating sanitation practices (like pruning and elimination of the infested parts) along with a scheduled fungicide spray regimen gives maximum control of powdery mildew in apple orchards.

Monitoring of orchards on a regular basis for the early discovery of infected shoots and regular removal of these inoculum sources are key elements of integrated disease management.



Powdery Mildew of Apple

Fire blight of Apple disease

Causal Organism

Apple fire blight is caused by the bacterium *Erwinia amylovora* (Burrill) Winslow et al. The pathogen is a member of the family Enterobacteriaceae and is among the most devastating of apple and other members of the **Rosaceae** family diseases (e.g., pear, quince).

Symptoms: The disease occurs in different parts of the apple tree:

Blossoms: Infected blossoms are water-soaked at first, later browning or blackening and shriveling, usually still attached to the spur. The whole blossom cluster can fall.

Twigs and shoots: The infection progresses from blossoms into developing shoots, resulting in the typical shepherd's crook bending of the tip of the shoot. Infected shoots present with blackened, shriveled, and killed tissue, appearing as if they had been fire-scorched.

Leaves: Infected leaves of shoots turn brown or black and become persistently attached to the tree.

Fruits: Fruits are immature with water-soaked lesions that become dark and sunken. These fruits later dry up and become black, persistently attached to the tree in the form of "mummies."

Cankers: Cankers can occur as perennial cankers on branches, twigs, or the trunk at infection areas. Cankers consist of sunken, split bark having dark, wet margins and ooze amber-colored bacterial exudate in humid weather.

Etiology

Erwinia amylovora is a Gram-negative rod-shaped bacterium with peritrichous flagella that provides it with motility.

The bacterium survives overwintering in the borders of cankers that develop during the earlier season. In spring, with the warm temperatures and resumption of growth

by the host, the bacteria become active and multiply at the margins of cankers. The disease causes sticky, bacterial ooze, which is the main inoculum.

This ooze is transmitted by rain splash, wind, insects (bees, flies, ants), and infected tools. It enters the host tissues through natural openings (nectaries, stomata, lenticels) or wounds from pruning, hail, insects, or mechanical injuries.

Warm temperatures (24°C-29°C) and humid conditions are conducive to development of disease, especially during and following blossoming time.

Disease Cycle

The bacterium survives winter in cankers on twigs, branches, or trunks. In spring, the bacteria multiply and ooze out from these cankers. This ooze is spread to blossoms and young shoots via rain, wind, and insects.

Once primary infections occur in blossoms, the bacteria spread internally to shoots, leaves, and fruits. Secondary spread continues through the season whenever conditions are favorable, especially after rains or storms that cause wounds.

Management

Cultural Practices

Remove and kill infected twigs and cankers at dormancy and growing season, making cuts at least 20-30 cm below symptoms. Clean tools between cuts (e.g., with 10% bleach or alcohol).

Choose resistant or tolerant cultivars when available.

Prevent overfertilization with nitrogen since dense growth is more prone to infection.

Enhance air movement by good spacing and pruning to limit humidity surrounding plant parts.

Chemical Control

Use copper-based bactericides (e.g., copper oxychloride) or antibiotics (e.g., streptomycin sulfate 100 ppm) at critical times like pre-bloom, bloom, and post-bloom stages, according to local rules and suggestions.

Sprays must be scheduled to correspond to blossom opening and repeated under risky weather (warm, wet).

- ✓ Removal and destruction of affected parts.
- ✓ Removal of blighted twigs
- ✓ Spray with Streptomycin 500 ppm.

Integrated Disease Management

An integrated program incorporates sanitation (pruning, canker removal), utilization of disease-free planting material, correct nutrition and irrigation management, and timely protective sprays.

Weather-based models may be employed to forecast at-risk time periods and inform the timing of sprays to avoid blossom infections.

Key Points

Fire blight infects and spreads very quickly in warm, humid weather, particularly during and following bloom.

Disease spread can be minimized by early detection and removal of infected tissues.

Timely intervention and regular monitoring can easily reduce fire blight losses to a considerable extent.



Fire Blight of Apple

Chapter-5

Peach Disease, Causal Organism, Symptoms, Etiology and Their Management

Leaf Curl of Peach

1. Causal Organism

The pathogen is a fungus called *Taphrina deformans*.

The fungus falls in the category Ascomycetes and mainly infects the young and growing leaves of peaches. It overwinters in dormant forms and becomes active during conducive conditions.

2. Symptoms

The most characteristic symptom is curling, thickening, and puckering of the young peach leaves, which get distorted and twisted.

Affected leaves first develop reddish or purplish color, which gradually becomes yellow to brown as the infection advances.

Prematurely fallen severely infected leaves greatly decrease the photosynthetic ability of the tree.

In a few instances, the young shoots also become swollen, stunted, and distorted.

The fruits can get blemishes, abnormal shape, and poor quality, and in high infections, they can fall prematurely, causing economic loss.

3. Etiology

Taphrina deformans fungus overwinters in the form of ascospores or conidia on the bark, twigs, and buds' surface.

Infection starts early in the spring, particularly under cold and damp weather (optimum temperature 10°C to 20°C).

The spores germinate and infect the new, soft, and expanding leaves when the bud break occurs.

The fungus develops intercellularly, leading to hypertrophy and hyperplasia of cells, and results in curling and thickening of the leaf.

4. Disease Cycle

The pathogen overwinters in a resting condition on bud scales and clefts of the bark of infected trees.

When favorable environmental conditions are favorable in spring, the resting spores become active and induce primary infection on new leaves and shoots.

The disease is monocyclic, i.e., it goes through only a single infection cycle in a season, and there is no secondary infection during the same cropping season.

5. Management

A. Cultural Practices

Inspect trees regularly and destroy infected shoots and leaves to limit inoculum.

Prune the trees during dormancy for enhanced air circulation and sunlight penetration, lessening humidity surrounding the leaves.

Prevent overhead watering or overwatering at bud break since it promotes fungal infection.

B. Chemical Control

Fungicide sprays during the dormant season (late winter or early spring) are most effective in disease control.

Apply Lime Sulfur (3%), Copper Oxychloride (0.3%), or Chlorothalonil (0.2%) prior to swelling of the buds.

If there is rain at the time of bud swelling, application of fungicide should be repeated to provide protection.

- Remove and burning of infected shoots.
- Spray the plants with Bordeaux mixture 1% or 0.1% Carbendazim.
- Spray Mancozeb 0.25 % at 20 days before harvesting.

C. Resistant Varieties

When available, farmers must select resistant or tolerant peach varieties as a preventive measure to minimize the occurrence of the disease.



Leaf Curl of Peach

Brown rot of Peach disease

1. Causal Organism

Brown Rot of Peach disease is produced by the fungal pathogen *Monilinia fructicola*.

This is an Ascomycota fungus that infects peach trees, but not only them. Other stone fruits, like plum, cherry, and apricot, can also be infected.

It is a virulent pathogen which grows in hot and humid conditions and results in rotting of blossoms, shoots, and fruits.

2. Symptoms

A. On Flowers

The infected flowers of peach exhibit wilting and browning.

These blossoms do not fall from the twigs and remain attached, and this serves as a persistent source of the fungal spores for subsequent infection.

This is known as blossom blight.

B. On Twigs and Shoots

The fungus is spread from infected flowers to new twigs and shoots, causing twig blight or canker development.

Cankers are sunken, dark brown lesions and, if they encircle the twig, can cause shoot dieback.

C. On Fruits

The most noticeable symptom is seen on the fruits in the form of soft, brown, circular spots in the ripening phase.

These brown spots spread quickly, and the infected fruit softens, shrivels, and later turns mummified.

Infected fruits under high humidity bear grayish-brown spore masses in concentric rings.

These mummified fruits usually hang on the tree and serve as a reservoir for next season's infection.

3. Etiology

Fungus ***Monilinia fructicola*** overwinters as mycelium or conidia in mummified fruits, twigs, and cankered tissues.

In spring, under favorable weather conditions, conidia are formed on such infected structures.

They are disseminated by wind, splashing rain, insects, or farm equipment.

Warm and humid weather encourages infection, particularly at a temperature of between 20°C and 25°C with a high relative humidity.

The fungus penetrates host tissues largely by wounds, insect stings, or natural plant openings like stomata.

Once it's established within the plant, it releases enzymes that dissolve the cell walls, resulting in rotting and tissue collapse.

4. Disease Cycle

The disease has a polycyclic disease cycle, that is, there are several infection cycles within a single growing season.

It overwinters as latent spores or mycelium within mummified fruits and twig cankers.

When spring sets in, the spores germinate and induce primary infection on young shoots and flowers.

In the growing season, secondary infections may repeatedly take place on growing fruits, especially during wet, warm weather, thereby increasing the intensity of disease.

5. Management

A. Cultural Practices

Destroy and remove all infected twigs, mummified fruits, and shed fruits during the dormant season to minimize inoculum load.

Prune and eliminate cankered branches to prevent the infection spreading and encourage new healthy development.

Ensure proper spacing of plants and carry out regular pruning of the canopy to maintain adequate air circulation and sunlight penetration, which reduces humidity.

Avoid fruit damage by careful picking and handling and timely pest management, since wounds provide entry sites for the fungus.

B. Chemical Control

To properly control the disease, use protective fungicidal sprays at the critical growth phases:

- ✓ Pre-bloom stage
- ✓ Full bloom stage
- ✓ Fruit development stage
- ✓ Pre-harvest stage
- ✓ Recommended fungicides are:
- ✓ Chlorothalonil (0.2%)
- ✓ Captan (0.2%)
- ✓ Propiconazole (0.1%)
- ✓ Carbendazim (0.1%)
- ✓ Fungicide sprays should be re-done particularly after rain or during humid weather to provide cover.

C. Biological and Resistant Methods

Application of biocontrol agents like *Trichoderma harzianum* has proven effective in the control of the fungal population.

Choose and plant peach selections exhibiting resistance or tolerance to brown rot, if such options exist in the area.

Promote integrated disease management (IDM) using a mix of cultural, chemical, and biological control techniques for sustainable disease management.



Brown Rot of Peach

Chapter-6

Strawberry Disease, Causal Organism, Symptoms, Etiology and Their Management

Leaf Spot of Strawberry disease

1. Causal Organism

Leaf Spot of Strawberry disease is caused by the *Mycosphaerella fragariae* fungus, whose asexual or anamorphic form is *Ramularia brunnea*.

It belongs to the Ascomycota division and mainly attacks the strawberry plant leaves.

Under optimum environmental conditions, it infects the rest of the plant, including petioles, stolons (runners), and caps of the fruit, causing extensive damage if not controlled in time.

2. Symptoms

The disease starts with the development of small, round, dark purple lesions on the upper side of young strawberry leaves.

As the disease advances, the center of the lesions will be pale gray or white, with the peripheral edge still being colored purple, and thus a characteristic "frog-eye" effect is produced, which is a diagnostic sign.

When infection is severe, the spots become many and merge, causing chlorosis (yellowing) and early droppings of the leaves, reducing photosynthesis and overall plant vigor.

Infected leaves become brittle, blotchy, and can also die prematurely, particularly in cold and humid conditions.

In later stages, petioles (leaf stalks), runners (stolons), and the caps of fruits can also be attacked by the fungus, developing narrow elongated purple to brown lesions, further decreasing plant health and yield.

Disease

3. Etiology

The fungus overwinters as dormant mycelium or conidia on diseased plant refuse or as latent infection on green strawberry plants.

Under favorable conditions, particularly low temperatures (15–25°C) and high humidity, air-borne conidia infect healthy leaves and start the infection process.

The spores of the fungus germinate on wet leaf surfaces and penetrate into the plant tissue either through stomata or direct penetration of the cuticle.

After entering the leaf, the fungus infects intercellularly, colonizing the tissue and producing typical spots.

Dew, rain, or irrigation water supplies moisture necessary for spore germination and infection success.

The disease is disseminated by wind-borne spores, rain splash, and infested tools or plant material, so sanitation is the key to control.

4. Disease Cycle

The fungus *Mycosphaerella fragariae* is polycyclic, meaning it will have several infection cycles within a single growing season.

It survives the winter in dead and diseased leaves as a source of primary inoculum during spring.

When temperature increases and humidity is greater in spring, the fungus gets activated and starts producing conidia.

These conidia are distributed to healthy young leaves, where new infections are caused.

During the growing season, if conditions are wet and humid, secondary infections develop very fast, causing epidemic outbreaks in poorly controlled fields.

Densely planted and lacking in air circulation, it forms a microclimate conducive to disease development and accumulation.

5. Management

A. Cultural Practices

Remove and destroy infected leaves, crop residues, and plant trash frequently after harvest or pruning to avoid the source of infection.

Don't employ overhead irrigation, which prolongs leaf wetness duration to suit the development of fungi.

Provide good drainage for the strawberry crop and prevent waterlogging.

Implement a crop rotation system by not cultivating strawberries in the same field perennially for two or more seasons.

Ensure proper space between the plants to facilitate good air circulation and minimize humidity within the canopy.

Employ certified, fungus-free strawberry runners or transplants at all times to avoid the inoculation of the fungus.

B. Chemical Control

Use protective fungicidal sprays during the crop season, particularly when rain or dew is common.

Start spraying at the initial infection, and repeat at 10–14 day intervals, as weather and disease pressure dictate.

Typical fungicides listed for control are:

- ✓ Chlorothalonil @ 0.2%
- ✓ Mancozeb @ 0.25%
- ✓ Copper oxychloride @ 0.3%
- ✓ Azoxystrobin @ 0.1%

Alternate fungicides with different modes of action to prevent resistance building up in the pathogen population.

C. Resistant Varieties

As far as possible, grow strawberry varieties resistant or tolerant to leaf spot disease.

Obtain advice from local agricultural extension offices or research centers for the most appropriate and resistant cultivars suitable for your area.

Include resistant cultivars with sound cultural and chemical practices to achieve effective and sustainable disease control.



Leaf Spot of Strawberry

Powdery mildew of Strawberry disease

1. Causal Organism

The disease Powdery Mildew of Strawberry is induced by the fungus *Podosphaera aphanis*, previously known as *Sphaerotheca macularis*.

This Ascomycota fungal pathogen infects different parts of the strawberry plant like leaves, flowers, fruits, stems, runners, and petioles.

Podosphaera aphanis is an obligate parasite, and as such, it can survive and multiply only on the living tissues of hosts and not on dead organic matter.

Fungus thrives in dry conditions with high humidity and leads to heavy losses in both quantity and quality when not controlled.

2. Symptoms

On Leaves

The symptoms begin as reddish or purplish patches on the lower side of young leaves.

These lesions eventually turn white to grayish powdery growth of the fungus, later extending to the upper leaf surface.

On progression of infection, leaves curl upwards at the margins, and leaves can become brittle, dwarfed, and exhibit twisted growth, hence impairing photosynthesis and the overall health of the plant.

On Flowers

Infected flowers usually don't open fully or turn brown, wrinkled, and withered, which leads to inadequate fruit set or deformed fruits.

On Fruits

Fruits, particularly the green or ripening fruit, can have a whitish powdery appearance, making them unsightly and unsellable.

Infected fruits can be small, dry, and poorly formed and usually seedy or leathery in appearance.

In acute infections, fruits can crack, become distorted, and even grow off-flavor, decreasing consumer acceptability and market price.

On Runners and Petioles

Powdery fungal development can also occur on stems, runners, and petioles, which become elongated, twisted, or weakened by infection, decreasing vegetative propagation and plant vigor.

3. Etiology

The causal fungus ***Podosphaera aphanis*** winter on cleistothecia (resting structures) or mycelium in quiescent buds, infected leaves, and field debris.

With favorable environmental conditions, such as warm days (15–27°C) with high humidity (60–80%), the fungus releases copious conidia (asexual spores) on diseased plant surfaces.

In contrast to most other fungal pathogens, free water is not necessary for the germination of powdery mildew spores. This enables the disease to develop in relatively dry conditions and makes it especially risky in covered cultivation systems such as polyhouses or greenhouses.

The fungus enters the plant tissues by puncturing epidermal cells, where it forms haustoria — specialized structures employed to get nutrients from host cells, weakening the plant.

4. Disease Cycle

The powdery mildew fungus overwinters in a quiescent state as mycelium or cleistothecia in infected leaves, buds, or debris.

During the subsequent season, with the increase in temperature and conducive humidity, ascospores are discharged from cleistothecia, and mycelium again develops, forming huge quantities of conidia.

Conidia are windborne and cause new infections on young leaves, stems, and flowers.

The disease develops swiftly under conducive conditions and is polycyclic, and more than one infection cycle will be completed during a single crop season.

Compact canopies, high humidity, and restricted air flow drastically enhance the growth rate of the disease, particularly in intensive culture systems.

5. Management

A. Cultural Practices

Select planting locations with proper air movement and sufficient sunlight to discourage fungal development and lower surrounding humidity.

Prevent the application of excessive nitrogen fertilizers, as they encourage lush, succulent growth, which is more susceptible to infection by fungi.

Practice routine sanitation by pulling out and destroying infected leaves, runners, and plant residues to minimize the main inoculum source.

Apply plastic mulch and drip irrigation systems to decrease leaf surface moistures and prevent spore germination-friendly conditions.

Ensure adequate plant spacing and regular pruning of runners to open up the plant canopy and enhance air circulation, which reduces humidity and infection risk.

B. Chemical Control

Start fungicide applications early in the season, particularly when warm, dry days and humid evenings are forecasted perfect for powdery mildew growth.

Spray fungicides preventatively and spray regularly at 7–10 day intervals during periods of high risk.

Fungicides recommended include:

- ✓ Wettable Sulfur @ 0.2% – efficient and cost-effective.
- ✓ Propiconazole @ 0.1% – a systemic triazole fungicide.
- ✓ Azoxystrobin @ 0.1% – a strobilurin fungicide with extensive activity.
- ✓ Potassium bicarbonate, neem oil, and biological fungicides (such as *Bacillus subtilis*) – beneficial organic options in low-to-moderate pressure situations.

Note: Rotate fungicides with varying modes of action to avert the establishment of resistance in the pathogen population.

C. Resistant Varieties

Whenever possible, plant resistant or tolerant strawberry cultivars that have lower powdery mildew susceptibility.

Consult with local horticultural research stations or extension offices for recommendations on cultivars specific to your area and favorable to your conditions of growth.



Powdery Mildew of Strawberry

Chapter-7

Potato Disease, Causal Organism, Symptoms, Etiology and Their Management

Late blight of potato disease

Causal Organism

Late blight of potato is the disease that's caused by the pathogen *Phytophthora infestans* (Mont.) de Bary. Even though many people mistakenly call it a fungus, in reality, it is an oomycete or water mold, belonging to the phylum **Oomycota**, class **Oomycetes**, order **Peronosporales**, and family **Pythiaceae**. This pathogen is one that's infamous in history, having caused the Irish Potato Famine during the 1840s and one that remains one of the most damaging potato diseases in the world.

Symptoms

On leaves

Small, water-soaked spots are seen on leaf margins or tips in the beginning. They develop quickly into large, brown to purplish-black irregular blotches. On the lower side of infected areas under cool, moist conditions, white, cottony fungal growth (sporulation) is observed. Infected leaves turn necrotic, wilt, and die early, leading to quick defoliation of the plant.

On stems

The infection spreads to the stems, which have dark brown to black, elongated lesions. These can girdle the stem and cause rotting, resulting in sudden death or collapse of the plant.

On tubers

The disease also infects the tubers, with purplish, sunken lesions on the surface. On cutting the infected tubers, the flesh from inside is reddish-brown and granular in nature. The tubers become soft and rotten by secondary infection with bacteria and other fungi, with a repulsive smell.

Etiology

Phytophthora infestans produces both asexual and sexual forms of reproduction.

During asexual reproduction, it bears sporangia on infected plant tissue, which can either germinate or discharge zoospores free-swimming spores that need water to swim and infect.

During sexual reproduction (when both A1 and A2 mating types co-occur), the pathogen bears thick-walled oospores, which enable it to overwinter in the soil and plant material for years.

The disease develops well under cool (10–20°C) and moist conditions, particularly when relative humidity is above 90% and leaf surfaces are wet for over 10 hours. These conditions favor rapid multiplication and development of the disease.

Disease Cycle

The pathogen survives for a long time mainly on infected seed tubers, cull piles, or volunteer plants left in the field.

On infected leaves and stems, sporangia develop during the growing season and are dispersed through wind, rain splash, and farm implements.

When the sporangia infect new potato plants under moist conditions, they discharge zoospores that infect leaves and stems.

Subsequently, in the harvest season, spores from infected leaves can infect tubers, particularly if the soil is moist, resulting in tuber blight and rot in storage.

Management

Cultural Practices

Always plant certified, disease-free seed tubers to prevent the introduction of the pathogen.

Implement crop rotation by not growing potatoes continuously in the same field.

Destroy and uproot volunteer potato plants and waste tubers to remove the sources of the pathogen.

Provide a good spacing and also ensure drainage in the field to minimize leaf humidity.

Harvest tubers under dry weather conditions only to avoid infection during lifting.

B. Chemical Control

Spray protective fungicides such as mancozeb (0.25%), chlorothalonil, or copper oxychloride every 7–10 days, particularly at initial crop stages.

During peak-risk periods, utilize systemic fungicides like metalaxyl + mancozeb, dimethomorph, cyazofamid, or fluopicolide for the prevention of internal infection.

Always rotate fungicides with different modes of action to minimize the possibility of resistance development.

Integrated Disease Management

Integrate sanitation practices, healthy planting material, and judicious use of fungicides for successful control.

Employ prediction systems such as BlightCast to optimize the timing of fungicide applications.

Destroy infected crop debris after harvest and do not leave residues in the field.



Late Blight of Potato

Early blight of potato disease

Causal Organism

Potato early blight is a debilitating foliar disease brought about by the fungal pathogen *Alternaria solani*.

Alternaria solani fungus is in the phylum Ascomycota, class Dothideomycetes, order Pleosporales, and family Pleosporaceae.

Alternaria solani is a necrotrophic organism, i.e., it kills the host cells and obtains nutrients from the dead tissue.

The fungus has a broad geographical distribution worldwide and infects both potato and tomato crops, resulting in yield loss and impact on quality.

Symptoms

On Leaves

Symptoms tend to first appear on the older, lower leaves of the potato plant.

The initial symptoms are small, dark brown to black circular spots.

The lesions spread and become larger with concentric rings that create a characteristic "target board" or "bull's-eye" appearance.

As the disease develops, tissue surrounding the lesions turns yellow (chlorotic), and in extreme cases, the entire leaf dries up, resulting in lost canopy and photosynthesis.

On Stems and Petioles

Infection can also occur on the petioles and stems, where it creates dark brown, dry, and sunken lesions.

These lesions can weaken the stem and result in breakage or lodging of the plant.

On Tubers

In certain instances, the fungus infects potato tubers, resulting in sunken, dark lesions on the surface.

Upon cutting open the tubers, the internal tissues can be corky, dry, and necrotic with a distinct margin between healthy and diseased tissue.

Secondary pathogens may also infect the wounds, causing soft rot or further decay during storage.

Etiology

The infection is brought about by the asexual spores (conidia) of *Alternaria solani*, which are multicellular, dark-colored, and possess both transverse and longitudinal septa.

The conidia are the major means of dispersal and can readily be transported by wind, rain splash, insects, or infected hands and tools.

At a temperature range of 24°C to 30°C in conducive environmental conditions along with the occurrence of leaf surface moisture (from dew, rain, or irrigation), the conidia germinate and cause infection on the plant.

The fungus survives the winter in the crop residue, diseased tubers, soil, and alternate host plants (such as tomato and solanaceous weeds), which serve as sources of primary inoculum for the succeeding crop season.

Disease Cycle

Alternaria solani exists in nature as diseased crop residues, volunteer plants, infected tubers, and soil-borne structures.

During early season growth, conidia formed on crop residues or volunteer plants become wind or rain-splashed to the lower foliage of potato plants.

Conidia germinate on the leaf surface and infect by way of natural openings (stomata) or minor wounds.

Upon entering the plant, the pathogen replicates fast and forms more conidia on the infected tissue, causing secondary infection cycles during the season.

If the weather is favorable (particularly warm and humid climates), the disease can progress rapidly, defoliating, stunted growth, and heavy yield loss.

Management

Cultural Practices

Always begin with certified, disease-free seed tubers to avoid introducing the pathogen.

Practice crop rotation with non-host crops such as cereals or legumes for 2–3 years, to minimize the inoculum in the ground.

Destroy and remove crop residues and infected plant material at harvest, to avoid overwintering of the disease.

Prevent overhead irrigation, which enhances leaf wetness, and irrigate early in the day to permit rapid drying of leaves.

Keep adequate spacing between the plants to promote good air circulation and suppress humidity in the canopy.

Chemical Control

Start fungicide applications preventively as soon as disease symptoms become evident or when weather conditions are conducive.

Apply protectant fungicides like mancozeb, chlorothalonil, or copper oxychloride at 7–10-day intervals.

In severe to moderate infection, switch over to systemic fungicides like azoxystrobin, boscalid, tebuconazole, or trifloxystrobin that have longer residual protection.

Disease free seed tubers should be used for planting.

Removal and destruction of infected plant debris should be done because the spores lying in the soil are the primary source of infection.

Very early spraying with Zineb or captan 0.2% and repeating it for every 15 – 20 days gives effective control.

The variety **Kufri Sindhuri** possesses a fair degree of resistance.

Always rotate between fungicides with a different mode of action to avoid developing resistance among the fungal population.

Integrated Disease Management

Adopt an integrated approach by integrating cultural practices such as field sanitation and crop rotation along with chemical protection using disease forecasting models.

Check the crop frequently, and use fungicides only when required, thereby saving on cost and environment.

Make use of disease-resistant or tolerant potato varieties, if available.



Angular Spot on Leaf



Concentric Rings



Infected Crop

Black scurf of potato disease

Causal Organism

Black Scurf of Potato disease is caused by the ground-living fungus ***Rhizoctonia solani*** Kühn, which is taxonomically placed within the phylum Basidiomycota, class Agaricomycetes, order Cantharellales, and family Ceratobasidiaceae.

This fungus is a saprophytic and weakly parasitic species, able to live for many years in the soil without a host.

The disease is mostly linked with the asexual or imperfect form of ***Rhizoctonia solani***, since the sexual phase (*Thanatephorus cucumeris*) seldom occurs in nature.

Although the pathogen does not invade deep into the potato tuber tissues, it infects the outer part of the tubers and underground plant parts like stolons and stems.

Symptoms

1. Tubers symptoms:

The most noticeable and diagnostic symptom of Black Scurf is the development of black, irregular, and crusty masses on the surface of harvested potato tubers. These black masses are called sclerotia, which are compact, hardened bodies of the fungus.

Sclerotia appear as if the tubers are soiled or dirty, but they cannot be easily washed off with water, which helps differentiate them from soil particles.

While these sclerotia do not directly harm the internal tuber flesh, they significantly decrease the aesthetic and commercial value of the produce, especially for the fresh market and export purposes.

2. Symptoms on Stems and Emerging Shoots

The potato plants from infected seed tubers tend to have delayed or weak sprouting, which is caused by rotting of the sprout or the eye area through fungal infection.

The emerging shoots can rot at the base, particularly in wet or cool soil, resulting in stand loss or uneven plant emergence in the field.

Infected plants can be stunted, chlorotic (yellowish), and show poor vigor as a result of impaired root and stem growth.

3. Underground Stem (Subsoil Parts) Symptoms

The infection could spread to the subsoil stem area where there are sunken cankers or brown to black lesions near the ground or soil line.

In extreme cases, these cankers girdle the stem, severing water and nutrient transmission, which causes wilting and death of the plant, especially in young plants.

This type of stem damage also predisposes the plant to secondary infection as well as stress conditions.

Etiology

The fungus ***Rhizoctonia solani*** exists in soil mainly as sclerotia, which are dense, melanized bodies that can tolerate adverse conditions like dry periods or absence of crops.

These sclerotia serve as the initial inoculum source, remaining dormant in soil or on diseased tubers until infection and germination conditions become favorable.

In contrast to most plant pathogens, ***R. solani*** does not form conidia or spores; it spreads and survives on vegetative mycelial growth and sclerotia.

The best conditions for the pathogen are cool and moist soil, particularly at spring planting, with an optimum temperature of 15°C to 20°C.

The fungus infects the plant at early development stages, like sprouting and emergence of the tuber, when tissues are young and very susceptible.

Disease Cycle

The disease cycle starts with the survival of the sclerotia in soil or on seed tubers for planting.

With favorable environmental conditions (correct temperature and moisture), these sclerotia germinate and develop hyphae, which infect potato sprouts or underground plant structures.

The fungus penetrates by natural openings or through direct penetration of the host tissue and infects the periderm of young tubers, stolons, and stems.

When the crop is mature, the fungus infects the developing tubers with new sclerotia on the surface, which are buried in the skin during tuber maturation.

At harvest time, the tubers can transport the sclerotia to storage or to the following growing season, closing the cycle.

In addition to this type of indirect soil-to-plant transmission, direct transmission through infected plant debris, volunteer plants, or dirty tools and equipment also occurs.

Year-round cropping of potato in one field without rotation causes the development of a large amount of inoculum in soil.

Management

1. Cultural Practices

Employ certificated, disease-free seed tubers which are exempt from black sclerotial masses on their surface.

Take care that the area of planting has well-drained, loose, and aerated soil since poorly drained or compact soils enhance the possibilities of infection.

Follow a 3 to 4-year crop rotation with non-host plants such as cereals, pulses, or oilseeds to minimize the inoculum in soil.

Deep plowing and soil solarization where possible should help minimize pathogen load.

Practice field sanitation by eliminating infected plant residues and volunteer potato plants from the field and vicinity.

2. Chemical Control

Seed tuber treatment prior to planting with contact or systemic fungicides like thiophanate-methyl (0.2%), pencycuron (0.25%), or fludioxonil can effectively minimize seed-borne inoculum.

Fungicides must be sprayed evenly over the seed tubers and dried before planting.

Foliar sprays do not control sclerotia very well, hence preventive seed treatment is more essential.

Soil application of fungicides is costly and not very effective unless applied as part of an integrated practice.

3. Biological Control

Application of biocontrol agents such as *Trichoderma harzianum* or *T. viride*, either soil application or seed dressing, may suppress populations of *R. solani*.

Application of organic amendments such as compost or farmyard manure with *Trichoderma* can increase soil microbial diversity and alleviate disease pressure.

4. Integrated Disease Management

A holistic approach involving the use of clean planting material, crop rotation, field sanitation, biocontrol agents, and fungicidal seed treatment works best in the control of the disease.

Prevent excessive watering or waterlogging, particularly at the seedling stage, to reduce stem and root infection.

Seed quality monitoring and removal of visibly infected tubers is required before planting, particularly in seed production systems.

Key Points

Black Scurf is not a pathological internal rot but is significant because of its cosmetic damage to tubers and its effect on inhibiting crop emergence and vigor.

The availability of sclerotia on tubers degrades market value and may inhibit seed quality.

Long-term control relies on rotation of crops, biocontrol organisms, certified seed, and field sanitation.

The disease is most easily controlled prior to planting because post-harvest treatments cannot remove sclerotia.

Trichoderma-based solutions provide an environmentally friendly, sustainable alternative for disease incidence reduction with no chemical residues.



Black Scurf of Potato

Leaf roll of potato disease

Causal Organism

Leaf Roll of Potato disease is caused by a virus named Potato Leaf Roll Virus (PLRV). PLRV is a phloem-limited virus belonging to the genus Polerovirus and family Solemoviridae. PLRV infects and gets restricted to the vascular tissues (phloem) of the plant. The virus is not seed-transmitted, but the disease is seed-borne through tubers, i.e., if infected tubers are employed as seeds, then the infected plants are also developed. The key method of transmission of PLRV is through aphid vectors, primarily the green peach aphid (*Myzus persicae*). These aphids spread the virus in persistent, circulative, and non-propagative fashion. Once an aphid acquires the virus, it retains the ability to transmit it for life, although the virus does not replicate within the aphid.

Symptoms

On Leaves

The most common and early symptom is the upward rolling of leaf margins, especially in the lower leaves of the potato plant.

The affected leaves become leathery, brittle, and exhibit a stiff texture, often standing upright instead of drooping.

With the advancement of the infection, yellowing (chlorosis) occurs, especially of older leaves.

Severely, the reverse of leaves can be purplish or reddish in color.

An important differential feature is that rolled leaves never unroll even after watering, in contrast to physiological rolling due to drought stress.

On Whole Plant

Infected plants are stunted, with diminished vigor and reduced growth.

The plant develops an erect and upright habit of growth, with conspicuously smaller leaves and less canopy cover.

When the infection is from a diseased tuber (primary infection), the disease occurs early in the crop and involves the whole plant.

When the infection is contracted later in the season through aphids (secondary infection), the disease occurs only on the top leaves and is localized.

On Tubers

There is drastic reduction in tuber formation, and the tubers are small, deformed, and also fewer in number.

When the infected tubers are cut, brown streaks or necrosis are visible in the vascular (phloem) ring, a characteristic feature.

These infected tubers can be a source of primary inoculum if they are planted during the next season.

Etiology

PLRV is confined to the phloem tissues and cannot travel freely in the plant. It is neither mechanically transmissible (in contrast to some other viruses) nor transmitted through the seed, but it is easily transmitted via tubers and aphid vectors. The virus is acquired during feeding on infected hosts by aphids. After a latent period of 12–24 hours, aphids become able to spread the virus to healthy hosts for their entire lifespan. Though, the virus does not infect inside the aphid—it only goes around and stays infectious. Hence, the infected tuber and aphid population are responsible for disease establishment and spread.

Disease Cycle

The disease cycle starts with the planting of infected seed tubers, which produce infected plants. These infected plants serve as virus reservoirs in the field. Aphids that feed on such plants pick up PLRV, and within a latent period of short duration, they become able to transmit the virus to healthy plants. Due to the persistent transmission of the virus, such aphids carry the virus for their lifespan. During migration and colonization of other potato plants, they cause further spread of the virus, especially at the early stages of the crop growth. The virus persists in latent in infected tubers, which are the main inoculum for the following season and thus perpetuate the cycle of disease.

Management

1. Cultural Practices

Epidemic potato appearance: Always plant certified, virus-free seed tubers to avoid bringing in PLRV to new locations.

Rogueing or destruction of diseased plants early from the field prevents virus sources.

Keep plant spacing optimum to promote air circulation and deter aphid colonization.

Reflective mulches or plastic films may prevent aphids from settling on the plants.

Planting should be avoided during peak aphid pressure or aphid migration times.

2. Vector Control

Establish frequent monitoring of aphid populations through yellow sticky traps or field scouting.

Use systemic insecticides like imidacloprid, dimethoate, or thiamethoxam at the beginning of the crop season to avoid aphid accumulation.

Avoid excessive use of insecticides to prevent resistance and keep populations of natural parasites such as ladybird beetles and parasitic wasps.

3. Seed Certification and Tuber Handling

Employ virus-indexed tissue culture tubers or nuclear seed stock which is certified free of viruses.

Keep tubers cool and dark to prevent aphid growth and virus spread.

Shun the use of small and suspicious tubers from previous infected crops as these may carry PLRV.

4. Resistant Varieties

Certain varieties of potatoes might have partial tolerance or resistance to PLRV.

Such varieties may be used particularly in high-risk areas to minimize disease effects.

5. Integrated Disease Management

Follow an integrated strategy using a combination of virus-free seed, early aphid control, proper field sanitation, and coordinated planting.

Spray insecticides on the basis of weather forecasting or during peak aphid flight periods.

Prevent staggered planting, which delays aphid exposure and enhances disease transmission.

Key Points

PLRV is a phloem-limited and systemic virus, transmitted primarily through aphids and diseased tubers.

The virus decreases plant vigor, induces leaf rolling, and has an enormous effect on tuber yield and quality.

The disease is most devastating when infection takes place early during the season.

There is no chemical treatment directly for the virus; hence, vector control and seed use that is healthy are very important.

A combination of preventive cultural measures, prompt aphid control, and virus-free seed application is the best way to control this disease.



Leaf roll of Potato Disease

Chapter-8

Cucurbits Disease, Causal Organism, Symptoms, Etiology and Their Management

Downy Mildew of Cucurbits (Cucurbitaceae)

Causing Organism

Cucurbits downy mildew is caused by a fungus-like oomycete pathogen, scientifically named as *Pseudoperonospora cubensis*.

It is a member of phylum **Oomycota**, which comprises a group of water molds that are different from true fungi but have the same modes of infection.

The pathogen is an obligate parasite, i.e., it needs a living host plant to grow and complete its life cycle. It cannot survive for extended durations independently in the soil without host tissue.

Its life cycle is closely associated with its host plants from the cucurbitaceous family, such as cucumber, pumpkin, melon, squash, and other related vegetables.

Symptoms

The first visible symptoms of downy mildew are light yellow, angular spots on the upper leaf surface of older leaves. The spots are usually bounded by veins, creating a mosaic or net-like appearance.

Later in the disease, a grayish to purplish fungal growth can be observed on the lower (undersurface) side of the leaves. This growth is the sporulation (reproductive stage) of the pathogen.

Infected leaf tissues turn yellow, wilt, and then die, causing blight and defoliation of leaves prematurely.

The disease usually progresses from lower older leaves first and then slowly upwards to the younger leaves.

In a severe infection, the photosynthetic potential of the plant is severely impaired, causing low fruit set, stunted plant growth, and high yield loss.

At times, very heavily infected fields can suffer from total crop loss if not addressed in a timely manner.

Etiology

Pseudoperonospora cubensis forms reproductive bodies referred to as sporangia, which are formed on small stalk-like structures named sporangiophores.

These sporangiophores protrude through the stomata (natural openings of leaves) in the lower surface of leaves, enabling the pathogen to spread its spores.

The sporangia get dispersed mostly by wind and splashing water, which enables the disease to spread quickly plant to plant.

When the environmental conditions are humid and cool, particularly at early morning dew, fog, or rain, the sporangia germinate and infect healthy leaf tissue.

Unlike other pathogens, *Pseudoperonospora cubensis* does not infect the fruits, but the leaf damage weakens the plant and impacts fruit quality, size, and marketability.

The most favorable temperature for disease development is between 15°C to 25°C, and it needs a relative humidity of above 85% to infect and germinate successfully.

Disease Cycle

The pathogen overwinters (survives) as mycelium or spores in contaminated plant debris remaining in the field or on living hosts of cucurbits in the warmer climates, including weeds or volunteer plants.

During the beginning of the growing season, in optimal environment conditions, the initial infection is when sporangia are discharged and infect new hosts.

Following infection, the pathogen enters its asexual life cycle, forming many new sporangia that are blown by wind and splashed by rain to cause secondary infections.

The cycle is repeated throughout the season, particularly on cloudy, wet, or humid days, to cause fast spreading of disease throughout the field.

Without effective control, downy mildew may spread rapidly, destroying the whole crop within weeks.

Management

1. Cultural Practices

Production of resistant or tolerant cucurbit varieties is one of the best methods to control downy mildew since it restricts infection and disease formation.

Provide enough space between plants to enhance air circulation, which decreases leaf wetness and relative humidity surrounding foliage.

Implement sound crop rotation practices not to continuously plant cucurbits in a single field every year but change their location periodically to minimize inoculum accumulation.

Implement drip irrigation or furrow irrigation rather than overhead sprinklers, since overhead watering causes increased leaf wetness, conducive to disease dissemination.

Provide irrigation in early morning hours to enable leaves to dry rapidly during the day.

2. Sanitation

Following harvest, destroy and remove all infected plant material, such as leaves, vines, and stems, to avoid overwintering of the pathogen and starting infection the following year.

Remove weedy cucurbit hosts or volunteer plants that can potentially shelter the pathogen between crops.

Don't plant new cucurbit crops in proximity to infected or newly infected fields to avoid transmission from adjacent sources.

3. Chemical Control

Use of bed system with wide spacing with good drainage and air movement and exposure to sun help to check the disease development.

Spray with Moncozeb 0.2 % or Chlorothalonil 0.2% or Difolaton 0.2% or Ridomil MZ 72 0.1%

Seed treatment with Apron SD 35 @ 2 g./kg. followed by spraying with Mancozeb 0.2% is effective in reducing the disease.

Start preventive sprays with contact fungicides such as mancozeb, chlorothalonil, or copper oxychloride when the weather is conducive to disease.

After symptoms have developed, use systemic fungicides such as metalaxyl (Ridomil), dimethomorph, cymoxanil, cyazofamid, or propamocarb, which are more penetrative in plant tissues and control the internal infection better.

Switch between fungicides that operate through different modes of action to prevent the buildup of fungicide-resistant strains of the pathogen.

Always adhere to recommended intervals and doses noted on fungicide labels for safe and effective control.

4. Biological Control

Biological products like *Trichoderma harzianum* and *Bacillus subtilis* have been found to be promising in inhibiting the growth of downy mildew pathogens.

These products compete with the pathogen, release antifungal metabolites, and can trigger plant resistance mechanisms.

Use of biopesticides or microbial formulations should be incorporated into regular schedules of disease management for sustainable and environmentally friendly control.



Downy Mildew of Cucurbits

Powdery Mildew of Cucurbits

Causal Organism

Powdery mildew of cucurbits is caused by two fungal diseases, i.e., ***Podosphaera xanthii*** (now *Sphaerotheca fuliginea*) and *Erysiphe cichoracearum*.

Of these, ***Podosphaera xanthii*** is regarded as the most widespread and destructive species, particularly in hot and tropical climates.

These two pathogens both belong to the Ascomycota phylum of fungi and are obligate parasites, which means that they are only able to grow and multiply on living host plants.

The fungi leave a thick, powdery white coating made up of fungal mycelium and spores on the surfaces of plants, especially leaves, stems, and occasionally on fruits.

They develop specialized feeding organelles known as haustoria, which invade the outer epidermal cells of the host to take up nutrients without instant death of the host cell.

This shallow growth and nutrient collection seriously compromises the plant, breaks physiological processes, and reduces crop yield.

Symptoms

The most common and distinctive symptom of powdery mildew is the development of white to grayish, powdery fungal growth on upper and lower leaf surfaces, petioles, stems, and sometimes young fruits.

Initially, growth occurs in small, circular, white spots that become increasingly larger in size and fuse to develop into large patches that cover the entire leaf surface.

Infected leaves tend to turn yellowish (chlorotic) near the infected regions, and eventually, they curl, wither, and drop early, lowering the overall photosynthetic activity of the plant.

With further development of the disease, the entire canopy of the plant becomes covered with this powdery surface, badly impairing the health and vigor of the plant.

In later stages, infection may lead to stunted growth, reduced flowering, small fruits, and poor quality fruits, resulting in high yield loss.

In contrast to downy mildew, powdery mildew favors warm and dry climatic conditions and is hence a serious problem in cucurbit production in open fields as well as controlled environments such as greenhouses and polyhouses.

Etiology

The powdery mildew fungi are spread by both sexual and asexual reproduction. Asexually, they form high numbers of conidia (asexual spores) on conidiophores, which are responsible for the powdery nature of the disease.

The conidia can easily be blown by wind and thus disperse the pathogen over long distances and infect the new host very quickly.

Interestingly, conidia do not need free water for germination compared to the majority of other fungal spores.

Moderate temperatures between 20°C and 30°C, and high humidity levels between 50% and 95%, are, however, appropriate for their germination and infection.

Once the spore comes in contact with a host, it germinates directly on the leaf surface, and the fungus penetrates epidermal cells by sending haustoria into them to acquire nutrients.

The fungus is superficial throughout its life, which means that the infection can be observed externally on the plant surface.

At adverse conditions, the pathogen may survive as resting bodies of chasmothecia (sexual fruiting structures), mycelium within plant debris, and hence overwinter and cause infection during the subsequent season.

Disease Cycle

The disease cycle of powdery mildew starts with the survival of the pathogen within crop residues, on volunteer cucurbit plants, or on alternate weed hosts during the off-season.

Throughout this stage, the fungus is either as chasmothecia or overwintering mycelium, depending on climatic factors.

Upon the advent of favorable weather conditions during the growing season, e.g., mild temperatures and moderate humidity, the resting structures give rise to conidia, which are the major source of infection.

These conidia are spread by air currents and infect neighboring susceptible cucurbit hosts, leading to the first infections (primary infections).

After the disease becomes established, the pathogen continues to grow asexually, producing new conidia that are distributed to cause secondary infections.

In tight plant canopies or poorly ventilated settings, particularly in greenhouses, the disease may explode, infecting entire crops in days.

The disease cycle persists as long as environmental conditions favor it, frequently necessitating repeated management controls throughout the crop growing season.

Management

1. Cultural Practices

Choose and cultivate cucurbit varieties with powdery mildew resistance or tolerance to minimize incidence of disease.

Provide adequate spacing between the plants to enhance air movement and lower the humidity near the leaves, which inhibits fungal growth.

Refuse to overuse nitrogen-containing fertilizers, as they encourage luxuriant, succulent plant growth, which is more subject to infection.

Practice crop rotation, particularly with non-host crops, to interrupt the disease cycle and avoid inoculum accumulation in the soil.

Destroy all infected plant material and debris at harvest time, as they may carry the fungus for the subsequent season.

Weed control should also be regular, since most alternate hosts serve as reservoirs for the pathogen.

2. Chemical Control

In early disease stages, use protective fungicides such as wettable sulfur, dinocap, or horticultural oils (e.g., neem oil).

For advanced infections, utilize systemic fungicides such as:

Triazoles (e.g., hexaconazole, propiconazole)

Strobilurins (e.g., azoxystrobin, trifloxystrobin)

Myclobutanil, which is a penetrative into the plant tissues and provides curative and protective activity.

Fungicide sprays should start at the onset of the first symptoms and be repeated at 7- to 10-day intervals, based on disease pressure and environmental factors.

Alternate between fungicides that have different modes of action in order to slow or prevent the buildup of resistance by the pathogen population.

Adhere to label directions and safety intervals for efficacy and safety.

3. Biological Control

Use biological control materials such as:

Ampelomyces quisqualis, a hyperparasite parasite of the powdery mildew fungus itself.

Bacillus subtilis, an antibiotic producer that competes with fungal pathogens.

Trichoderma spp., which inhabits plant surfaces and suppresses fungal growth by mycoparasitism.

These biocontrol agents need to be used preventatively or at early infection stages for optimal efficacy and can be included in an organic or integrated program.

Reapplication may be required to keep them effective in the field.

4. Integrated Disease Management

Integrate resistant cultivars, good agronomic practices, timely application of fungicides, and biologicals for a complete disease control strategy.

Regular field monitoring for initial signs of disease, and undertake management at the earliest to avoid spread.

Adopt sanitation measures, particularly in protected cultivation systems, to minimize disease accumulation and avoid disease outbreaks.

Educate farm laborers and employees on the identification of early signs and the significance of clean cultivation practices.



Powdery Mildew of Cucurbits

Chapter-9

Onion Disease, Causal Organism, Symptoms, Etiology and Their Management

Purple Blotch of Onion disease

Causal Organism

Purple Blotch of Onion disease is caused by a fungal pathogen *Alternaria porri*. This fungus is a member of the phylum Ascomycota and class Dothideomycetes, a group of several economically significant plant pathogenic fungi.

Alternaria porri is classified as a necrotrophic fungus, which means it kills the host tissue before feeding on the dead cells, leading to tissue decay and destruction.

The pathogen is known to survive in infected crop debris left in the field after harvest and is commonly found in onion-growing areas across the globe.

It targets mostly the leaves and seed stalks of onion crops but also infects other *Allium* species, e.g., garlic and leeks.

Under suitable conditions, the pathogen may lead to huge economic loss through decreased yield and quality of onion bulbs.

Symptoms

The initial symptoms of Purple Blotch manifest as minute water-soaked lesions on the older leaves or seed stalks of the onion crop.

These early lesions rapidly increase and become oval to elongated purplish macules, characteristically bordered by a yellow halo. This pronounced coloration names the disease as Purple Blotch.

As the infection progresses, the lesions can form concentric rings, a diagnostic sign characteristic of *Alternaria* species, making them target-like.

In serious infections, a few lesions merge to form confluent lesions, causing girdling of the leaf, which limits the flow of nutrients and water. This causes

bending, wilting, and drying of the leaf, often beginning from the tip and extending downwards.

Disease-affected seed stalks can become weakened and wilt, causing reduced seed production.

The infected plants are usually disadvantaged by a loss of green leaf area, which restricts photosynthesis and eventually contributes to poor bulb formation, size reduction, and yield reduction.

The disease begins on older leaves at the base and works its way up to younger leaves, especially during hot and humid weather, and therefore is a prime concern in the late growth stages of the crop.

Etiology

Alternaria porri multiplies asexually by forming conidia (spores) on conidiophores, which are stalk-like bodies specialized for this purpose.

The conidia are typically dark in color, multicellular, with both transverse and longitudinal septa (divisions) and a distinctive shape.

The spores are easily scattered by wind, rain splash, water used for irrigation, and also by infected implements or hands, so the disease can travel long distances.

The conidia germinate upon contact with free moisture like dew, rain, or overhead irrigation water.

Infection and germination are promoted by favorable temperatures of 20°C to 30°C and a relative humidity of more than 90%, resulting in the disease being most common during the monsoon or in poorly ventilated circumstances.

The disease spreads through natural openings such as stomata or wounds on the leaf surface and starts colonizing the tissue, causing necrosis.

Conditions such as high plant population densities, inadequate air movement, and prolonged leaf wetness highly promote development and quick progress of the disease.

The fungus can also survive between growing seasons in the form of mycelium or spores in infected plant debris or on contaminated seeds, which act as the primary source of inoculum for future infections.

Disease Cycle

The disease cycle of *Alternaria porri* begins when the pathogen overwinters or oversummers in infected plant debris or on contaminated seeds.

When conditions in the environment are favorable, the fungus infects dead tissue, producing conidia that are carried by wind, rain, or irrigation water to healthy onion plants.

The initial infections most commonly occur on the lower, older leaves, particularly following rain or heavy dew, which furnish the moisture needed for germination of the spores.

After infection is established, the fungus forms additional conidia on infected tissue, which in turn become a source of secondary infections of the same growing season.

Secondary spread is particularly quick under wet and warm conditions, resulting in epidemic situations in the field if not controlled.

The disease may recycle several times within a season, especially in heavily planted fields or where drainage is poor, and result in increasingly damaging injury to the crop if interventions are not initiated in a timely manner.

Management

1. Cultural Practices

Start with the use of certified disease-free seed or transplanting, which prevents bringing the fungus into the field.

Adopt a proper crop rotation procedure, preventing repeated cultivation of onion or garlic in the same field for two successive seasons, which creates inoculum accumulation.

Provide adequate spacing between the plants to enhance air movement and minimize leaf humidity, preventing spore germination and infection.

Use drip irrigation practices instead of overhead irrigation to prevent leaf wetness and minimize leaf wetness periods.

Following harvest, remove and destroy all infected plant debris, including leaves, stalks, and culls, to avoid overwintering of the fungus in the field.

Do not enter the field when leaves are damp, as this has the potential to spread the spores mechanically through clothing, tools, or equipment.

2. Chemical Control:

Begin fungicide application at the initial onset of disease symptoms and repeat at frequent intervals to guard new growth.

Apply contact fungicides like:

Disease free bulb should be selected for planting

Seeds should be treated with Thiram @ 4 g/kg seed.

The field should be well drained.

Three foliar sprayings with Copper oxychloride 0.25 % or Chlorothalonil 0.2 % or Zineb 0.2 % or Mancozeb 0.2 %.

Mancozeb (multi-site contact activity)

Copper oxychloride, effective to prevent initial infection.

For systemic and curative effect, use fungicides like:

Tebuconazole, Propiconazole, or Hexaconazole (triazole family)

Azoxystrobin, Trifloxystrobin (strobilurins), which provide broad-spectrum and extended protection.

Repeat fungicide sprays at 10–14-day intervals, varying frequency with weather and disease pressure.

Always alternate fungicides of various modes of action to avoid the buildup of fungicide resistance in the pathogen population.

Always adhere to label guidelines, i.e., correct dosage, method of application, pre-harvest interval to ensure safe and effective disease management.

3. Biological Control

Apply biological control organisms such as:

Trichoderma harzianum – a fungus that inhibits *Alternaria porri* growth by competing with it and producing antibiosis.

Pseudomonas fluorescens and *Bacillus subtilis* – disease-suppressive bacteria that favor plant health.

These agents may be applied as seed treatments, foliar sprays, or soil drenches to increase plant defense and lower disease pressure.

Biocontrol products are environmentally friendly and can be combined with other methods as part of sustainable management.

4. Field Hygiene and Resistant Varieties

If available, utilize moderately resistant onion varieties with tolerance to purple blotch disease.

Practice rigorous sanitation by clearing volunteer onion plants, alternate host weeds, and residues that can carry the pathogen.

Use balanced nutrition, particularly with sufficient levels of potassium and micronutrients, to enhance plant resistance and minimize disease severity.

Integrated Disease Management (IDM)

A successful IDM approach entails a blend of resistant varieties, cultural control, opportune use of fungicides, biological control, and constant field monitoring.

Carefully observe weather patterns, including temperature and humidity, to forecast potential outbreaks and schedule fungicide application accordingly.

Educate farmers and field laborers to recognize initial symptoms, appreciate the role of clean cultivation, and implement timely management practices.

With the incorporation of all the tools at hand, purple blotch of onion can be managed successfully to realize greater yield, improved bulb quality, and economic gain.



Purple Blotch of Onion

Stemphylium Blight of Onion disease

Causal Organism

Stemphylium blight of onion is incited by a fungal pathogen ***Stemphylium vesicarium***.

The fungus belongs to Ascomycota phylum, class Dothideomycetes, and is one of the renowned necrotrophic pathogens, as it kills host plant cells and then survives on dead tissues.

The fungus infects mainly the foliage (leaves) of onions and is usually found in association with or after purple blotch disease, which is caused by *Alternaria porri*.

It survives and lasts in the environment by producing conidia (asexual spores) and mycelium that can be found in infected plant debris, soil, or seeds for quite a long time.

Under favorable environmental conditions, i.e., high humidity and moderate temperature, the pathogen is activated and causes infection.

Symptoms

The disease manifests initially as minute, water-soaked, light brown lesions on mature onion leaves, particularly toward the tips or margins.

These lesions enlarge and become elongated, sunken, spindle-shaped spots with a purplish-brown center and surrounded by a yellowish halo.

With progression of disease, these spots become confluent, resulting in widespread blighting of the leaves, particularly from leaf tip downwards.

Infected leaves wilt and collapse later on, substantially decreasing the photosynthetic area of the plant.

Severe infection results in premature death of whole foliage, and this severely impacts development of bulbs, causing smaller size, quality, and extensive yield loss.

Disease symptoms are more pronounced when occurring together with purple blotch, which commonly serves as an entry point or predisposes the tissue to Stemphylium invasion.

Etiology

The pathogen organism, ***Stemphylium vesicarium***, multiplies asexually by forming conidia that are multicellular, dark, and oval to oblong in shape.

These conidia develop on brief conidiophores that develop from the surface of infected tissues in humid conditions.

The fungus needs high relative humidity (over 90%), leaf surface free moisture, and moderate temperatures of 20°C to 30°C for effective germination and infection.

The conidia germinate on wet leaves and infect the plant either through natural openings like stomata or through wounds due to insect injury, other pathogens, or mechanical damage.

The disease is positively associated with high canopy density, overhead watering, inadequate air circulation, and extended leaf wetness, each of which facilitates spore production and dispersal.

Disease Cycle

The pathogen persists out of crop season as mycelium and conidia in infected crop residues remaining in the field.

It also persists in soil and onion seeds infected by the fungus, serving as a source of primary inoculum.

Under suitable weather conditions, for example, warm and damp conditions, the resting fungal structures become active and yield conidia, which get dispersed by wind, water splash, insects, and infected tools or hands.

These conidia start primary infections, and the disease spreads quickly with secondary infections as conidia are produced constantly on newly infected tissue.

The cycle is repeated during the cropping season, particularly under conditions of high humidity and temperature changes, leading to widespread epidemics in unprotected field.

Frequently, purple blotch infection acts as a predisposing factor, softening leaf tissues and inducing *Stemphylium vesicarium* colonization.

Management

Cultural Practices

Several cultural practices need to be adopted to control *Stemphylium* blight.

Utilize certified, disease-free seeds to avoid introduction of the pathogen into the field.

Practice crop rotation with non-host crops like cereals for a minimum of 2 to 3 years in order to break the disease cycle.

Prevent overhead irrigation, particularly during late nights, since this extends leaf wetness and promotes fungal infection.

Use drip irrigation as an alternative to minimize canopy humidity.

Sustain optimal plant spacing to facilitate satisfactory air circulation and faster drying of leaf surfaces.

Destroy and dispose of infected plant residues and volunteer onions after harvest to minimize carry-over of the fungus into the next year.

Chemical Control

Chemical fungicides work best when used preventively or at the first evidence of disease.

Spraying preventative fungicides like Mancozeb (0.25%), Copper oxychloride (0.3%), or Chlorothalonil (0.2%) can suppress initial infections.

For systemic and curative purposes, Propiconazole (0.1%), Tebuconazole, Azoxystrobin, or Trifloxystrobin can be applied at 10–12 day intervals, particularly under favorable environmental conditions.

Make alternate use of fungicides belonging to different chemical classes to avoid pathogen resistance.

Timely application of fungicides based on weather forecasts, particularly prior to high humidity or rain.

Biological Control

Biological controls like *Trichoderma harzianum*, *Pseudomonas fluorescens*, or *Bacillus subtilis* may be applied through seed treatment or foliar spray as part of an integrated control strategy.

These antagonists interfere with the development of *Stemphylium vesicarium* by giving off antifungal metabolites or competing with it for space and nutrients.

Biocontrols work best when used at the beginning of the season or as a preventative and are also eco-friendly.

Nutrient Management and Resistant Varieties

Use resistant or tolerant onion cultivars where such varieties are available and have inherent lower susceptibility to *Stemphylium* blight.

Minimize the application of nitrogenous fertilizers, which encourage lush, juicy growth that is more susceptible to infection by fungi.

Use balanced fertilization with adequate potassium and phosphorus, which increase disease resistance in the plants.

Integrated Disease Management (IDM)

The most eco-friendly method of management for Stemphylium blight of onion is by Integrated Disease Management (IDM).

This entails an integration of cultural management practices, utilization of resistant varieties, timely application of fungicides, biological controls, and effective monitoring.

Conduct regular field inspection to identify early signs and begin control action early.

Educate field personnel and farmers on how to detect disease symptoms and maintain good field sanitation and hygiene.

Through the application of IDM measures, the occurrence and severity of the disease can be minimized considerably, hence enhancing the quality and yield of onion crops and reducing chemical reliance.



Stemphylium Blight of Onion

Chapter-10

Chili Disease, Causal Organism, Symptoms, Etiology and Their Management

Anthracnose of Chili Disease

(A Fungal Disease of Chili Fruits)

Causal Organism:

Anthracnose of chili is mostly caused by the fungal pathogen *Colletotrichum capsici*.

Sometimes *Colletotrichum truncatum* also occurs as a causal organism.

This pathogen is of the phylum Ascomycota, class Sordariomycetes, and order Glomerellales.

The fungus forms asexual spores named conidia, which are hyaline (colorless), unicellular, and typically curved or falcate (sickle-shaped).

Symptoms

The disease primarily infects chili fruits, but leaves and stems can also be infected in extreme cases.

Early symptoms are in the form of small, depressed, round dark spots on mature or green fruits.

As the infection develops, the lesions get larger and join together to form large, depressed lesions that have concentric rings.

Pinkish or salmon-colored spore masses may develop at the center of these lesions under humid conditions.

Fruits that are very heavily infected can rot, dry up, and drop prematurely, both lowering yield and market value.

Infected fruits' seeds could also germinate poorly and lead to seedling blight during the subsequent planting period.

Etiology

The fungus overwinters in infected plant residues, seeds, and soil and acts as the main source of inoculum.

Conidia (spores) are transmitted by rain splashes, irrigation water, wind, insects, and infected farming equipment.

Optimum temperatures between 25°C and 30°C, in combination with high humidity (above 80%), facilitate the expression of the disease.

High plant canopy cover and extended leaf wetness enable quick transmission and secondary infections.

Disease Cycle

The pathogen remains in the environment in the form of dormant mycelium or conidia in plant residues, soil, and seeds.

The conidia germinate and infect young plant tissue via natural avenues such as stomata or mechanical wounds upon the onset of suitable environmental conditions.

The fungus grows in the tissue and forms new conidia that are spread to adjacent plants.

This results in cyclic secondary infections during the growing season.

Management:

1. Cultural Practices:

Plant disease-free seeds certified and grow resistant or tolerant crop varieties, if any.

Rotate crops with non-host crops (e.g., cereals or pulses) for 2–3 years to minimize soil-borne inoculum.

Destroy and remove infected plant residues at harvest.

Prevent overhead sprinkling and ensure good plant spacing to enhance air movement and lower humidity.

2. Seed Treatment:

Soak seeds with Carbendazim (0.2%) or Thiram (0.3%) for 30 minutes before planting.

This reduces seed-borne infection and offers protection at the early stages.

3. Chemical Control:

Spray Carbendazim (0.1%), Mancozeb (0.25%), or Chlorothalonil (0.2%) every 10–15 days from the early fruiting stage.

Rotate fungicides to avoid developing resistance.

4. Biological Control:

Apply biocontrol agents such as *Trichoderma harzianum* or *Pseudomonas fluorescens* for seed or soil application.

These microbiological organisms suppress the disease-causing pathogen and enhance plant health.

5. Integrated Disease Management (IDM):

Utilize all the methods—resistant varieties, good cultural practices, seed treatment, scheduled fungicide sprays, and biological controls—for effective and sustainable management of the disease.



Anthrachnose of Chili

Leaf curl of chilies disease

(A severe viral disease of chili plants)

Causal Organism:

Leaf curl of chilies is caused mainly by a virus, Chili Leaf Curl Virus (ChiLCV).

It is a Begomovirus belonging to the family Geminiviridae.

It is a single-stranded DNA virus, meaning its genetic material is in the form of only one strand of DNA.

The virus does not spread via seeds but is carried by an insect vector, i.e., the whitefly (*Bemisia tabaci*).

Symptoms:

The most conspicuous symptom of this disease is curling of leaves that is either upwards or downwards.

The infected leaves become smaller than the normal ones, leathery in texture, and are often puckered or wrinkled on the surface.

Leaf margins roll, and the leaf shape is quite distorted.

Infected plants of chili show stunted growth as a result of the virus interfering in physiological processes.

The plants are also bushy, brought about by over-production of shoots and branches.

Dropping of flowers is the most frequent occurrence, leading to poor fruit set and yield.

Even if fruits form, they are usually small, deformed, and often not suitable for sale to the market.

In later stages, the plant is chlorotic (yellowish) and in poor health throughout, and yield is significantly lowered.

Etiology

The pathogen responsible for the disease is Chili Leaf Curl Virus, which is a whitefly-borne begomovirus.

The virus does not get transmitted via seeds but is spread between plants by the feeding process of the whitefly.

Whiteflies pick up the virus upon feeding on the infected plant, and they carry the virus in their body in a persistent state.

When such infected whiteflies feed on a healthy chili plant, the virus is injected into the plant tissue, and infection begins.

The disease becomes very severe with rapid spread during hot and dry weather conditions, which support the proliferation and activity of whiteflies.

Moreover, the virus persists in alternate host crops like tomato, tobacco, and some solanaceous wild weeds, which serve as reservoirs or hosts of the virus between crop cycles.

Disease Cycle:

The Chili Leaf Curl Virus persists between crop cycles in infected crop residues and in alternate host plants like tomato, tobacco, and weeds (e.g., *Datura*, *Solanum* species).

The whitefly vector is responsible for carrying out the disease cycle by acquiring the virus during feeding on an infected host and subsequently transferring it to healthy plants.

The virus remains within the body of the whitefly and is passed during later feeding on healthy chili plants.

Once inside the plant, the virus replicates in plant cells and spreads systemically through the vascular tissues, causing characteristic symptoms of the disease.

Since the whiteflies keep on hopping from plant to plant, particularly in a favorable environment, the disease spreads very quickly in the field and to the adjacent areas.

Recurring presence of vector populations and alternate host plants allows the virus to complete its life cycle and remain prevalent through the seasons.

Management Practices

In order to control Leaf Curl of Chilies effectively, an integrated method incorporating cultural, mechanical, biological, and chemical means must be followed:

1. Cultural Methods

Always transplant healthy, virus-free seedlings to the field.

Remove and destroy diseased plants as soon as they show symptoms, to avoid further transmission of the virus.

Practice field sanitation through frequent weeding and removal of alternate host plants like *Datura*, *Solanum*, and others that carry the virus and whiteflies.

Successively crop-rotate with non-host crops like cereals, legumes, or other unrelated ones to terminate the disease cycle.

Place yellow sticky traps in the field to trap and observe the whitefly population.

2. Resistant Varieties:

Plant chili varieties that are resistant or tolerant to leaf curl virus, if any such variety is available from agricultural research centers or seed companies.

3. Mechanical and Physical Control:

Protect nursery beds and young plants with insect-proof netting or screen houses to avoid whitefly attack in an initial stage.

Use reflective mulches of silver or plastic around the base of chili plants, which reflect light and repel whiteflies.

4. Biological Control:

Encourage and maintain natural whitefly enemies in the field, including:

Ladybird beetles (*Coccinellids*), which are whitefly egg and nymph predators.

Green lacewings (*Chrysoperla* spp.), which are predators of soft-bodied insect pests.

Parasitic wasps (*Encarsia* spp.), which parasitize whitefly pupae and suppress their population.

5. Chemical Control:

When whitefly numbers are above threshold levels, use systemic insecticides to control the vector. Highly recommended choices are:

Imidacloprid at 0.3 ml per liter of water.

Thiamethoxam at 0.25 g per liter of water.

Spiromesifen or Diafenthiuron are the alternatives for whitefly control.

Initiate the application of insecticides from the nursery stage itself, particularly in regions with a past record of viral infections.

Keep on spraying the main field every 10 to 15 days, depending on trapping and field observations for whitefly activity.

Rotate insecticides with various modes of action at all times to avoid resistance buildup in the whitefly populations.



Leaf Curl of Chilies

Chapter-11

Turmeric Disease, Causal Organism, Symptoms, Etiology and Their Management

Leaf spot of Turmeric diseases

(A widely prevalent foliar disease of turmeric plants resulting in decrease in yield as well as quality of rhizomes)

Causal Organism:

Turmeric leaf spot is a fungal disease caused mainly by the pathogen *Colletotrichum capsici*.

The fungus belongs to the class Deuteromycetes, or Fungi Imperfecti, as there is no known sexual stage present in its life cycle.

It asexually reproduces by forming huge masses of conidia, which are minute-sized spores that infect at an alarming rate under optimum conditions.

In certain areas, *Taphrina maculans* or *Cercospora* spp. have been found to develop leaf spot symptoms in turmeric as well, but *Colletotrichum capsici* is the most prevalent and economically significant pathogen of the disease.

Symptoms

The disease usually starts with the development of small, water-soaked spots on the upper leaf surface.

The early spots are generally round or irregular and are slightly depressed in appearance.

With the progression of infection, these spots enlarge and turn brown to dark brown in color, frequently with a well-defined yellow halo surrounding them.

As time passes, single spots coalesce or merge together, creating big areas of necrotic (dead) tissues on the leaf blades.

Infested leaves become very dry from around the margins, first curl upwards or downwards, and eventually wither and are shed from the plant.

Pre-mature defoliation decreases the plant's capacity to photosynthesize effectively, resulting in poor growth and a major decrease in rhizome yield.

In badly infected crops, the disease can cause economic losses through quality and quantity loss of the turmeric crop.

Etiology

The fungus pathogen *Colletotrichum capsici* overwinters from one season to another in infected residues of the crop, soil, and on stored turmeric rhizomes for sowing.

In the crop season, the fungus sporulates conidia, which are disseminated primarily through wind currents, splashing rain, irrigation water, or insects.

The disease spread is greatly facilitated by environmental factors like high relative humidity (more than 80%), temperatures between 25°C and 30°C, and frequent or long duration of rainfall.

Other factors facilitating the disease are close plant spacing, inadequate field aeration, and overhead irrigation, all of which establish conducive microclimates for fungal development.

The disease can also persist in its latent stage in the infested rhizomes or dried plant material and act as a source of infection during the next growing season.

Disease Cycle

The disease cycle of leaf spot starts with the persistence of the fungus in infested crop residues and turmeric planting materials (rhizomes) remaining in the field or stored after harvest.

When optimal environmental conditions prevail particularly warm temperatures and high humidity the conidia germinate and infect the healthy leaves via natural openings like stomata or wound-like lesions generated by insects or mechanical injury.

After the fungus penetrates the leaf tissue, it multiplies and spreads internally, forming the characteristic spots and necrotic lesions.

The fungus forms secondary conidia on the infected tissues, which are spread to nearby healthy plants by water splashes or wind, causing secondary infections.

The infection cycles can recur throughout the growing season, especially when the weather is high in humidity and rainfall.

The disease becomes more severe if not controlled early, and infected plant debris repeats the cycle the following season.

Management Practices

To manage the leaf spot disease of turmeric effectively, an integrated disease management (IDM) approach must be followed by integrating cultural, biological, and chemical methods.

1. Cultural Methods

Plant only disease-free, clean, and certified rhizomes to prevent the inoculation of the pathogen into the field.

Plant sparsely to promote good air movement among the plants, which tends to minimize leaf wetness duration.

Remove and kill infected leaves and plant waste regularly to limit the source of fungal inoculum.

Use crop rotation with non-host crops such as pulses or cereals to disrupt the disease cycle.

Provide good drainage in the field, as long soil moisture and water standing encourage disease development.

Do not use overhead irrigation, particularly during humid weather, since it encourages disease distribution by splash dispersal.

2. Resistant Varieties

Where possible, select varieties or cultivars of turmeric with moderate resistance or tolerance to leaf spot disease.

These types have the potential to minimize the impact of the disease and ensure yield stability with modest disease pressure.

3. Biological Control

Use of biocontrol products like *Trichoderma harzianum*, *Trichoderma viride*, or *Pseudomonas fluorescens* in the soil and as leaf spray can significantly suppress the fungal pathogen population.

These useful microbes compete with the pathogen and can inhibit its growth by generating antifungal compounds or causing systemic resistance in the plant.

Use biocontrol agents at planting time and at frequent intervals for long-term effect.

4. Chemical Control

For successful chemical control, start fungicide spraying as soon as the first disease symptoms are noticed, especially when weather conditions favor the development of disease.

Fungicides recommended are

Carbendazim at 1 gram per liter of water, a systemic fungicide useful in controlling early infection.

Mancozeb at 2.5 grams per liter of water, a contact fungicide that prevents spore germination.

Chlorothalonil at 2 grams per liter, another protective fungicide with a wide spectrum of action.

Fungicide sprays should be repeated after 10–15 days as per the intensity of the disease and weather.

Rotate fungicides that have different modes of action to prevent resistance from developing in the pathogen instead of sticking with the same product all season.



Leaf Spot of Turmeric

Chapter-12

Wheat Disease, Causal Organism, Symptoms, Etiology and Their Management

Rust of Wheat Disease

Causal Organism:

Rust disease in wheat is caused by various species of fungi of the genus **Puccinia**, which are obligate parasites. This means that these fungi need a living host to survive and finish their life cycle.

There are three main types of rust diseases in wheat, and each has a different species of **Puccinia** as its cause:

Brown Rust or Leaf Rust is brought on by *Puccinia triticina*. It essentially occurs on the leaves and is the most prevalent form of rust in most wheat-growing regions.

Yellow Rust or Stripe Rust is brought on by *Puccinia striiformis f. sp. tritici*. The rust is prevalent in cold and wet climatic zones and results in yellow stripe-like markings on the leaves.

Black Rust or Stem Rust is due to *Puccinia graminis f. sp. tritici*. This is regarded as the most pernicious race because it can infect stems and trigger serious lodging as well as yield loss.

All these species are members of the phylum Basidiomycota and are characterized by their capability to form several types of spores, such as urediniospores and teliospores, that enable them to survive and propagate under field conditions.

Symptoms

Brown (Leaf) Rust appears as small, oval to round orange-brown pustules on the top surface of wheat leaves. The pustules occur randomly and can fuse together in severe infection. Infected leaves turn premature gray and cause photosynthesis to be diminished and grain yield severely reduced.

Yellow (Stripe) Rust forms bright yellow, finger-shaped pustules that occur in thin, parallel stripes, following the direction of the leaf veins. These occur on

leaves and leaf sheaths. Yellow (Stripe) Rust prefers cooler and wetter environmental conditions. As infection goes deeper, the leaves become yellow, and eventually dry up.

Black (Stem) Rust is identified by reddish-brown elongated pustules that form on the wheat stem, leaf sheaths, and occasionally on leaves. Pustules turn black later as teliospores form. It is very damaging and can lead to weakening or lodging of the plant and ultimately to severe yield loss if not controlled.

Etiology

The life cycle of rust fungi starts with the formation of urediniospores, which are spores of asexual origin that are responsible for secondary spread.

The spores are light in weight and can be transported easily by wind over lengthy distances.

In humid conditions (more than 90%) and at moderate temperatures of 10°C to 20°C, the urediniospores germinate and infect the wheat plant by entering through stomata.

After entering, the fungus infects the leaf tissues, causing pustules to form that discharge further spores, thereby propagating the infection cycle.

Later in the growing season, the fungus forms teliospores, which are thick-walled and allow the fungus to survive through the winter or in crop residues and soil.

Puccinia graminis, the causal agent of stem rust, has a lifecycle on two hosts and is therefore referred to as heteroecious. It needs an alternate host, e.g., barberry (*Berberis* spp.), to fulfill the sexual reproduction stage.

The rusts progress quickly once suitable environmental conditions continue, particularly cool and wet conditions with free leaf moisture due to dew, irrigation, or rain.

Management

1. Resistant Varieties

One of the best and cost-effective ways of controlling rust diseases is the planting of rust-resistant wheat varieties.

Highly recommended resistant varieties are HD 2967, PBW 725, WH 1105, and many more appropriate for geographical conditions. These varieties lower the chances of epidemics and reduce chemical spray dependency.

2. Cultural Practices

Proper sowing of wheat at the right time can avoid the period of infection.

The elimination of volunteer wheat plants (those plants that resprout from seeds dropped during harvest) and alternate hosts such as barberry assists in the disruption of the disease cycle.

Evading the application of excess nitrogen is also relevant since dense growth caused by high nitrogen tends to promote rust development.

3. Chemical Control

At first rust symptom onset, fungicide treatment should be carried out with Propiconazole 25 EC @ 0.1% or Tebuconazole @ 0.1%.

If the infection persists or weather conditions continue to favor disease spread, a second spray should be carried out after 12–15 days.

Promote uniform spraying across the complete foliage for effective protection.

4. Monitoring and Forecasting

Scheduled field scouting and strict observation of the crop, particularly during the critical growth stages, aid in early detection.

Farmers must remain informed via agricultural advisories and rust forecast systems to undertake timely control activities.

Utilization of mobile apps and weather-related prediction systems can be helpful for disease forecasting and fungicide timing.



Rust of Wheat

Loose Smut of Wheat Disease

Causal Organism

Loose smut of wheat is caused by a fungal disease organism named ***Ustilago tritici***, also synonymously called ***Ustilago nuda tritici***.

The fungus is placed in the phylum Basidiomycota and class **Ustilaginomycetes**, a class of smut fungi that are well known for causing destructive attacks on cereal crops.

The pathogen is defined as an internal seed-borne fungus, i.e., it stays inside the seed embryo. Even though infected, the seeds are healthy in appearance externally and do not display any symptoms, and hence detection at the sowing level is practically impossible without a laboratory test.

Symptoms

The loose smut symptoms are not evident until the flowering phase of the wheat plant, normally when the ear or spike is coming out of the boot leaf.

The infected spikes come out earlier than the healthy spike and are discovered to be totally replaced by a black, powdery mass of fungal spores known as smut sori.

These spores, referred to as teliospores, are loose, powdery, and not covered, and thus the disease is named "loose smut."

Soon after the diseased spike comes out, easily the black spore mass is carried away in the wind, and only the naked center axis (rachis) of the spike remains.

Physically, the rest of the plant also appears green and healthy, hence it also contributes to delayed or missed detection of the disease at its initial stage.

Etiology

The life cycle of the loose smut pathogen is initiated with the infection of the healthy wheat flowers' ovary during the crop's flowering phase of the previous season.

It is at this point that airborne teliospores of infected plants are deposited on the exposed florets of healthy plants and germinate, facilitating penetration of fungal hyphae into the developing embryo and ovary.

The fungus enters dormancy in the embryo of the infected seed and stays dormant until the seed is planted the next season.

When the infected seed germinates, the dormant mycelium is activated and grows systemically with the emerging seedling, infecting all parts of the plant including the growing point.

With maturity of the wheat plant, the fungus is inside the plant tissues and comes to the reproductive structures, replacing the normal floral organs with black spore-bearing sori.

At flowering, these loose spores are discharged in the air and can infect newly produced healthy adjacent flowers, thereby completing the disease cycle.

Loose smut does not spread by soil or residues of the crop. It is transmitted only by infected seeds, and hence it's a highly seed-borne disease.

Management

1. Utilization of Disease-Free Seed

Prevention of loose smut spread can best be achieved by planting only certified and disease-free seeds.

Seeds must come from trustworthy, certified organizations, and planting farm-saved seeds from loose smut-infected fields in the past is strictly avoided.

Seed health testing on seed production chains should regularly be practiced to cut off the source of infection.

2. Seed Treatment with Systemic Fungicides

Chemical seed treatment with systemic fungicides is the most effective control method against loose smut.

Seed must be treated with a mixture fungicide like Carboxin 37.5% + Thiram 37.5% (marketed as Vitavax Power) at 2 to 2.5 grams per kilogram of seed.

Alternatively, the use of Tebuconazole 2% DS may be done at 1.5 grams per kilogram of seed.

These fungicides enter into the seed coat and eliminate the latent fungal mycelium in the embryo, averting systemic fungus development in the field.

3. Hot Water Seed Treatment (Traditional Method)

Alternatively, as a non-chemical method, the hot water seed treatment can be employed, particularly under regions where the application of fungicides is limited.

In this technique, seeds are initially cold-soaked in 4–6 hours in cold water and then hot-water soaked at 52°C for a precise 10 minutes.

The temperature is adequate to destroy the internal fungal mycelium but should be carefully controlled since too much heat or longer exposure would harm the viability of the seeds.

4. Production of Resistant Varieties The use of wheat varieties resistant to loose smut is a long-term and environmental friendly approach to minimize disease occurrence.

Resistant varieties can be different in different areas, therefore, farmers should seek advice from agricultural extension officers or universities for proper recommendations.

5. Field Sanitation and Crop Rotation

Though the disease is seed-borne and not soil-borne, field sanitation helps minimize the pressure of fungal inoculum.

Practicing crop rotation with non-host crops such as pulses or oilseeds can indirectly minimize the accumulation of the disease by preventing repeated exposure of wheat to infection sources.

6. Avoidance of Early Sowing

In areas where loose smut is endemic, extremely early sowing might expose wheat flowers to airborne spores at vulnerable stages.

Delaying sowing slightly within the recommended window can help avoid overlap with spore dispersal, thus minimizing the chances of fresh infection.



Loose Smut of Wheat

Karnal Bunt of Wheat Disease

Causal Organism:

Karnal bunt of wheat is the result of the fungal pathogen *Tilletia indica*, also scientifically known as *Neovossia indica*.

This disease causative pathogen is a member of the phylum Basidiomycota and class Ustilaginomycetes, the same groups that smut fungi belong to.

The disease was originally detected in the city of Karnal in the state of Haryana, India, in the year 1931 and was subsequently named "Karnal bunt" based on its origin.

Contrary to diseases like loose smut that are totally seed-borne and systemic, Karnal bunt is both seed-borne and soil-borne, and the infection is localized, that is, it infects only particular grains within an ear of wheat but does not spread systemically in the plant.

Symptoms

The symptoms of Karnal bunt usually become evident at the grain filling to maturity stage of wheat crop.

On the outside, infected ears usually look normal, and hence early detection of the disease is extremely hard without physically opening and inspecting individual grains.

Infected spikes of wheat usually have only a few infected grains, and the infection is partial, where only a part of the grain is infected rather than the entire grain.

The diseased grains typically become shriveled, discolored, and deformed in shape, and typically dark brown to black color.

There is a characteristic and offensive fishy or putrid odor emitted from infected grains because of the presence of the chemical compound trimethylamine, which is a diagnostic characteristic of Karnal bunt.

The infected part of the grain is usually overlaid by a black powdery mass of pathogen spores known as bunt balls, which are made up of teliospores, the resting spores of the pathogen.

In heavily infected crops, the quality of harvested grain is compromised, rendering it unfit for use or export despite the total loss of yield being only moderate in quantity.

Etiology

The teliospore, being a thick-walled fungal spore, is the main source of inoculum of Karnal bunt. Due to this, the pathogen can live for many years in the crop residue and soil.

Apart from survival in soil, the pathogen can also survive as seed-borne infestation, where teliospores stick to the seed surface or dwell in grains' cracked and fractured portions.

Infection occurs between the ear emergence to flowering phase of the wheat crop, which is the most vulnerable period for disease development.

During this phase, the teliospores in the soil germinate and form primary sporidia, which in turn produce secondary sporidia.

These waterborne or airborne secondary sporidia are carried by wind, irrigation, or rain splashes to the wheat flowers, where they infect the forming ovaries.

The environmental factors conducive to infection are temperatures between 15°C and 25°C and relative humidity in excess of 70%, which enhance germination of sporidia and penetration of the floral organs.

Following infection, the fungus colonizes the grain tissue in development, in which it replaces some of the endosperm with an black sooty mass of teliospores.

These teliospores subsequently become the major source of inoculum for the subsequent cropping season and thus perpetuate the disease cycle.

In contrast to loose smut, Karnal bunt is not systemic and the fungus will not infect the vascular tissue of the plant. It infects the individual grain directly via floral tissues.

Management

1. Employment of Disease-Free and Clean Seed

One of the best ways to control Karnal bunt is through the use of certified, disease-free seed that is tested and cleared for teliospore absence.

Wheat seeds planted from previously infected fields should be avoided, as this will heighten the risk of seed-borne and soil-borne transmission.

Placing clean seed aids in the disruption of the disease cycle and lowers the initial inoculum load in the field.

2. Fungicide Treatment of Seeds

Although seed treatment is not very effective in controlling Karnal bunt as opposed to loose smut (because the infection is non-systemic), it can still be useful in minimizing surface contamination of teliospores.

Seed dressing with systemic fungicides like Carboxin + Thiram or Tebuconazole is likely to offer some protection.

Treatment must be done at the rate recommended, typically 2–2.5 g/kg of seed, to provide effective cover and safety.

3. Cultivation of Resistant Varieties

Production of Karnal bunt-resistant or tolerant wheat varieties is an extremely effective and sustainable control practice.

The varieties like HD 2009, PBW 502, WH 542, and other local lines of adaptable characters have been bred with moderate to high resistance to the pathogen.

Use of resistant cultivars serves to decrease the level of infections and also saves the crop from the risk of rejection in the market in regions where the quality for exports is an issue.

4. Cultural Practices

Good agricultural practices can significantly lower the occurrence of Karnal bunt.

Do not sow late, as this gives the wheat spikes greater exposure to cool and moist environments while flowering, which is congenial to pathogen infection.

Deep summer ploughing after harvest buries teliospores and infected residues deep into the soil, thereby lowering the surface inoculum.

Crop rotation with non-host crops like legumes and oilseeds prevents the repetitive accumulation of the pathogen in the same field every year.

5. Field Sanitation

Field hygiene is very important in reducing the spread of the disease.

Roguing (excision) of infected ear heads from seed production fields must be carried out immediately upon observing infection.

Destruction or burial of infected plant residues ensures the teliospores cannot survive and disperse during the following season.

6. Regulatory and Quarantine Measures

Karnal bunt is a quarantine disease in numerous countries, and the detection of even low levels of infection in seed lots can result in export rejection.

Thus, rigorous field inspections, laboratory analysis, and certification by government and seed regulating agencies are necessary in seed marketing and production.



Karnal Bunt of Wheat

Powdery Mildew of Wheat Disease

Causal Organism

Wheat powdery mildew is caused by the fungal pathogen ***Blumeria graminis* f. sp. tritici**, formerly classified as *Erysiphe graminis* f. sp. tritici.

The fungus falls under the phylum Ascomycota and class Leotiomycetes, a fungal group notorious for inciting powdery development on host surfaces.

Blumeria graminis f. sp. tritici is an obligate biotroph, which lives only on host plant tissues and derives its nutrition directly from the plant using feeding structures known as haustoria.

The pathogen is strongly host-specific and infects wheat and closely related grasses only.

Symptoms

The first visible symptom of the disease is the production of small white powdery spots on the upper surfaces of newly emerged young wheat leaves.

These dusty patches are composed of surface mycelium and asexual spores (conidia), making the leaf appear to be covered in dust or flour.

As the disease progresses, white patches merge to create large irregular blotches, commonly covering extensive areas of the leaf blade.

Underlying infected tissues can become chlorotic (yellowish) and eventually necrotic (brown to black) due to cellular injury from fungal growth.

In advanced infection, the disease becomes systemic and moves to other above-ground plant parts, including leaf sheaths, stems, and even to ear heads, severely compromising photosynthetic ability.

The infected leaf becomes yellowish, dries prematurely, and drops, resulting in weak plant growth and loss of grain filling and yield.

Etiology

The pathogen overwinters mostly in the form of cleistothecia (sexual fruiting structures) or as mycelium and conidia on infected plant residues, grassy weeds, and volunteer wheat plants.

During the beginning of the cropping season, airborne conidia that are discharged from infected materials fall onto sound wheat leaves and establish new infection.

Conidial germination is not necessary with free water but occurs best under cool temperatures (15°C to 22°C) and at high relative humidity (more than 60%).

After settling on the host surface, the conidia germinate and develop germ tubes that invade the epidermal cells and set up haustoria for feeding.

The pathogen multiplies quite quickly and travels from an infection to another via several infection cycles during a single season (polycyclic nature), leading to epidemic outbreaks under conducive weather conditions.

Abundant crop canopies and overuse of nitrogen can further contribute to disease development by raising the humidity and encouraging profuse vegetative growth.

Management Strategies:

1. Cultivation of Resistant Varieties:

The most cost-effective and sustainable way of controlling powdery mildew is the application of resistant or moderately resistant wheat varieties.

A number of varieties like PBW 343, WH 542, and HD 2329 have demonstrated good resistance at field level, but local recommendations may differ.

2. Implementation of Cultural Practices:

Early sowing of wheat decreases the period for which the plants are exposed to favorable conditions for fungal growth.

Keeping minimum plant spacing optimal enhances air movement in the crop canopy and lowers the local humidity levels.

Volunteer wheat plants and grassy weed hosts from adjacent fields and bunds are prevented from carrying diseases over from one season to another by eradicating them.

Use balanced fertilizer applications—overapplication of nitrogen must be avoided since it enhances leaf succulence and vulnerability to mildew.

3. Chemical Control Measures:

Protective fungicide sprays need to be applied at the first appearance of powdery mildew.

Apply Wettable Sulphur @ 0.2% or Dusting Sulphur @ 20–25 kg/ha as a low-cost and effective measure of control.

If there is moderate to heavy infestation, systemic fungicides like Tebuconazole @ 0.1%, Propiconazole @ 0.1%, or Triadimefon @ 0.1% can be employed.

Repeat spray after 10–15 days if weather conditions persist in favor of disease development.

4. Biological Control Options:

Certain mycoparasites like *Ampelomyces quisqualis* have been observed to parasitize the powdery mildew fungus and suppress its growth.

However, commercial use of such biocontrol agents is still limited and mainly under experimental or greenhouse conditions.

5. Monitoring and Forecasting:

Conduct regular field inspections, especially during tillering to flowering stages, to catch early signs of the disease.

Exclude overhead watering or regular spraying during cold, wet conditions since these can create microclimates conducive to disease transmission.

Use forecasting models and tools available to make predictions about peak risk times for infection and schedule fungicide treatment accordingly.



Powdery Mildew of Wheat

Chapter-13

Sugarcane Disease, Causal Organism, Symptoms, Etiology and Their Management

Red Rot of Sugarcane Disease

Causal Organism:

Red rot of sugarcane is caused by a fungal pathogen, *Colletotrichum falcatum* Went.

The fungus belongs to the phylum Ascomycota, class Sordariomycetes, order Glomerellales, and family Glomerellaceae.

It is regarded as a facultative parasite, which can survive either as a parasite on living sugarcane plants or as a saprophyte on dead plant material in the soil.

This disease is ranked as one of the most economically significant and destructive diseases of sugarcane cultivation, particularly in tropical and subtropical areas such as India.

It is particularly infamous for inducing serious yield loss as well as a major impact on juice quality and sugar recovery and hence represents a serious threat to the sugar industry.

Symptoms

The early symptoms of red rot tend to begin with yellowing and drying of the topmost leaves of the sugarcane crop, generally referred to as "top drying".

As the disease advances, leaf tips and margins turn reddish or brown, and the coloration creeps downward, causing the leaf sheath to dry gradually.

Diseased canes might look healthy from the outside but, when slit open along the length, the inside tissue shows a characteristic red coloration interspersed with white streaks or blotches.

This red and white speckled pattern within the stalk is diagnostic and indicative of red rot infection.

The diseased tissue can have a sour, fermented, or alcoholic odor, which signifies decomposition of internal tissues.

In older stages, the pith tissue disintegrates and becomes hollow, causing stalk weakening, and acervuli (minuscule black dot-like fungal structures) can be seen on the rind or cut surfaces of the stalk.

Premature lodging (falling) of canes, reduction in the quality and quantity of juice, and complete loss of canes in infested clumps in extreme cases result from the disease.

Etiology

The pathogen *Colletotrichum falcatum* overwinters mainly as conidia or as latent mycelium in infected cane setts, crop trash, and ratoon canes.

The infection is seed-borne, i.e., the infection is transmitted into the new crop through infected planting material (setts).

The pathogen infects the sugarcane plant through wounds, nodal buds, or cut ends of the sett during germination and establishment.

Beneficial environmental conditions for disease occurrence are high humidity (over 90%), moderate to high temperatures (25–30°C), poorly draining soils, and close planting which hinders air circulation.

Upon entry into the host plant, the fungus infects the vascular bundles, disrupting the plant's water and nutrient transport, ultimately leading to tissue discoloration and rot.

The pathogen may spread within and between fields by way of infected tools, irrigation water, wind-borne conidia, ratooning, and infected setts.

The genetic heterogeneity present in the pathogen population enables it to breach host resistance, thus necessitating incessant monitoring and rotation of resistant varieties.

Management:

1. Use of Disease-Free and Healthy Planting Material:

Always employ certified, disease-free sugarcane setts harvested from red rot-free fields.

Avoid planting setts that have been procured from ratoon crops or fields that have had a history of red rot.

2. Hot Water or Aerated Steam Treatment

Treat the planting material in hot water at 50°C for 30 minutes, or in aerated steam at 52°C for 1 hour to kill the fungus living inside the setts.

3. Cultivation of Resistant Varieties

Grow resistant or moderately resistant varieties of sugarcane, e.g., Co 86032, Co 0238, Co J 64, Co 775, CoS 767, and CoS 8436, based on regional guidelines.

Replacement of sugarcane varieties is recommended periodically to avoid the development of new races of virulent pathogens.

4. Cultural and Sanitation Practices

Destroy and pull out infected clumps at the time of detection by uprooting and burning them in order to avoid further infection.

Don't ratoon in fields infested with red rot.

Practice effective crop rotation using non-host crops such as legumes or cereals to intervene in the disease cycle.

Provide effective drainage, proper spacing of plants, and balanced application of fertilizers to support plant growth and minimize susceptibility.

5. Chemical Control Measures:

Treat cane setts prior to planting with systemic fungicides like Carbendazim (0.1%), Thiram (0.25%), or Mancozeb to kill seed-borne infection.

Foliar sprays will not be effective after systemic infection has taken place, therefore preventive treatments of setts should take precedence.

6. Biological Control Methods:

Trichoderma viride or *Trichoderma harzianum* can be used as sett dips or soil conditioners to repress red rot fungus.

These biological control inputs induce plant health and compete with or inhibit *C. falcatum* growth in the rhizosphere.

7. Frequent Field Monitoring and Tool Disinfection

Conduct regular scouting for signs of red rot, particularly during hot and humid weather.

Destroy immediately any clumps appearing suspicious and disinfect tools and equipment used in cutting or harvesting with formalin or sodium hypochlorite solution.

Conclusion:

Red rot is a destructive sugarcane disease that has the potential to completely destroy fields if not managed effectively. A combination of preventive practices

such as resistant varieties, hot water treatment, the use of clean planting material, field sanitation, and integrated biological and chemical control must be practiced in order to effectively manage it and ensure safe sugarcane productivity.



Red Rot of Sugarcane

Smut of Sugarcane Disease

Causal Organism

Fungal disease smut of sugarcane is caused by a pathogen *Sporisorium scitamineum*, previously identified as *Ustilago scitaminea*.

This fungus belongs to the phylum Basidiomycota, class Ustilaginomycetes, and order Ustilaginales, known commonly for causing smut diseases in crops.

The parasite is biotrophic in nature, i.e., it lives solely on living host tissues for its growth, reproduction, and to finish its life cycle.

After infecting the host, it gets systemic, traveling along the internal tissues of the sugarcane plant and remaining concealed for many years before showing any noticeable symptoms.

Symptoms

The most obvious and diagnostic feature of smut in sugarcane is the development of a long, black, whip-like structure, referred to as the smut whip, which grows out from the growing point or the terminal bud of the shoot.

At first, this smut whip is silvery-gray in color and has a thin, filmy covering. With the advance of the disease, the covering bursts and discharges a cloud of black powdery spores, or teliospores, causing further spreading of the disease.

The diseased sugarcane plants display stunted growth, i.e., the plant gets reduced in height considerably, and the leaves become spindle-shaped and pale green in color.

The stalks (canes) develop into thin, weak, and poorly developed ones with a significant decrease in internode length, thus giving rise to poor cane quality and yield.

The infected plants further exhibit excessive tillering, i.e., a lot of side shoots are produced, which makes the plants appear bushy, also known as "bunchy top."

In cases of heavy infections, the plant doesn't produce even any harvestable or millable cane, resulting in total economic loss to the farmer.

Symptoms typically become apparent when the crop is approximately 3–4 months old, at the stage of active vegetative growth.

Etiology / Disease Cycle

Smut fungus survives mainly in the bud area of infected seed canes, i.e., setts, in the form of resting teliospores.

These are very resistant and can stay alive in the soil or on planting material for months, all the while awaiting suitable conditions.

Infection occurs primarily at the time of planting or early germination, when the spores enter through the bud scales of the seed cane.

Following germination, the spores give rise to a promycelium, which further develops into sporidia. These sporidia infect the growing buds and develop a systemic infection in the developing shoot.

This systemic stage proceeds internally without any symptoms for a significant period and hence proving challenging to recognize in the early stages.

After the fungus has invaded the apical meristem (growth point), it induces the production of smut whips, which are the external reproductive organs of the disease.

High temperatures (30–35°C), low humidity, light or sandy soils, delayed planting, and the cultivation of susceptible sugarcane varieties are conducive to the disease.

Spores emitted by mature whips are readily carried by wind, rain splash, water of irrigation, or farm equipment and cause quick and extensive secondary infection in adjacent plants or fields.

Management Strategies

1. Use of Disease-Free Planting Material

Always choose and transplant certified, healthy setts of seeds from smut-free fields.

Never use setts from previously infected or adjacent infected fields because they can have dormant spores.

2. Treatment with Hot Water or Aerated Steam

Treat the seed setts before planting by immersing them in hot water at 50°C for 30 minutes, or by subjecting them to aerated steam at 52°C for 1 hour.

This kills the latent spores inside the buds effectively and thus prevents initial infection.

3. Application of Resistant or Tolerant Cultivars

Grow smut-resistant or moderately resistant sugarcane varieties, like Co 86032, Co 92005, Co 0238, and CoC 671, or varieties suggested by local agricultural research stations.

4. Chemical Seed Treatment

Drench the seed canes with Carbendazim (0.1%) or Thiram (0.25%) for 10–15 minutes before sowing to kill surface spores and guard against early infection.

Seed setts may also be dipped in 1% Bordeaux mixture for further protection.

5. Adoption of Cultural Practices

Avoid delay in planting, because it results in higher disease incidence with warmer temperatures and spore accumulation.

Practice crop rotation with non-host crops such as pulses or oilseeds to minimize inoculum accumulation in the field.

Provide sufficient row-to-row spacing and avoid over-use of nitrogenous fertilizers, which enhance succulent growth and expose the crop to disease attack.

6. Removal and Destruction of Infected Plants

Regularly check the field and remove and burn infected clumps and smut whips immediately to reduce the dissemination of infectious spores.

7. Biological Control Measures

Spray biocontrol agents such as *Trichoderma viride* or *Pseudomonas fluorescens* onto the soil or sett surface. These microorganisms are antagonists, inhibiting the growth and establishment of the smut pathogen.

8. Sanitation and Tool Hygiene

Maintain the field free and clean of sugarcane stubbles, volunteer weeds, and wild host plants.

Disinfect knives, cutters, and tools employed in cutting as well as planting constantly to eliminate mechanical transmission of spores.



Smut of Sugarcane

Wilt of Sugarcane of Disease

Causal Organism

Wilt of sugarcane is caused primarily by the fungal pathogen *Fusarium sacchari*, although in certain instances *Cephalosporium sacchari* and *Fusarium moniliforme* might also be involved.

The fungus is a member of the phylum Ascomycota, class Sordariomycetes, and order Hypocreales.

It is a soil-borne facultative parasite, able to endure in the soil and on infected plant litter for extended periods.

The pathogen penetrates the plant vascular system and induces clogging of water conduction, which results in wilting symptoms.

Symptoms

The disease typically manifests during the formative to grand growth phase of sugarcane and is marked by progressive yellowing and wilting of the upper leaves.

As the disease advances, rolling and drying of the leaves from the tip towards the base can be noticed.

Infected canes indicate stunted shoot growth, spindly shoot morphology, and diminished tillering.

Reddish-brown internal tissue discoloration (vascular bundles and pith) can be seen when the infected stalk is opened.

In severe cases, the pith turns dry, fibrous, and hollow, and at times has a sour, fermented smell from secondary infection.

Infected canes are light with low juice content and low sugar recovery.

Etiology

The fungus remains in between crops in infected setts, debris of the crop, and soil in the form of chlamydospores or mycelium.

It is infected through roots or setts, particularly through wounds or natural pores.

Conditions favorable for infection are poor drainage, heavy soils, high temperature (30–35°C), and moisture stress.

Once within the host, the fungus infects the vascular system, disrupting water and nutrient movement.

The disease can be latent for a few years and manifests when plants are stressed, especially because of water deficiency or lack of nutrients.

It is disseminated in the field by infested planting material, soil transfer, and water used for irrigation.

Management

1. Use of Disease-Free Planting Material

Employ healthy, certified seed canes from healthy fields free from wilt.

Avoid planting canes from fields with a known wilt history.

2. Hot Water Treatment

Treat setts in hot water at 50°C for 30 minutes or in aerated steam at 52°C for 1 hour to eliminate internal fungal inoculum.

3. Resistant Varieties

Cultivate resistant or tolerant sugarcane varieties like Co 86032, Co 0238, and Co 0118 where available.

Replace varieties from time to time to avoid the buildup of pathogen races.

4. Cultural Practices

Provide adequate drainage and prevent waterlogging, particularly in heavy soils.

Implement crop rotation with non-host crops such as pulses or cereals to minimize pathogen load in the soil.

Prevent ratooning in severely wilt-infected fields and ensure optimum plant spacing for improved aeration.

5. Field Sanitation

Destroy and burn infected plant debris and stubbles after harvest.

Clean equipment and tools to prevent the spread of pathogens through mechanical transmission.

6. Chemical Control

Dip setts in Carbendazim (0.1%) or Thiram (0.25%) for 10–15 minutes before sowing to minimize sett-borne infection.

Drenching of soil at the base of infested clumps with fungicides might be useful in the initial stages but will often be of limited value once the disease is systemic.

7. Biological Control

Sow setts or soil with *Trichoderma viride* or *Pseudomonas fluorescens* in order to control the wilt pathogen biologically.

These useful microbes suppress or compete with *Fusarium* spp. and enhance plant health.

8. Irrigation and Nutrient Management

Supply sufficient and timely irrigation to avoid moisture stress.

Avoid over-nitrogen fertilization, which encourages disease.

Balanced fertilization with potassium and micronutrients enhances resistance to wilt in plants.



Wilt of Sugarcane

Grassy shoot of Sugarcane Disease

2. Causal Organism:

The grassy shoot disease of sugarcane is caused by a **phytoplasma**, which is a small, wall-less, and pleomorphic prokaryotic cell.

It is a member of the class Mollicutes, a class of bacteria that are devoid of a firm cell wall and are very minute in size.

Due to their minute size and lack of cell wall, phytoplasmas cannot be seen using ordinary light microscopes and are typically identified using molecular techniques, such as PCR-based assays.

The phytoplasma responsible for this disease resides in the phloem tissues of the sugarcane plant, where it disrupts normal physiological and developmental processes, especially those related to hormone balance and cell division.

3. Symptoms:

The most typical symptom of grassy shoot disease is the profuse sprouting of auxiliary (lateral) buds at the plant base of sugarcane.

This leads to the development of a heavy bunch of shoots, making the plant bushy or grassy in appearance, which has given the disease its name.

The leaves on the diseased shoots become pale yellowish and narrow, soft, and exhibit signs of chlorosis and decreased photosynthetic activity.

The internodes get abnormally shortened, which makes the plant look weak and stunted in structure.

In the severely infected plants, the growth of normal sugarcane stalks is entirely prevented, and the plants become economically unproductive.

The disease further impairs flowering; most of the infected plants will not flower, or they have abnormal and sterile flowers.

As the disease further advances, the whole sugarcane clump becomes a cluster of delicate, grass-like shoots, which do not develop into marketable canes.

4. Etiology

The pathogen, **phytoplasma**, systemically moves through the plant phloem tissue where it increases and travels within the vascular system.

The major source of disease is using infected seed setts (vegetative planting material), which carry the pathogen inside.

Besides seedborne transmission, the disease is also transmitted by insects, and a key insect vector is the leafhopper insect *Proutista moesta*.

The leafhoppers become infected with the phytoplasma when feeding on diseased sugarcane plants and subsequently transfer it to healthy plants on their saliva while feeding.

Upon introduction into a healthy plant, the phytoplasma propagates and interferes with normal metabolic and hormonal processes, particularly those in apical dominance, causing hyperlateral bud growth and cane suppression.

5. Management:

Cultural Control:

Prevention of the development of grassy shoot disease can be achieved by the use of certified disease-free seed canes from trusted sources.

Periodic roguing (removal) and destruction of the infected plants from the field is necessary to reduce the spread of the disease in a crop cycle.

Implementation of a crop rotation system and discontinuation of constant cultivation of sugarcane on the same land can help disrupt the disease cycle.

Field hygiene by eliminating weeds and crop residues is required since these could act as host or breeding site for the vector insects.

Physical Control:

The disease also can be controlled by hot water treatment of seed setts.

It is done by soaking the sugarcane setts in 50°C water for 30 minutes, which kills the phytoplasma without killing the buds.

This is the most effective and highly recommended pre-planting treatment for grassy shoot disease.

Chemical Control:

To control the insect vectors transmitting diseases, insecticides like Monocrotophos (0.05%), Acetamiprid, or Imidacloprid can be sprayed at regular intervals.

These insecticides reduce the population of the leafhoppers, thus inhibiting the spread of the *phytoplasma* in the crop.

During the initial phase of infection, antibiotics like Tetracycline or Oxytetracycline can be sprayed either by soaking the setts or by injection directly in the plant.

Nevertheless, the application of antibiotics under field conditions is generally restricted by cost, environmental issues, and practicality.

Resistant Varieties:

Where feasible, farmers should plant resistant or tolerant sugarcane varieties, which have been bred and selected by agricultural research organizations.

Such varieties may not entirely stop infection but help considerably mitigate the severity of disease and sustain acceptable yield levels under endemic circumstances.



Grassy Shoot of Sugarcane

Pokkah Boeng of Sugarcane Disease.

Pokkah Boeng of Sugarcane is an important fungal disease that mainly infects young and actively growing sugarcane plants, especially at the early growth stages.

The name "Pokkah Boeng" comes from the Javanese language, in which it means "twisted or deformed leaves," which is one of the main characteristics of the disease.

Pokkah Boeng disease is mainly due to a fungal pathogen *Fusarium verticillioides*, which was previously known as *Fusarium moniliforme*.

Under certain situations, additional related species like *Fusarium sacchari* and *Fusarium subglutinans* can also contribute to disease development, particularly under certain environmental or variety conditions.

These fungi are members of the phylum Ascomycota and class Sordariomycetes and are characterized by their competitiveness in soil and plant residues.

The pathogen is an opportunistic and soil-borne microorganism, which thrives in warm and moist conditions that support its sporulation and infection cycle.

3. Symptoms:

Initial symptoms of Pokkah Boeng typically begin on the youngest unfolding leaves, where chlorotic streaks or yellow patches begin to appear close to the leaf margins.

As the disease progresses, the margins of the leaf become twisted, crinkled, or wavy, making the leaf irregularly shaped, boat-shaped, or fan-shaped.

This leaf morphological change is where the disease gets its name since "Pokkah Boeng" denotes distorted or irregular foliage.

The leaf tissues will split or tear, especially along the folds, and the leaf tips will become shredded, broken, or ragged and seem scorched or burnt.

In harsh infections, the causative fungus might infect the growing point or apical meristem, causing dead heart to form.

The central shoot wilts and is readily removed in such situations, leaving the plant unable to develop further.

Infected plants normally show shortened development, inadequate tillering, and scrawny or immature canes, all of which cause a high decline in cane quality and yield.

This disease is most common during the monsoon season or where there is high humidity and temperature of 25°C to 35°C, which favors quick disease growth.

4. Etiology:

Pokkah Boeng is brought on by *Fusarium* species that can survive for long periods in soil and on sugarcane crop residues.

The fungus releases airborne spores (conidia), which are wind and rain-dispersed and have the potential to infect healthy sugarcane plants.

These spores germinate in ideal environmental conditions, especially when relative humidity is high and the temperatures range within the optimal levels.

The fungus infects the plant mainly on the leaf surface, but it has the ability to infect through mechanical injury or insect feeding-induced wounds.

Upon entry into the plant, the fungus can systemically infect inner tissues, such as the growing point, and cause more severe damage.

Close plant stands and inadequate air movement in the field may also further contribute to the disease by keeping high humidity within the crop canopy.

Young sugarcane plants' fast vegetative growth during the rainy season also provides the perfect environment for the fungus to grow and propagate.

5. Management:

Cultural Practices:

For effective management of Pokkah Boeng, healthy and disease-free seed canes should be obtained from certified sources.

Nitrogenous fertilizers should not be overused since vegetative growth is more favorable with lush growth, where fungal infection takes place more easily.

Proper plant spacing helps to facilitate proper air circulation so that leaf wetness and humidity are minimized, restricting fungal spread.

There should be regular inspections of fields to detect early signs, and any plants or leaves with distortion should be removed immediately and burned to prevent the spread of the disease.

Using crop rotation and not planting monoculture of sugarcane in the same field year by year can cut down on the accumulation of the disease in the soil.

Chemical Control

Application of system fungicides like Carbendazim (0.1%) or Propiconazole (0.1%) at the onset of symptoms is advised.

These fungicides have the ability to penetrate leaf tissues and inhibit the growth of the fungus.

Two to three sprays with a 10–15 day interval during the initial growth stage are effective enough to manage the disease.

Besides, pre-planting seed sett treatment by immersion in fungicidal solution such as Carbendazim or Thiram for 10–15 minutes may significantly lower initial infection levels.

Resistant Varieties:

Growing resistant or moderately tolerant sugarcane varieties ranks among the most efficient and cost-effective disease control methods over the long term.

Agricultural research stations and universities usually suggest certain varieties with field resistance to Pokkah Boeng in various agro-climatic regions.

Field Sanitation and Sanitation:

At the close of the crop season, all contaminated crop residues should be removed and destroyed through deep ploughing or burning to eliminate the spore source.

Correct field sanitation procedures such as managing insects and avoiding plant wounds minimize entry points for the disease and reduce the subsequent crop's pressure.



Pokkah Boeng of Sugarcane

Chapter-14

Mustard Disease, Causal Organism, Symptoms, Etiology and Their Management

Alternaria blight of Mustard disease

One of the most economically significant and most common fungal diseases that infect mustard crops is **Alternaria Blight of Mustard**.

It attacks predominantly the leaves, stems, and siliquae (seed pods) of the mustard plant and leads to tremendous loss in seed yield and oil content, especially under favorable environments.

This disease is prevalent in temperate and wet weather conditions, particularly at the flowering and pod development phases, and thus poses a great threat to growers of mustard.

2. Causal Organism:

The disease is due to the fungal pathogen *Alternaria brassicae*. The disease can also be caused by *Alternaria brassicicola* in certain instances.

Both fungi are members of the phylum Ascomycota and class Dothideomycetes, which specialize in the production of dark, multicellular spores known as conidia that facilitate quick disease transmission.

Alternaria brassicae has been categorized as a necrotrophic pathogen, where it kills the host tissue and then feeds on dead plant tissue, leading to quick tissue deterioration and damage.

3. Symptoms:

The initial visible symptom of Alternaria blight manifests as small, circular to irregularly shaped dark brown or black spots on the older leaves of the mustard plant.

As the disease advances, the spots become progressively larger and show concentric rings or zonations, which impart a target board-like appearance—a characteristic symptom of Alternaria infection.

As the disease progresses to be more severe, these leaf spots merge into big necrotic patches, causing yellowing, early drying, and defoliation of leaves, thus significantly impairing the photosynthetic capacity of the plant.

On the stems, symptoms of the disease appear as elongated, sunken, dark brown to black spots, which can result in weakening, cracking, or even breakage of the stems in extreme conditions.

The siliquae (seed pods) also show symptoms, as black, sunken, circular lesions which can cause shriveling, cracking, and seed abortion.

Infected seeds can be discolored, shriveled, and lightweight, resulting in poor germination and lower quality when planting the following season.

Generally, infected mustard plants have stunted growth, poor or delayed flowering, low seed setting, and eventually substantially lower yield and oil content.

4. Etiology

The *Alternaria brassicae* pathogen can survive for long periods in infected plant debris, seeds, and soil, mostly as mycelium and conidia.

It is a soil-borne, seed-borne, and air-borne fungus that allows it to survive between seasons and travel rapidly when environmental conditions are favorable.

Airborne conidia, dispersed by rain splash, wind, and mechanical abrasion, are the main means of spreading.

Cool and humid weather, particularly temperatures between 18°C to 27°C and relative humidity of more than 80%, are well-suited for spore germination, penetration, and colonization of plant tissues.

Dense canopy fields, over-fertilization with nitrogen, or frequent overhead watered fields can develop high-humidity microclimates that encourage further disease development.

Intensive cultivation of mustard or any other cruciferous plants without crop rotation is the reason for the accumulation of pathogens in the soil, which in turn enhances disease intensity in the following crop.

5. Management:

Cultural Control:

Healthy and disease-free certified seeds should be sown to control *Alternaria* blight effectively.

Do not use seeds from fields that had been infected earlier, as they can have the pathogen inside or on the surface.

Practice crop rotation by interchanging mustard with non-host crops such as cereals or legumes for a period of at least 2–3 years, which decreases inoculum levels in the soil.

Undertake deep summer ploughing after harvest to cover crop residues, thus subjecting fungal spores to high temperature and desiccation, decreasing their viability.

Maintain optimal plant spacing at sowing to promote air movement and minimize humidity within the crop canopy.

Shun excessive use of nitrogenous fertilizers, since these stimulate abundant vegetative growth that is highly prone to infection.

Sowing in time, ideally at the beginning of the season, enables the crop to avoid the best conditions for the disease during flowering and pod formation.

Chemical Control:

Chemical controls are effective if used at the early stage of appearance of symptoms.

Seed treatment with fungicides like Thiram or Captan in the dose of 2–3 grams per kilogram of seed aids in destroying seed-borne inoculum.

For the control of foliar infections, preventative sprays with fungicides like Mancozeb (0.25%), Chlorothalonil (0.2%), or Propiconazole (0.1%) must be started at the initial appearance of the disease.

These sprays must be repeated every 10–15 days, and 2 to 3 repetitions will usually be sufficient as a function of disease pressure and weather.

Fungicide rotation is recommended to avoid resistance in the pathogen population.

Biological Control

The application of biological control agents like *Trichoderma harzianum* or *Pseudomonas fluorescens* as seed treatment or soil application can control early infection by inhibiting the pathogen.

These biocontrol agents also induce plant defense responses and assist in soil health improvement and overall plant health.

Resistant Varieties:

Development of resistant or tolerant varieties is one of the best and cost-effective long-term measures to control Alternaria blight.

Varieties such as Varuna, Kranti, Pusa Bold, and other region-specific resistant lines are suggested by agricultural research centers due to their moderate to high resistance to the disease.



Alternaria Blight of Mustard

White Rust of Mustard Disease

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The disease has also been known by the name White Rust of Mustard.

It is a damaging and highly contagious disease that infects mustard as well as other members of the **Brassicaceae (Cruciferae)** family crop, including radish and cabbage.

White rust is actually not a rust disease, though the name might be misleading, but is instead a fungus-like oomycete pathogen that causes significant economic losses to its host by affecting seed yield and quality.

The disease mostly affects the leaves, stems, floral organs, and siliquae (pods) of the mustard plant.

Under conditions of favorable environment, it causes systemic infection that leads to malformation and sterility in reproductive structures, finally inhibiting pod and seed development.

2. Causal Organism:

Causal organism of white rust is a fungal-like oomycete ***Albugo candida***.

It belongs to the class Oomycetes, phylum Oomycota, which is a class of organisms other than true fungi and yet phylogenetically closer to algae.

Albugo candida is an obligate parasite, requiring a living host plant to fulfill its life cycle and cannot exist on its own.

A variety of cruciferous crops are infected with this pathogen, with mustard being among the hosts most severely infected.

3. Symptoms

The first observable symptoms are on the lower surface of leaves as white, elevated, blister-like pustules.

The pustules contain sporangia, which are asexual spores of the pathogen.

On the upperside of infected leaves, there can be accompanying yellowish spots or chlorotic patches.

As the disease advances, pustules burst open, spewing forth white powdery mass of spores, imparting the disease its distinctive "white rust" appearance.

One of the most typical and debilitating symptoms is the formation of hypertrophied (swollen), distorted inflorescences, commonly referred to as "staghead" or "cabbage head".

These stagheads are caused by hyperplasia and hypertrophy of floral organs, resulting in their swelling, coiling, and sterility, hence the inhibition of pod formation.

In infections that are systemic, the disease is transmitted through the vascular system to stems and flower buds and results in extreme distortions, abnormal growth of tissues, and tumor-like growths.

Such infections can cause absolute sterility and extreme declines in seed yield and quality.

4. Etiology:

The pathogen *Albugo candida* forms oospores (thick-walled resting spores) through which it can survive for extended periods under adverse environmental conditions.

These oospores remain in the soil, plant residue, and seeds infected by the pathogen and act as the main source of inoculum for future cropping seasons.

Under cool (13°C–25°C) and wet conditions (RH greater than 80%), the oospores germinate to develop sporangia, which get dispersed by wind, rain splash, irrigation water, or direct contact between diseased and healthy plants.

Sporangia germinate to release zoospores, which move in a water film and infect plant tissue through stomata or wounds.

High plant population, high frequency of irrigation, overcast weather, and year-round cropping of crucifers are the major factors that promote spread and accumulation of disease in the field.

Infections are particularly serious when the pathogen infects the reproductive organs during the flowering and pod-setting phase.

5. Management

Cultural Control

Use of certified seed of disease-free origin is a must in order to prevent introduction of primary inoculum into the field.

Crop rotation is to be adopted by cultivating non-host crops like cereals or legumes for 2–3 years to disrupt the disease cycle.

Deep summer ploughing uncovers and kills resting oospores in the soil.

Avoid overhead watering to minimize leaf wetness and ensure optimal plant spacing to facilitate good air movement and minimize humidity within the crop canopy.

After harvest, destruction and removal of infected plant residues (particularly staghead-bearing branches) is advised to reduce the carry-over of oospores.

Chemical Control:

Treatment of seed with systemic fungicides like Metalaxyl + Mancozeb (2–3 g/kg seed) will avert early infection from seed-borne inoculum.

Foliar spray with Mancozeb (0.25%), Metalaxyl (0.1%), or Ridomil Gold at pre-flowering stage and again after 10–15 days may control the progress of the disease during the cropping season.

These fungicides exert both protectant and curative action, particularly when integrated into a system for integrated disease management.

Biological Control:

Bio-agents such as *Trichoderma harzianum* and *Pseudomonas fluorescens* can be used on seeds or soil to suppress the viability of inoculum in the soil.

These organisms are inhibitory to the pathogen, they compete with the pathogen for space and nutrients, and they induce systemic resistance in plants.

Resistant Varieties

Growing resistant or moderately resistant mustard varieties is a reliable and sustainable approach.

Crop varieties such as Varuna, Kranti, and Pusa Jaikisan have developed appreciable field resistance against white rust and should be grown in regions where it is endemic.



White Rust of Mustard

Downy mildew of Mustard disease

Downy mildew is a pathological, destructivenon-fungal disease that infects mustard and many other crucifer crops.

This disease is found more commonly in areas facing cool and wet climatic situations, particularly at the initial stages of crop growth.

Downy mildew infects mainly the leaves, which lowers the green leaf area directly affecting the process of photosynthesis, leading to poor seed setting, lower quality, and heavy losses in yield.

In very serious instances, the disease is able to spread systemically across the plant, leading to stunting of growth and deformed reproductive structures.

2. Causal Organism

The causal agent behind the disease is an oomycete species called ***Peronospora parasitica***, which is now commonly known by its synonym *Hyaloperonospora parasitica*.

It is a member of the class Oomycetes in the phylum Oomycota, a water mold group of organisms.

Though long regarded as a fungus, *Peronospora parasitica* is more closely allied in the line of evolution with algae and therefore a fungus-like organism.

It is an obligate parasite, or it cannot exist nor carry on its life cycle in any other way than by being parasitic on a living host plant.

3. Symptoms

Downy mildew in mustard is first observed on the top side of the leaves as pale green to yellowish, irregularly shaped spots.

The lesions gradually increase in size and become necrotic or chlorotic as the infection progresses.

On the lower, corresponding surface of the leaf, there is a grayish-white to purplish downy growth that consists of a mat of sporangiophores each producing sporangia – the pathogen's reproductive organs.

With the progression of the disease, infected leaf tissues will shrivel, dry, and die, causing leaf blight in extreme cases.

Young seedlings or young plants can show systemic infections, leading to dwarfing, distorted growth, and leaf malformation.

In good weather, the disease also infects floral organs, pods, and young siliquae, leading to twisting, deformation, or premature abortion.

Such infections ultimately lessen plant vigor, affect seed setting, and lead to considerable economic losses.

4. Etiology

The primary source of inoculum for this disease is the soil or plant debris containing the oospores – the thick-walled, durable resting spores of the pathogen.

These oospores can survive in adverse environmental conditions and remain viable in the soil for several cropping seasons.

Additionally, the pathogen may be seed-borne, thereby initiating primary infection in the next crop season.

At cool (10°C to 20°C) and humid (greater than 85% relative humidity) temperatures, the pathogen quickly forms air-borne sporangia that spread by wind and infect new leaves.

Infestation is through the stomata on the lower leaf surface, particularly where there is prolonged wetness of the leaves by dew, fog, or sprinkler irrigation.

Contributing conditions such as close spacing, delayed planting, and poor crop rotation contribute to the build-up and dissemination of the disease in mustard crops.

5. Management

Cultural Practices

Employing certified, pathogen-free seeds will prevent the reintroduction of the pathogen into clean plots.

Employ a cropping rotation plan, excluding cruciferous crops for a minimum of 2 to 3 years, and cultivate cereals, pulses, or legumes to terminate the disease cycle.

Opt for early planting of mustard to miss peak times of disease-favorable environmental conditions.

Implement deep summer ploughing post-harvest to bring and kill resting oospores in the soil.

Practice optimal plant spacing to allow air circulation and minimize humidity in the crop canopy.

Prevent overhead irrigation techniques and provide sufficient field drainage to avoid excessive leaf wetness conducive to pathogen growth.

Chemical Control

Treat seeds with a fungicide blend such as Metalaxyl + Mancozeb (for example, Apron XL @ 2–3 g/kg seed) prior to sowing to destroy seed-borne inoculum.

Start foliar sprays of systemic and protective fungicides at the onset of disease.

Good fungicide choices include:

- ✓ Metalaxyl + Mancozeb @ 0.2%
- ✓ Copper oxychloride @ 0.3%
- ✓ Chlorothalonil @ 0.2%

Repeat foliar sprays 10–12 days apart, particularly if cool and humid conditions continue, to be effective in disease control and stop its further spread.

Biological Control

Use bioagents such as *Trichoderma harzianum* or *Pseudomonas fluorescens* as seed or soil treatments to minimize levels of soil-borne inoculum.

These desirable microbes not only inhibit pathogen activity but also enhance plant resistance and general soil health.

Resistant Varieties

prefer planting resistant or moderately resistant cultivars like Varuna, Kranti, and Pusa Jaikisan, which have been known to exhibit tolerance against downy mildew in field conditions.

Choosing regionally recommended cultivars boosts the efficacy of other control options and promotes integrated disease management.



Downy Mildew of Mustard

Chapter-15

Cotton Disease, Causal Organism, Symptoms, Etiology and Their Management

Anthracnose of Cotton disease

Anthracnose is a devastating fungal disease of cotton that drastically affects plant health and yield under hot and humid environmental conditions. Anthracnose can infect cotton plants at any growth stage from seedling emergence to boll formation and infects different parts of the plant such as cotyledons, stems, leaves, and bolls.

1. Causal Organism

The causal organism of anthracnose of cotton is the fungal pathogen *Colletotrichum gossypii*.

This fungus is a member of the class Deuteromycetes (Fungi Imperfecti) and is characterized by its ability to form acervuli, specialized asexual fruiting bodies on which conidia are borne.

The conidia are dark-colored, curved or sickle-shaped, and are commonly formed in a pink to orange gelatinous mass when humidity is high.

In tropical and subtropical areas, a more virulent strain referred to as *Colletotrichum gossypii* var. *cephalosporioides* has been seen, which results in a more serious infection.

2. Symptoms

The disease first shows up in the seedling phase, when it leads to damping-off or seedling blight. Infected seedlings form dark lesions on the stem base, which lead to collapse and death.

Cotyledons (first leaves) form small, round, sunken spots that tend to enlarge and coalesce, leading to drying and shedding of the leaf tissue.

As the disease advances, dark brown to black lesions form on petioles and stems, girdling the plant at times and resulting in lodging or breaking.

Irregular, dark spots are formed on leaves and branches on older plants.

Infected bolls show sunken, circular, dark lesions, and under conditions of high humidity, the lesions are covered with pinkish spore masses.

Infected bolls may not mature well, dry up inside, rot, or stay closed at harvest, leading to inferior lint quality and less viable seed.

3. Etiology

The pathogen mostly survives on infected seeds, crop residue, and soil as mycelium and conidia.

Seed-borne is the largest primary inoculum source in new infestations.

Secondary infection develops by means of wind-borne spores, splashing rainfall, insects, and infected equipment and tools.

The disease flourishes in high temperatures of 20–30°C, relative humidity above 85%, and is particularly so during rainy seasons.

Repeated growing of cotton in the same field, infected seed sowing, heavy sowing, inadequate drainage, and mechanical injury to the plants favor quick disease development.

4. Management

Cultural Practices

Use only certified, disease-free seeds harvested from healthy fields to prevent injecting the pathogen.

Implement crop rotation with cereals or legumes for 2–3 years to disrupt the pathogen life cycle.

Destroy and burn or bury infested crop residues after the crop is harvested to minimize fungal survival within the field.

Ensure proper plant spacing to enhance air movement and decrease excess moisture.

Prevent wounding of stems and bolls at the time of field operations and sow the crop on time to ensure better crop health.

Avoid waterlogging and proper drainage to minimize favorable conditions for fungal growth.

Seed Treatment

Treatment of seeds with Carbendazim @ 2 g/kg or with a combination of Thiram + Carbendazim will destroy surface and internal seed infections.

Alternatively, hot water seed treatment at 52°C for 30 minutes effectively minimizes internal seed-borne inoculum without seed viability loss.

Chemical Control

Use fungicides like Carbendazim (0.1%), Mancozeb (0.25%), or Copper oxychloride (0.3%) at the initial onset of symptoms.

Repeat sprays every 10–14 days, particularly during conducive disease conditions.

Destroy and remove highly infected plants or bolls to curtail spread and reinfection.

Resistant Varieties

Wherever possible, plant anthracnose-resistant or tolerant cotton varieties, which provide good protection and minimize chemical use.

Adopt local extension guidelines for farming to choose locally recommended varieties that have demonstrated resistance against anthracnose.



Anthracnose of Cotton

Vascular Wilt of Cotton Disease

Cotton vascular wilt is a chronic and severe fungal infection that greatly reduces cotton productivity by interfering with the plant's water-conducting system. The disease infects the vascular (xylem) tissues of the plant, causing progressive wilting, yellowing of the leaves, stunting of the plants, and severe yield loss. It is more common in areas with high temperatures and dry weather, particularly where cotton crops are continuously planted without rotation.

1. Causal Organism : The disease is due to a soil-borne fungal pathogen *Fusarium oxysporum* f. sp. *vasinfectum*.

The pathogen belongs to the phylum Ascomycota and is in the class Hyphomycetes, which is known for its asexual reproduction structures.

It forms three kinds of spores:

- ✓ Microconidia, which are single-celled and small.
- ✓ Macroconidia, which are multi-celled and fusiform.
- ✓ Chlamydospores, thick-walled resting spores that enable the fungus to live under unfavorable conditions for many years in soil or plant remains.

The fungus is cotton-specific and can live in soil for many years without a host plant. It infects the cotton by colonizing xylem vessels, thus preventing water and nutrient transport and resulting in the typical wilt symptoms.

2. Symptoms

The initial visible symptom of vascular wilt in cotton is yellowing and drooping of leaves on one side (unilateral wilting) of the plant.

Early in the disease, lower leaves exhibit interveinal chlorosis, i.e., the spaces between the veins become yellow and the veins green.

Later, affected leaves turn marginal browning, dry, brittle and fall off, usually leaving only the terminal shoot at the tip.

The diseased plants exhibit symptoms of stunted growth, smaller vigor, fewer branches, and inhibited boll development, all of which decrease the overall yield.

At later stages of the disease, the wilting is irreversible, and the whole plant may die, particularly under conditions of drought stress or starvation.

One of the diagnostic features is brown staining of vascular tissues, particularly prominent in case of the main stem or roots cut longitudinally to exhibit brown streaks running along the xylem.

3. Etiology (Development of Disease)

The fungus lives through in soil and infected plant residues in a resistant form called chlamydospores, which can survive for many years.

The infection is initiated by the primary infection, which takes place through root tips, wounds, or natural openings in roots, predominantly under conducive environmental conditions.

Root knot nematodes (*Meloidogyne* spp.) usually act synergistically by injuring roots and facilitating the entry and spread of the pathogen, thereby increasing the severity of the disease.

Once it has entered, the pathogen spreads in the root cortex and further advances in the xylem vessels, where it secretes toxins and creates mycelial masses that are responsible for mechanical clogging of water circulation.

Favorable conditions for disease occurrence are high temperatures between 25°C and 30°C, alkaline soil, low soil fertility, and poor soil drainage.

Repeated monocropping of cotton, particularly with vulnerable varieties, leads to high inoculum accumulation in the soil, causing higher disease intensity over a period.

4. Management

A. Cultural Practices

Grow resistant or tolerant varieties of cotton suggested by agricultural institutions for wilt-susceptible zones.

Practice crop rotation by planting non-host plants such as cereals or legumes for a minimum period of 2–3 years to shatter the disease cycle.

Improve the structure and drainage of the soil by adding organic matter and adopting correct tillage measures to prevent waterlogging.

Keep the soil optimally fertile, especially with proper potassium fertilization, to optimize the resistance of the plant against wilt.

Adopt field sanitation by removing and destroying all the plant residues after harvest to prevent the survival of pathogens.

Prevent root mechanical injuries during intercultural practices to reduce infection sites.

B. Biological Control

Treat the soil with bio-agents such as *Trichoderma viride*, *Pseudomonas fluorescens*, and *Bacillus subtilis*, which inhibit *Fusarium* populations through competition, antibiosis, and mycoparasitism.

Use *Trichoderma*-based bioformulations as seed treatment, nursery treatment, or soil application to enhance root health and natural resistance.

Repeated applications of beneficial microbes enhance soil microbiota and diminish pathogen load

C. Chemical Control

Seed treatment with Carbendazim or Thiram + Carbendazim at the dose of 2–3 grams per kilogram of seed can be effective in removing surface-borne inoculum and safeguarding the seedlings from infection at an early stage.

Fungicide soil drenching is not usually very effective because the pathogen is deeply rooted but can be utilized as a supporting measure in nurseries or in case of localized outbreaks.

In nematode-infested fields, use nematicides or plant nematode-resistant varieties since nematodes aid in disease entry and exacerbation.

D. Integrated Disease Management (IDM)

Take a holistic IDM strategy that integrates:

- ✓ Utilization of resistant cultivars,
- ✓ Crop rotation with non-hosts,
- ✓ Biocontrol agents for seed and soil treatment,
- ✓ Balanced fertilization, and
- ✓ Monitoring and early intervention based on disease scouting.

Early detection of early symptoms and application of suitable control methods can avoid spread and minimize economic losses.



Vascular Wilt of Cotton

Black Arm of Cotton Disease Expanded more and make sentence format

Black Arm of Cotton is a lethal bacterial disease causing severe damage to the aerial parts of the cotton plant. It most often appears on the stem, petiole, branches, leaves, and bolls. It is notorious for causing black, elongated lesions on the stem and branches resembling "arms," which is why it is called Black Arm of Cotton. These lesions cause stem breakage, boll shedding, and wilting of leaves, and drastically lower the fiber yield and quality. The disease flourishes under warm, wet weather conditions and is particularly harsh in the initial growth phases of cotton.

1. Causal Organism Causal organism of Black Arm disease of cotton is the bacterium *Xanthomonas citri subsp. malvacearum*, previously recognized as *Xanthomonas campestris pv. malvacearum*.

This bacterium is of the phylum Proteobacteria, class Gammaproteobacteria, and order Xanthomonadales.

It is a motile, rod-shaped, Gram-negative bacterium with a single polar flagellum to facilitate movement.

The disease-causing agent is seed-borne but also lives in infected plant material debris and soil for long times, particularly when environmental conditions are favorable.

2. Symptoms

The signs of Black Arm disease depend on the age of the plant at infection and the infected area.

On Cotyledons and Seedlings:

On young seedlings, the bacterium results in water-soaked, tiny, round spots on cotyledons (seed leaves) and the first true leaves.

Infected seedlings can be stunted, yellowish, and under certain conditions can die prematurely, particularly in high humidity and low air movement.

On Leaves:

On mature leaves, the disease causes angular leaf spots, normally bordered by the veins of the leaf.

The spots start as water-soaked lesions and later turn dark brown to black as they mature.

Where infection is extensive, the lesions merge, resulting in leaf blight, leaf curl, and finally premature defoliation (leaf shedding).

On Petioles and Stems:

One of the most distinctive signs of the disease is black, elongated, sunken lesion formation on petioles, veins, and the stem.

The black lesions are arm-like, and this is how the disease derives its common name—Black Arm.

The lesions will gird (encircle) the petiole or stem and cause breakage, resulting in wilting or killing of the involved shoot or leaf.

On Bolls

On immature, growing bolls, water-soaked lesions form that later turn sunken and black.

The diseased bolls can remain unopened, become internally rotted, or yield discolored, low-quality lint.

In extreme cases, the boll can be shed prematurely, leading to heavy loss of yield.

3. Etiology

The disease is seed-borne in most cases, i.e., infected seeds are the main source of inoculum in new plantings.

The bacterium also exists on crop residues, cotton trash, and soil particles for a period of months.

It gets transmitted from one plant to another by splashing rain, irrigation, wind, insects, and infected tools or human handling.

Entry of the pathogen to the plant is either through natural openings like stomata and hydathodes, or wounds due to insects, pruning, or mechanical damage.

The disease is promoted by hot (25–35°C) and humid (above 80% relative humidity) weather, particularly during the initial stages of crop development.

Poor air movement, high-density planting, and the growing of susceptible cultivars also promote quick disease spread.

4. Management

Black Arm of Cotton control is best achieved through an integrated program using cultural, chemical, and biological means supplemented with the planting of resistant cultivars.

A. Cultural Practices

Always plant certified and disease-free seeds to prevent bringing the pathogen onto the field.

Rotate the crop with non-host crops like cereals and legumes to minimize the carryover of inoculum in the soil.

Destroy and uproot infected crop residues and plant debris at harvest time to minimize the main source of infection.

Keep optimal spacing between plants to promote more airflow and minimize leaf wetness, which inhibits the growth of disease.

Shun overhead irrigation during disease seasons to avoid spurring water splash and bacterial transmission.

B. Host Resistance

Plant resistant or tolerant cotton cultivars, particularly *Gossypium hirsutum* hybrids bred for resistance to bacterial blight.

Plant Bt cotton hybrids with additional genes for resistance to Black Arm and other bacterial diseases.

C. Chemical Control

Treat seeds with Streptocycline @ 100 ppm or with Ceresan/Thiram fungicides @ 2.5 to 3 g/kg seed prior to sowing to destroy the seed-borne inoculum.

In susceptible areas, spray a mixture of Streptocycline (100 ppm) and Copper oxychloride @ 0.25% at the vegetative stage of crop growth.

Repeat sprays at 10–15 days' intervals, particularly in humid or rainy weather when disease risk is high.

Begin chemical control as soon as the first symptoms are observed to avoid secondary spread to other plants.

D. Biological Control

Application of plant growth-promoting rhizobacteria (PGPR) like *Pseudomonas fluorescens* has been found to decrease disease severity.

These desirable microbes not just suppress the growth of *X. citri* subsp. *malvacearum*, but also enhance plant health, root development, and tolerance to other stress factors.

E. Integrated Disease Management (IDM)

IDM combines the integrated practice of several management tactics including:

- ✓ Resistant cultivars,
- ✓ Clean and treated seeds,
- ✓ Timely cultural practices,
- ✓ Regular monitoring,
- ✓ Careful application of bactericides,
- ✓ Destruction of infected plants and residues.

Field scouting on a regular basis facilitates early detection to enable timely action to limit the outbreak from spreading before incurring immense losses.



Black Arm of Cotton

Chapter–16

Guava Disease, Causal Organism, Symptoms, Etiology and Their Management

Wilt of Guava Disease.

Guava wilt is a very destructive vascular disease affecting chiefly the roots and inner conducting tissues (xylem) of guava. It causes sudden wilting, gradual yellowing of leaves, and finally the death of the plant. The disease is serious in poorly drained orchards or where guava is planted continuously without rotation, and it is a severe constraint to guava production in most areas.

1. Causal Organism

Guava wilt is caused mainly by a fungal pathogen that lives in the soil named *Fusarium oxysporum f. sp. psidii*.

Fusarium oxysporum f. sp. psidii belongs to the Ascomycota phylum, class Sordariomycetes, and Hypocreales order.

It forms thick-walled, long-surviving chlamydospores that can survive for many years in the soil or plant rubbish even without a host plant.

Under certain environmental conditions, the disease will also be caused by a complex of pathogens such as *Macrophomina phaseolina*, *Cephalosporium* species, and *Rhizoctonia solani* that can increase the severity of disease and make diagnosis and control more difficult.

2. Symptoms

Symptoms on Leaves:

The disease will usually initiate with yellowing of the leaves, especially starting from the older leaves at the bottom of the plant and migrating progressively upwards.

As the illness advances, wilting and leaf curling are manifested, particularly at the warmer times of the day.

In due time, the infected leaves fall off prematurely, leading to partial or full defoliation of the tree.

Branch and Stem Symptoms

Infected branches usually have dieback symptoms, starting from the tip and advancing downwards.

On cutting open infected branches or the stem, there is a characteristic brown to black discoloration of vascular tissues, which is the sign of internal vascular blockage.

In later stages, splitting and cracking of bark on the main limbs and trunk.

Root symptoms

The root system of infected plants indicates extreme weakening with discoloration, rotting, and drying of the main roots and feeder roots.

Fine feeder roots tend to be the first ones to perish, but greater roots turn dark, brittle, and rotting.

When the vascular system is progressively clogged, the plant cannot take in water and nutrients, and it progresses to irreversible wilting and plant death.

General Symptoms

The disease can result in a sudden collapse or wilting of apparently healthy trees, often during conditions of water stress or hot weather.

In highly infected orchards, several plants can die in patches, resulting in enormous economic losses, particularly during the hot and dry summers.

3. Etiology (Disease Cycle and Development)

The *Fusarium oxysporum f. sp. psidii* pathogen is able to survive in the soil or infested plant residues in the form of chlamydospores, which have the ability to endure for numerous years.

The root is the main infection mode, particularly through wounds or natural openings, like those created through drought, nematode infection, or mechanical injury.

Once inside the plant, the fungus infects the xylem vessels, where it grows as mycelium and spores that physically clog the water-conducting tissue.

Besides physical obstruction, the fungus can also secrete toxins and enzymes that destroy tissues, further obstructing water and nutrient transport.

Environmental conditions such as excessive soil temperature (28–35°C), poor drainage, waterlogging, and low soil organic matter highly promote the development and advancement of disease.

Monoculture activities involving continuous cropping of guava and the lack of crop rotation, which cause the accumulation of pathogens in the soil, also contribute to the advancement of the disease.

4. Management

A. Cultural Practices

Do not plant guava in wilt-infected or poorly drained soils.

Rotate the crop with non-host crops like cereals or legumes for a minimum of 2 to 3 years to shatter the disease cycle.

Remove and eliminate all the infected and wilted plants immediately to avoid further disease spread.

Improve soil health and structure with the incorporation of well-decomposed organic compost or manure, which promotes microbial activity and inhibits pathogens.

Provide proper plant spacing and prevent over-irrigation to prevent root stress and discourage disease development.

B. Resistant Varieties

Graft wilt-tolerant rootstocks of *Psidium friedrichsthalianum* on guava scions, which exhibit strong resistance to *Fusarium* wilt.

Employ planting materials from certified, disease-free mother plants or nurseries.

C. Chemical Control

Use Carbendazim (0.1%) or Thiophanate-methyl as soil drenches close to the root zone to check fungal growth.

Treat seedlings in the nursery by immersing their roots in fungicide solutions (e.g., Carbendazim at 1 g/L) prior to transplanting.

Include *Trichoderma harzianum* and neem cake in the soil to lower soil-borne inoculum and enhance root health.

D. Biological Control

Employ biological control agents like *Trichoderma viride*, *Trichoderma harzianum*, and *Pseudomonas fluorescens* at the time of land preparation.

These useful microbes compete with *Fusarium* for nutrients and space and also secrete antibiotic substances inhibiting the pathogen.

Removal of bioagents, particularly repeated, enhances soil microbiome diversity and facilitates long-term suppressive environment against the pathogen.

E. Integrated Disease Management (IDM)

Practice an integrated approach involving soil solarization (placing moist soil under transparent polythene sheets for 4–6 weeks in summer) to destroy resting spores.

Integrate cultural, chemical, and biological control practices with resistant planting stock to promote improved disease control.

Monitoring of orchards routinely, particularly at times of high risk (hot and dry seasons), facilitates early detection and resultant timely action.

Maintain sound agronomic and orchard management practices to promote improved overall tree vigor and improve resistance against wilt.

Conclusion

Wilt of guava is a severe and perennial soil-borne disease that is a significant threat to guava cultivation in most tropical and subtropical countries. Multifaceted management is necessary, involving early detection, planting with clean and resistant planting material, right soil and orchard management, and the correct application of fungicides and biocontrol agents. Sustainable control of guava wilt can only be achieved through the long-term adoption of integrated disease management (IDM) practices and maintaining overall orchard hygiene and plant health.



Wilt of Guava

Anthracnose of Guava Disease

Expended more and make sentence format

Anthracnose of guava is a serious, economically significant fungal disease that infects the fruits but also the leaves and twigs of guava trees. It results in fruit blemishes, premature fruit shedding, and high post-harvest losses, especially under hot and humid weather conditions during flowering and fruit development phases.

1. Causal Organism

The disease is mostly caused by the fungal pathogen ***Colletotrichum gloeosporioides***.

This fungus is classified under the phylum Ascomycota, class Sordariomycetes, and order Glomerellales.

Colletotrichum gloeosporioides is an asexual spore-producing fungus that yields conidia, which are fusiform (spindle-shaped), single-celled, and hyaline. It is these conidia that are capable of spreading the disease at a fast rate, particularly in high humidity.

In certain regions, closely related species such as ***Colletotrichum psidii*** and ***Colletotrichum acutatum*** have also been found to create the same symptoms on guava.

2. Symptoms

On Leaves:

Small, brown to dark brown, irregular lesions on the upper surface of mature and young leaves.

Lesions can expand and join together under humid conditions, forming extensive necrotic patches.

Infected leaves can become severely infected, curl, dry, and drop early, causing partial defoliation.

On Fruits:

Most typical symptom is the development of sunken, circular, dark brown to black spots on the skin of immature and mature fruits.

These lesions tend to form concentric rings, which result in them having a characteristic "anthracnose" look.

Under wet conditions, pink to orange masses of spores can form in the center of the lesion through sporulation of the fungus.

Fruits infected with the pathogen tend to become hard, rot internally, and fall prematurely, rendering them unfit for sale.

On Branches and Twigs:

On occasion, the pathogen leads to tiny sunken cankers or dieback of young twigs.

Under heavy infection, especially in nurseries or during rainy weather, twig blight might occur.

3. Etiology (Disease Development and Cycle)

The fungus overwinters in the orchard as dormant mycelium or conidia on infected plant structures like twigs, dropped leaves, and mummified fruits.

When environmental conditions are conducive, i.e., temperatures between 25–30°C and relative humidity greater than 85%, the resting spores germinate and create fresh infections on tender, young leaves, twigs, and fruits.

The major means of dissemination are air-borne conidia, rain splash, insects, and mechanically via tools employed while pruning or harvesting.

The pathogen frequently enters the host plant via natural openings like stomata or insect-insect and abraded wounds.

Latent infections are prevalent, especially in fruits, where the pathogen infects at early development stages but is latent until ripening or storage and causes post-harvest rot.

4. Management

A. Cultural Practices

Remove infected twigs regularly, drop fruits and leaves, and dispose of the infected matter to reduce the source of inoculum.

Enhance air circulation and sunlight penetration by regular canopy management to lower the humidity levels within the orchard.

Prevent overhead irrigation, particularly at fruit development phases, as this enhances leaf wetness and promotes spore transmission.

Regularly clean equipment and tools to ensure orchard hygiene.

B. Resistant Varieties

These include some wild *Psidium* species and specific guava cultivars like Allahabad Safeda, Lucknow 49, and certain selections from Florida. Additionally, interspecific hybrids and rootstocks developed from crosses with wilt-resistant wild relatives like *Psidium molle* show promise.

Always obtain planting material from certified, disease-free nurseries.

C. Chemical Control

Application of systemic fungicides like Carbendazim (0.1%), Thiophanate-methyl (0.1%), or Propiconazole (0.1%) at 10–15 day intervals during flowering and fruit development can substantially cut down disease incidence.

Alternate fungicides belonging to different chemical classes to prevent resistance buildup.

Spray pre-harvest fungicide application for the control of latent infections and post-harvest rot.

D. Biological Control

Trichoderma harzianum and *Pseudomonas fluorescens* may be applied to the soil or leaves as biocontrol agent to check the pathogen.

Use neem-based bio-pesticides or plant extracts (e.g., garlic, turmeric) as supplementary protective treatments at low to moderate disease pressure.

E. Post-Harvest Management

Dip harvested fruits in hot water (50–55°C) for 5–10 minutes to minimize surface and latent infections.

Dipping fruits in a solution of Carbendazim (0.1%) extends shelf life.

Keep guava fruits in cool, well-ventilated, and dry storage conditions to slow symptom expression and decrease spoilage.

F. Integrated Disease Management (IDM)

Integrate orchard sanitation, planting with disease-free material, selective fungicide application, biocontrol, and post-harvest treatments to ensure long-term sustainable control. Keep track of weather patterns (particularly humidity and temperature) to time fungicide sprays appropriately to minimize unnecessary sprays.

Conclusion

Guava anthracnose is a very destructive fungal disease that substantially affects fruit quality and storage life. Warm and humid conditions promote its dissemination, and latent infections are usually expressed after harvest. A balanced and integrated management system integrating cultural, chemical, biological, and post-harvest practices is needed for successful management and reduction of crop as well as economic losses.



Anthracnose of Guava

Chapter–17

Banana Disease, Causal Organism, Symptoms, Etiology and Their Management

Panama wilt of Banana Disease

Panama wilt of banana is a destructive and chronic soil-borne fungal disease that has inflicted immense harm to banana plantations globally. It mostly targets the vascular system of banana plants, producing symptoms like yellowing, wilting, and death. The disease has the potential to annihilate full plantations and is particularly virulent in warm, humid conditions. It is a critical threat to commercial banana production, particularly cultivars such as Cavendish.

1. Causal Organism

Banana Panama wilt is caused by the fungal pathogen *Fusarium oxysporum* f. *sp. cubense* (Foc).

The causal agent is a member of the phylum Ascomycota, class Sordariomycetes, and order Hypocreales.

The fungus produces three forms of spores:

- ✓ **Microconidia** – small, unicellular spores involved in short-distance transmission.
- ✓ **Macroconidia** – large, multicellular spores that contribute to infection.
- ✓ **Chlamydospores** – waxy, very resistant spores that can make the fungus live in soil for over 20 years without even a banana crop.

There are various physiological races of the pathogen, with Race 1, Race 2, and the extremely virulent Tropical Race 4 (TR4) being the most important.

Race 1 primarily infects varieties such as 'Gros Michel'.

Tropical Race 4 infects a more extensive number of cultivars that include the worldwide preponderant 'Cavendish' group, so it is the deadliest strain.

2. Symptoms

Foliar Symptoms

The disease first appears as yellowing of older (lower) leaves, which starts from the leaf margins and moves towards the midrib.

As the disease spreads, such yellow leaves shrivel and drape down along the pseudostem, creating a skirt-like effect around the plant.

Eventually, young leaves will also yellow, and the plant is stunted and weak.

Internal Symptoms

One of the most diagnostic signs is reddish-brown or dark brown vascular bundle discoloration observed when the pseudostem or rhizome is split longitudinally open.

This vascular discoloration reflects obstruction of water transport as a result of fungal colonization.

Plant and Fruit Symptoms

Premature wilting and drying of infected banana plants occur, usually before the bunch is mature.

The fruit bunches can be reduced in size, deformed, or even absent.

Plants severely affected do not recover, and entire mats (clusters of banana plants) can become extinct in the field.

3. Etiology (Disease Development and Cycle)

The fungus persists in the soil as chlamydospores, which are tolerant of unfavorable conditions and can survive for decades.

Infection is initiated when the fungus enters the roots, usually through wounds or natural openings.

Upon entering the plant, the pathogen spreads to the xylem tissues and creates enzymes, toxins, and gums that clog water transport, causing wilting.

Being a vascular wilt, it interferes with the internal water and nutrient transport of the plant.

The disease is mainly transmitted by:

- ✓ Infected planting material (infected suckers or rhizomes),
- ✓ Contaminated soil or water,
- ✓ Tools, farm equipment, and footwear,

- ✓ Flooding or irrigation management that transports soil particles.
- ✓ Warm temperatures (27–30°C) and excessive soil moisture predispose to rapid progress of the disease.

The pathogen is not transmitted through seed but primarily through vegetative propagating materials.

4. Management

A. Cultural Practices

- ✓ Don't cultivate in areas previously affected by Panama wilt.
- ✓ Utilize only certified and healthy tissue culture plants or suckers.
- ✓ Frequently remove and destroy infected plants in order to minimize inoculum.
- ✓ Adopt good field sanitation, such as clean tools and limited field access.
- ✓ Improve drainage of soil to avoid waterlogging, which allows for infection.
- ✓ Implement crop rotation with non-host crops like rice or sugarcane for 4–5 years to suppress pathogen population in soil.

B. Resistant Varieties

'FHIA-01', 'FHIA-18' (Honduras hybrids)

'GCTCV-119', 'GCTCV-218' (Cavendish selections resistant to TR4)

Local varieties like 'Poovan' and 'Pisang Awak' are resistant to Race 1.

Grafting weak cultivars on to resistant rootstocks is an experimental approach to minimize risk.

C. Chemical Control

Fungicides will not be effective in managing **Fusarium** wilt after infection of the plant.

But drenching of soil with systemic fungicides such as Carbendazim (0.1%) or Thiophanate-methyl (0.1%) can give temporary protection at early stages.

Pre-plant treatment of suckers or tissue culture plants with fungicides or biocontrol agents is advised.

D. Biological Control

Use of disease-suppressing microbes such as:

- ✓ *Trichoderma harzianum*
- ✓ *Pseudomonas fluorescens*
- ✓ These antagonists infect the root zone and stop the pathogen by competition, antibiosis, or induced resistance.
- ✓ The addition of neem cake and other organic amendments can stimulate microbial activity and inhibit Fusarium.

E. Quarantine and Sanitation

Strict quarantine is to be implemented to avoid spread of TR4 to unaffected regions.

Clean and disinfect boots, tools, and vehicles prior to entering new plantations.

Create buffer zones and restricted entry zones in infected fields.

F. Soil and Nursery Management

Soil solarization with transparent polythene during summer can be used to minimize inoculum in nurseries.

Keep nurseries in TR4-free areas, and produce planting material under net houses with sterilized soil.

Conclusion

Panama wilt of banana is among the most dangerous and enduring threats facing world banana production. Its soil-borne status, extended survival period, and restricted chemical control measures render it a recalcitrant disease to be controlled once it has gained a foothold. The application of resistant cultivars, clean planting materials, rigorous sanitation, and biological agents provides the foundation for disease control.

There needs to be an integrated disease management (IDM) strategy uniting cultural, biological, and quarantine methods to limit the spread and protect banana plantations for generations to come.



Panama Wilt of Banana

Bacterial wilt of banana

Bacterial wilt of banana, also referred to as Moko disease or banana blood disease, is a fatal and highly infectious disease of the banana plant that causes systemic wilting, profound vascular discoloration, rotting of fruit, and eventually death of

the plant. The disease is of profound economic significance in tropical and subtropical banana-producing areas worldwide.

1. Causal Organism:

The cause agent of bacterial wilt of banana is a bacterium named ***Ralstonia solanacearum***, previously identified as *Pseudomonas solanacearum*. The infecting strains are typically Race 2, and the involved biovars are either Biovar 1 or Biovar 3, varying with geographic region.

Ralstonia solanacearum is a Gram-negative, rod-shaped, soil-borne bacterium with aerobic respiration and motility by a polar flagellum. It is a member of class Betaproteobacteria and family Burkholderiaceae. It has a broad host range and infects more than 200 crops such as banana, tomato, potato, eggplant, and ginger.

2. Symptoms:

The earliest signs of bacterial wilt in banana leaf start with a sudden wilting of young leaves, particularly innermost or center leaves, which immediately

yellow and subsequently fold. Yellowing and wilting move outward from the center to the older leaves. As the disease progresses, entire leaves wilt and dangle along the pseudostem, making the plant wilted and lifeless.

When the pseudostem, peduncle, or corm is cut, a yellowish or milky bacterial ooze that usually drips from the vascular tissues can often be seen, a diagnostic symptom of the disease. The vascular bundles, internally, have a conspicuous reddish-brown or dark brown discoloration, especially in the pseudostem, leaf sheath, and corm. In a few instances, the fruit is rotting, with darkened, water-soaked pulp and, at times, seed-like structures within seedless cultivars.

Another common diagnostic technique is the bacterial streaming test, where infected fresh-cut tissue is introduced into clear water and cloudy white streams of bacteria exude, showing systemic infection. The disease infects not only the mother plant but also its suckers and even the fruit bunches, ultimately causing the plant to collapse completely.

3. Etiology

The disease is caused when *Ralstonia solanacearum* infects the banana plant via wounds on the root, corm, suckers, or flowering tissues. The entry points may be natural wounds, insect feeding sites, or mechanical wounds.

Insect vectors like stingless bees, fruit flies, and flower insects are important in the transmission of the bacterium via the male flower parts. Upon entry of the plant, the bacterium establishes a colony and spreads via the xylem vessels and multiplies at high rates and clogs water and nutrient transport, causing wilting and subsequent death of the plant.

The pathogen survives for extended periods in soil, infected crop residues, planting materials, and contaminated irrigation water. It is both water-borne and soil-borne, and spreads quickly through infected hands, suckers, farm tools, and contaminated suckers.

Environmental conditions that are conducive to the development of disease are warm temperatures between 28°C and 35°C, high relative humidity, waterlogged soils, and open wounds. The ability of the pathogen to survive in the soil and water increases the difficulty in managing it.

4. Management:

A. Cultural and Sanitary Practices:

Successful control of bacterial wilt starts with the utilization of clean and disease-free planting material, like certified tissue-cultured banana plants or

suckers from disease-free fields. Field hygiene should be ensured by regular monitoring of banana mats and the removal and destruction of infected plants and their adjacent suckers.

The excision plant material will either be burned or buried deeply to inhibit the growth of the pathogen in the soil cycle again. Pruning and pronging equipment will be disinfected regularly using 1% sodium hypochlorite (bleach) to prevent mechanical transmission of the pathogen.

Farmers should avoid the use of suckers from already infected fields and adopt crop rotation with crops that are not hosts, e.g., maize, beans, or grasses. Adequate installation of field drainage systems prevents excess soil moisture, which favors the spread of disease.

B. Eradication and Exclusion Measures:

In heavily infected areas, quarantine or field exclusion measures should be adopted to prevent the spread of infected planting material or soil from one place to another. Infection-infested mats with all suckers attached should be eradicated at the earliest by identification.

Trap cropping with susceptible bananas can be used to detect and eliminate infections early, thereby reducing further spread.

C. Biological and Physical Control:

In propagation or nursery space, solarization of the soil (diluting moist soil under clear plastic during heat waves) may be used to suppress the bacterial load.

While no commercial biological control products are yet firmly established, preliminary investigation into healthy microbes like *Pseudomonas fluorescens* and *Bacillus subtilis* has demonstrated promise in inhibiting the pathogen under controlled conditions.

D. Host Resistance:

Generation and utilization of tolerant or resistant banana cultivars present a sustainable strategy for disease control. Certain wild *Musa* species and breeding lines like FHIA-01, FHIA-17, and Pisang Awak indicate partial resistance to *Ralstonia solanacearum*. Some cultivars employed in East African Highland Bananas also display field-level tolerance in certain areas.

E. Vector Control:

To avoid insect-mediated transmission, de-budding of the male flower bud (also referred to as de-budding) following fruit setting is advised since these are primary points of entry for bacteria brought in by insects.

Plastic or mesh bagging of the fruit bunch can also act as a physical barrier by minimizing the possibility of insect visit and mechanical damage.

Conclusion

Bacterial wilt of banana is a devastating, rapidly spreading disease that endangers commercial and subsistence production of banana. As chemical control is not very effective, management of the disease is based on preventive approaches such as the use of clean planting material, sanitation, management of vectors, and, where feasible, use of resistant cultivars.

An integrated disease management (IDM) strategy, which incorporates cultural practices, biological control, quarantine enforcement, and farmer education, is still the best way to reduce yield losses and make banana production sustainable in affected areas.



Bacterial Wilt of Banana

Bunchy Top of Banana Disease

(A very destructive viral disease that results in stunted plant growth and the telltale "bunchy" leaves on banana plants)

1. Causal Organism:

Bunchy Top of Banana is caused by a very infectious virus known as the Banana Bunchy Top Virus (BBTV).

The virus is a member of the genus **Babuvirus** in the family Nanoviridae and consists of circular, single-stranded DNA.

BBTV is phloem-restricted, or in other words, it stays within the phloem tissue of the banana plant, where it multiplies and inflicts damage.

The main vector for transmitting BBTV is the banana aphid, whose scientific name is *Pentalonia nigronervosa*.

The virus is both persistent and circulative in transmission, i.e., once infected in the aphid, it is present inside the body of the insect and can transmit to other healthy plants during its lifespan.

Notably, BBTV is not seed-, mechanically transmitted-, nor soil-borne-transmitted and its spread relies virtually totally on infected planting materials and aphid vectors.

The utilization of infected suckers is one of the primary causes of infecting the virus onto new banana plantations.

2. Symptoms:

The most typical symptom of the disease is the "bunchy top" feature, where a number of small, narrow, and erect leaves appear tightly bunched at the tip of the plant.

During early infection, dark green spots or streaks may be seen on the lower surface of the leaf petiole and midrib.

Young developing leaves become stiff, brittle, narrow, and erect, and they can show marginal chlorosis, inward curl, or twisted appearance.

The lamina (blade of the leaf) of such leaves becomes small in size, giving the plant top a rosette-like or crown-like look.

In heavily infected plants, development is totally retarded, and no flower or fruit bunch is produced by the plant.

Fruits in lightly infected plants are small, deformed, or underdeveloped and hence not fit for commercial purposes.

One of the characteristic features of BBTV infection is the "Morse code" pattern, which is seen as brief, dark green dashes or lines on veins and petioles.

The disease has a tendency to spread to suckers, mother plants, and surrounding healthy banana plants if due care and sanitation practices are not adhered to.

3. Etiology

BBTV disease cycle starts when banana aphids feed on infected plants and become carriers of the virus from the phloem tissue.

After entering the aphid's body, the virus is systemic and persistent, which enables it to be transmitted along the lifespan of the vector to healthy banana plants during feeding.

Upon being introduced into a healthy plant, the virus replicates in the cells of the phloem and is distributed through the vascular system of the plant.

The virus disrupts normal nutrient transport, causing abnormal growth, leaf deformation, and fruiting halt.

BBTV is not persistent outside the aphid or plant, hence cannot be transmitted by tools, soil, water, or plant residues.

Its spread is, however, greatly facilitated by the transmission using infected suckers or vegetative propagules, as is the case with conventional banana farming practices.

Environmental factors that encourage the rapid build-up of disease include high temperatures, high humidity, and close plantation spacing, all of which also favor the multiplication of aphids.

The under-surface of banana leaves usually has banana aphid colonies, enabling uninhibited passing of the virus over time.

4. Management

A. Cultural Practices

Always utilize virus-free and tissue-culture grown banana plants that have been sourced from certified nurseries in order to avoid virus introduction during planting.

Make frequent field inspections for identifying infected plants at initial stages.

Immediately pull and destroy all the plants that exhibit bunchy top symptoms, along with their suckers, to prevent serving as a source of inoculum.

Do not utilize suckers from fields with known BBTv history, even where they are symptomless.

Implement stringent sanitation practices such as field cleaning, adequate spacing, and eradication of weeds likely to harbor aphids.

Implement roguing by pulling out complete infected mats (main plant and its suckers) to stop local disease transmission.

B. Vector Control:

Monitor aphid population regularly through the use of yellow sticky traps and visual inspection, particularly on the lower leaf surface.

Use systemic insecticides such as Imidacloprid, Thiamethoxam, or Acetamiprid to manage aphid populations effectively.

In integrated farming systems, employ botanical insecticides like neem oil (Azadirachtin) to inhibit aphid feeding and reproduction.

Horticultural mineral oils applied can serve as a physical barrier, disrupting virus acquisition and aphid transmission.

Aphid management must be persistent and planned, especially during hot and humid weather when populations are liable to grow at a geometric rate.

C. Quarantine and Eradication:

In BBTv-free areas, impose strict quarantine measures to avoid the movement of contaminated planting material.

Create buffer zones or disease-free belts around areas infected with the disease in order to stop the spread to surrounding plantations.

Where disease is endemic and prevalent, total elimination of banana plantations followed by field fallowing for a few months could be required for successful control of the disease.

D. Resistant Varieties:

No banana variety has complete resistance to BBTv at present, but some wild species and hybrids exhibit partial resistance or tolerance.

Efforts are being concentrated on breeding programs and genetic engineering methods for developing BBTv-resistant cultivars.

Utilizing tolerant cultivars in conjunction with clean planting material and control of vectors can yield partial protection in high-risk environments.

F. Integrated Disease Management (IDM)

A balanced management strategy combining cultural, biological, chemical, and regulatory practices provides optimal protection against BBTV.

IDM encompasses: use of certified clean planting material, rigorous monitoring and roguing, consistent aphid control, and community-level disease surveillance.

Train extension workers, farmers, and nursery personnel to identify early symptoms of the disease, apply control methods, and report the occurrence of outbreaks early.

Promote area-wide management practices through cooperation by farmers in an area to suppress aphid populations and curtail disease spread as a group.

Conclusion

Bunchy Top of Banana is rated as among the most economically crippling viral banana disease globally.

The disease results in severe stunting, deformed leaf growth, and complete loss of yield, resulting in significant losses in income for the growers of bananas.

Since there is no good cure once the plant is infected, the best approach for disease control is prevention.

It can be done by following the use of virus-free planting material, early diagnosis and removal of infected plants, and strict control of aphid vectors.

Long-term disease management is also contingent on enforcement of quarantine, organized farmer involvement, and the development of resistant varieties with the aid of modern biotechnology and breeding programs.

Sustainable banana production in BBTV-susceptible areas needs concerted efforts at the field, community, and policy level.



Bunchy Top of Banana

Chapter–18

Papaya Disease, Causal Organism, Symptoms, Etiology and Their Management

Foot Rot of Papaya Disease

(A deadly soil-borne disease that infects the basal stem or collar area of papaya plants, frequently leading to plant wilting, collapse, and heavy yield loss.)

1. Causal Organism:

Papaya foot rot is caused primarily by a fungal-like pathogen *Phytophthora palmivora*.

It is classified under the phylum Oomycota, class Oomycetes, and order Peronosporales.

In contrast to the true fungi, *Phytophthora* species are generally known as water molds, as they prefer to live in wet environments and release motile spores.

Phytophthora palmivora flourishes most optimally in hot, moist, and poorly drained soils, and tropical and subtropical regions are most suitable for its growth.

The disease is caused by the pathogen reproducing and propagating through multiple forms of spores such as sporangia, which discharge zoospores (motile spores) and oospores, which are long-term survival in soil.

Other associated species, like *Phytophthora parasitica* and *Phytophthora nicotianae*, occasionally cause foot rot in papaya under certain environmental conditions.

2. Symptoms:

The disease usually starts with the formation of water-soaked lesions at the base of the papaya stem or collar area, normally a little above the soil level.

These lesions get larger, become dark, and become sunken as the infection advances, ultimately resulting in the softening, rotting, and cracking of the stem tissues.

One of the first symptoms seen on infected plants is yellowing and drooping of leaves, which can be confused with water stress or nutritional deficiency.

In sunny weather, these conditions quickly develop further to produce sudden wilting of the plant.

In most instances, the infected stem tissues develop a putrid smell, largely because of the invasion of secondary bacterial organisms.

As the disease progresses, the whole plant can collapse and die, especially in fruiting phases when the stress on plants is greater.

Under moist conditions, a white, cottony mycelial growth made up of pathogen mycelium and spores may be observed on the infected base of the stem, showing active sporulation by the pathogen.

3. Etiology

The causal pathogen, *Phytophthora palmivora*, is soil-borne and may be able to survive for many years in the soil as chlamydospores, oospores, or dormant mycelium.

Infection normally begins when zoospores are washed out by rain or irrigation and infect wounded or softened tissue around the papaya stem base.

The zoospores are swimming and are able to swim in water films present in the soil and thus the disease is particularly common in wet or water-logged and poorly drained fields.

The pathogen is easily spread from plant to plant with rain or irrigation splashing water.

High soil moisture, temperatures of 25°C-30°C, and relative humidity greater than 85% enhance the disease.

After entering the host plant, it quickly colonizes the vascular tissues, blocking the movement of water and nutrients, which leads to wilting and eventual killing of the plant.

Repeated growth of papaya in the same field or neglect of removal of infected residues results in a soil inoculum buildup, which enhances the risk and severity of subsequent infections.

4. Management:

A. Cultural and Preventive Practices:

Successful management of foot rot starts with the choice of well-drained fields and the avoidance of low areas or water-logged situations where the pathogen will prosper.

Raising seedbeds or planting on mounds is suggested to enhance drainage and minimize moisture in the root zone and collar area.

Precaution is taken to prevent mechanical damage to the stem base while weeding or intercultural practice, as the wound is an entry point of the pathogen.

Remove and destroy the infected plants and plant residues to avoid the spread of the disease.

Ensure application of balanced fertilizers so that there is less use of nitrogen, which promotes the growth of succulent tissues more susceptible to infection.

B. Resistant Varieties:

While full resistance is absent, some papaya varieties like 'Pusa Delicious', 'Red Lady', and 'Surya' are tolerant or partially resistant to foot rot under field conditions.

Growing such varieties as part of an integrated approach to disease management can serve to reduce crop losses.

C. Chemical Control:

The disease may be controlled by soil drenching with systemic fungicides like Metalaxyl, Mefenoxam, or Phosphonate-based fungicides (e.g., potassium phosphite) on the base of the plant.

Use of copper-based fungicides like Copper oxychloride on the collar area protects the stem from initial infection.

To avoid the development of resistance to fungicides, fungicides with various modes of action should be rotated.

D. Biological Control

Biological control provides a green and eco-friendly solution.

The application of beneficial microorganisms like *Trichoderma harzianum* or *Pseudomonas fluorescens*, used in the form of soil amendments, seedling dips,

or root zone treatments, has proven to be useful in reducing the population of the pathogen.

Multiple applications at the early stages of the crop can establish a balance of healthy microbes within the rhizosphere, providing an antagonistic habitat for *Phytophthora*.

E. Integrated Disease Management (IDM):

To achieve long-term control, there should be adoption of an Integrated Disease Management (IDM) approach that integrates cultural practices, chemical and biological control, and resistant varieties.

Monitoring in the field should be enhanced during rainy periods or when there are high humidity levels, and all disease management measures should be done preventively, not curatively.

Conclusion:

Papaya foot rot is a serious disease threat to papaya production, particularly in tropical and subtropical areas with high humidity and poorly drained soils.

If left uncontrolled, it can result in the death of the plant, complete crop loss, and major economic losses to farmers.

A combined strategy with proper drainage, preventive use of fungicides, employment of resistant or tolerant cultivars, biological control agents, and cultural sanitation is necessary to minimize disease pressure.

Early detection, frequent monitoring, and prompt removal of infected plants are necessary to avert the disease from spreading and becoming uncontrollable.



Foot Rot of Papaya

Leaf curl of Papaya Disease

(A destructive viral disease of papaya that results in leaf distortion, stunted plant growth, low fruiting, and economic losses.)

1. Causal Organism:

The Leaf Curl Disease of Papaya is brought about by a DNA virus called the Papaya Leaf Curl Virus (PaLCuV).

It is a member of **Begomovirus** genus within the family Geminiviridae.

It is a monopartite virus with a single circular single-stranded DNA (ssDNA) genome encapsidated within geminate (twin) particles.

The virus is not seed- nor mechanically transmitted, which distinguishes it from some other plant viruses.

Rather, it is naturally transmitted through whitefly vectors, primarily the ***Bemisia tabaci*** species, which pick up the virus during feeding on infected papaya plants.

The whitefly transmission of the virus is continuous and circulative, i.e., the virus moves within the insect before transmitting to healthy plants.

In most instances, the disease is exacerbated by the presence of betasatellite DNA molecules (like DNA-β), which maximize symptom expression and facilitate the virus to replicate more vigorously in host tissues.

2. Symptoms:

The first symptoms appear on the young, developing leaves, which are curled upwards, crinkled, and puckered, leading to a severely distorted leaf shape.

The infected leaves become thick, leathery, and brittle, and they tend to develop obvious vein thickening visible to the naked eye.

A reduction of leaf size can be noticed, and their natural flat shape is lost.

As the disease becomes advanced, plants exhibit stunted growth as a result of reduced internode length and less branching, thereby having a bushy appearance overall.

Leaves may develop interveinal chlorosis (yellowing between the veins) or yellow edges, also impacting the photosynthetic surface.

Seriously infected plants are dwarfed and unproductive and may even fail to form few or no flowers and fruits.

If fruits form, they tend to be small, irregularly shaped, and deformed and tend to drop prematurely before maturity.

Young transplants and seedlings are particularly susceptible and can die prematurely if infected early in development.

3. Etiology (Disease Cycle and Development):

Papaya Leaf Curl Virus (PaLCuV) persists and disseminates in papaya-growing areas primarily through its insect vector, the whitefly *Bemisia tabaci*.

When whiteflies feed on an infected plant, they become infected with the virus, which multiplies in their body and becomes a constituent of their circulatory system.

The virus is circulative and persistent in the vector, i.e., whiteflies are able to transmit it to several healthy plants while feeding during days or even weeks.

When in a healthy papaya plant, the virus becomes mobile and moves in the phloem tissue, infecting the plant's transport system and spreading systemically.

The virus then starts to disrupt the normal processes of the plant, particularly its growth and leaf formation, resulting in the symptoms seen.

Environmental conditions conducive to disease development are high temperatures (25–35°C), high humidity, and an infestation of large whitefly populations.

The disease is more acute during hot and dry periods when whitefly activity is most intense.

The virus survives in the field via secondary hosts, which comprise several vegetable crops like tomato, chili, brinjal, and cotton, as well as numerous wild weed hosts, that act as reservoirs for both the virus and the vector.

4. Management:

A. Cultural and Preventive Measures:

Begin every time with healthy, virus-free planting material. Plant seedlings raised in protected nurseries or net houses to avoid early infection.

Use recommended spacing in the field to achieve a good air circulation and limit the movement of whiteflies between crops.

Destroy (rogue) infected plants as soon as symptoms are observed to avoid further infection of neighboring healthy plants.

Employ crop rotation and do not cultivate papaya in succession with recognized host crops such as tomato, chili, cotton, and brinjal.

Ensure efficient weed control within and around the field since weeds serve as alternate hosts for the virus and whiteflies.

B. Vector Control (Whitefly Management):

Monitor whitefly populations regularly through field surveys and yellow sticky traps to monitor adult insect activity.

Spray systemic insecticides like Imidacloprid (0.005%), Thiamethoxam, or Dimethoate at the initial stage of infestation.

Apply insecticidal soaps or botanical sprays such as neem oil for environmentally safe control, particularly in home gardens or organic systems.

Minimize the use of nitrogen fertilizers, which promote dense foliage and thus attract more whiteflies.

C. Biological Control:

Promote the occurrence of natural whitefly enemies, including ladybird beetles (Coccinellidae), green lacewings (*Chrysoperla* spp.), and parasitic wasps like *Encarsia* spp., which feed on whitefly nymphs and adults.

Employ entomopathogenic fungi like *Beauveria bassiana* or *Verticillium lecanii* by foliar sprays to biologically control whitefly populations.

D. Resistant Varieties:

At present, there is no papaya variety commercially available that is fully resistant; however, there are genotypes with tolerance that develop less severe disease.

Research and breeding schemes are underway to generate virus-tolerant and whitefly-resistant papaya varieties, which will be a big breakthrough in disease management.

E. Integrated Disease Management

Implement holistic management approaches through the integration of vector control, cultural practices, and use of healthy seedlings.

Encourage farmer awareness of early detection of symptoms, whitefly life cycle, and preventive practices.

Impose quarantine regulations in disease-free zones to restrict the transfer of infected plant material and whiteflies.

Integrate biological, chemical, and ecological methods for managing the disease in a sustainable manner in the long term. Leaf Curl of Papaya is a crippling and economically significant viral disease, which drastically curtails the productivity and profitability of papaya farming.

It is whitefly-borne and could spread quickly, particularly under optimum environmental conditions and with poor crop management.

Since there is no chemical treatment for the viral disease, prevention is the best option, based on stern vector control, clean planting material, and implementation of integrated disease management techniques.

Ultimately, long-term success in its management relies on its detection at early stages, consciousness, and concerted action by farmers, extension personnel, and researchers.



Leaf Curl of Papaya

Mosaic of Papaya Disease

(A prevalent and economically damaging viral disease that impacts growth, appearance, and yield of papaya plants)

1. Causal Organism:

Mosaic disease of papaya is mainly caused by Papaya Ringspot Virus (PRSV). It is a member of the genus Potyvirus and family Potyviridae.

Papaya Ringspot Virus (PRSV) is a positive-sense, single-stranded RNA virus enveloped by a flexuous rod-shaped protein coat.

There are two known biotypes of the virus—PRSV-P (which affects both papaya and cucurbits) and PRSV-W (which affects cucurbits alone), and it is the biotype PRSV-P that causes mosaic disease in papaya.

The virus is not seed-transmitted, but is transmitted very quickly by aphid vectors like *Aphis gossypii* and *Myzus persicae* in a non-persistent way.

This implies that aphids pick up and spread the virus within minutes after sucking sap but fail to harbor the virus for long periods of time.

2. Symptoms:

The disease begins with the development of light green mottling and mosaic-like discoloration of the young leaves of affected papaya plants.

As the disease advances, leaves are distorted, exhibiting puckering, curling, and blistering of the leaf surface.

The leaf lamina becomes smaller, and clearing of the veins or vein banding is commonly seen.

Stunted growth with reduced internodes, making the plant bushy and deformed, occurs in infected plants.

In advanced stages, typical concentric rings or ring spots may be seen on the fruit surface, especially in PRSV-infected fruits, thus the name "ringspot" virus.

The infected fruits usually develop into irregular shapes, coarse textures, and do not ripen well, which results in poor market value and yield loss.

Young plants infected at an early stage usually do not flower and fruit and lead to total crop loss.

3. Etiology

The Papaya Ringspot Virus persists within the infected host plants like papaya and alternate hosts like cucurbits and weed species.

Aphids, while feeding on infected host plants, pick up the virus on their stylet (mouthparts) and can subsequently transfer it rapidly to uninfected plants in short feeding probes.

As the virus is transmitted in a non-persistent manner, aphids have the ability to spread it extensively in short visits to numerous plants.

Inside the plant, the virus is carried through the phloem (vascular tissues), where it replicates and disrupts the normal physiological processes of the plant.

Warm and humid weather (25–35°C) and massive aphid populations largely promote the quick development and transmission of the disease.

If papaya is continually cultivated in the same location without vector control, the virus would become endemic, resulting in recurring and widespread outbreaks.

4. Management:

A. Cultural and Preventive Practices:

Use only healthy, virus-free seedlings acquired from guaranteed and certified nurseries at all times.

Do not plant papaya near or following cucurbit crops (such as pumpkin, bottle gourd, cucumber) because they are alternate hosts to PRSV and aphids.

Destroy infected plants immediately when the disease is noticed to limit the source of infection.

Ensure proper plant spacing to limit the migration of aphid vectors from plant to plant.

Implement frequent weed management, particularly of crops such as *Amaranthus*, *Chenopodium*, and cucurbit weeds, which are virus reservoirs and provide habitat for aphids.

B. Vector Control

Monitor aphid populations regularly and use systemic insecticides like Imidacloprid, Thiamethoxam, or Dimethoate at early growth stages.

Apply reflective mulches (aluminum or silver foil) around young plantations to keep aphids away by reflecting sunlight.

Place yellow sticky traps at crop canopy level to trap and attract flying aphids.

Shun overuse of broad-spectrum insecticides since they can destroy natural enemies of aphids.

C. Biological Control:

Conservation and promotion of natural predators of aphids such as ladybird beetles (Coccinellidae), lacewings (Chrysopidae), and hoverflies (Syrphidae).

Use biological insecticides like *Beauveria bassiana* or *Verticillium lecanii*, which are potent fungal diseases against aphids.

D. Resistant Varieties:

No papaya varieties are available commercially that are fully resistant to PRSV. A few tolerant or moderately resistant genotypes have been found in research studies.

Transgenic papaya lines with viral coat protein genes have proved to be resistant in the trials but are restricted due to regulatory and acceptance problems in most countries.

E. Integrated Disease Management (IDM):

Implement an integrated strategy that involves cultural, chemical, biological, and mechanical approaches to control both the virus and aphid vectors.

Educate farmers and nursery personnel in the early detection of the disease, clean cultivation, and vector management techniques.

Implement quarantine practices to avoid bringing infected planting material or infested vector plants into PRSV-free areas.

Mosaic disease of papaya due to Papaya Ringspot Virus is a serious threat to papaya production and can lead to heavy loss in yield and quality of fruit.

Since the virus is transmitted by vectors and is not curable once it enters the plant, prevention and vector management must be given priority.

Successful management involves the use of clean planting materials, frequent monitoring, suppression of vectors, elimination of infected plants, and integrated pest and disease management.

Endemic and susceptible areas need to have constant surveillance, education to farmers, and agro-good practices to reduce the effect of PRSV in affected and vulnerable areas.



Mosaic Disease of Papaya

Chapter–19

Pomegranate Disease, Causal Organism, Symptoms, Etiology and Their Management

Bacterial Blight of Pomegranate Disease

(A highly destructive bacterial disease of pomegranate leaves, stems, and fruits, causing heavy yield and quality losses)

1. Causal Organism:

Bacterial blight of pomegranate is caused by a gram-negative bacterial pathogen named *Xanthomonas axonopodis* pv. *punicae*.

This pathogen does not retain the Gram stain crystal violet stain and has rod-shaped morphology.

The bacterium is mobile and it is capable of moving using the aid of one polar flagellum, enabling it to disperse effectively on plant surfaces and within plant tissues.

It is a member of the *Xanthomonas* genus of bacteria that causes numerous blight diseases in a broad host of plants.

The disease organism can survive on infected plant refuse, branches, cuttings, or even in the bark of saplings and these become the main source of inoculum next growing season.

Optimum conditions for quick development of disease are warm weather and high humidity, usually prevalent during monsoon or in heavy plantings, poorly ventilated orchards.

2. Symptoms:

The first symptoms of bacterial blight on the leaves are small, water-soaked, and translucent spots that subsequently develop to be dark brown to black color and are typically surrounded by a wide yellow halo.

These spots can merge to leave irregular patches of necrosis, particularly along the leaf margin or between veins, leading to early yellowing and defoliation.

Infected young twigs and shoots have longitudinal, dark brown or black streaks or lesions that ultimately cause dieback or browning of branches and compromise the plant's fruit-bearing capacity.

On the fruits, exceedingly characteristic symptoms take the form of elevated, oily, dark lesions, which crack open and allow exposure of the inner tissues and hence render the fruits unfit for sale or consumption.

In heavy infections, the spots on the leaves and fruits can increase in size and become confluent to give large infected areas, leading to extensive damage, fruit fall, and tremendous reduction in yield and market value.

3. Etiology

The bacterial blight pathogen overwinters mainly in infected plant debris, like fallen leaves, cut shoots, and bark of recently infected trees.

The bacterium penetrates new host tissues from natural plant openings such as stomata, or by wound paths produced by pruning, insect feeding, hail, or mechanical wounding.

Inside the plant, the bacteria quickly double in number within the intercellular spaces of leaf and stem tissues, and in so doing, release toxins and enzymes that kill plant cells, necrose tissues, and occlude vascular tissues, resulting in wilting and blackening.

Secondary spread of the bacterium is primarily caused by splashing rainwater, irrigation water, infected pruning tools, insects, and human handling.

The climatic conditions most suitable for disease development are temperatures ranging from 25–35°C, high relative humidity, and frequent rain or overhead irrigation.

Inadequate sanitation of the orchard, canopy density, and poor pruning contribute to the problem further by developing a microclimate that favors survival and multiplication of the bacteria.

4. Management: A. Cultural Practices

Firstly, it is necessary to employ certified disease-free planting material and nursery saplings to avoid introducing the disease into the orchard.

Ensure proper field sanitation by eliminating and destroying all infected plant parts, such as twigs, dropped leaves, and diseased fruits, on a regular basis.

Diseased branches must be pruned in dry weather to reduce bacterial spread and must have pruning cuts treated with copper-based bactericides to close the wounds and eliminate any bacteria.

Implement correct spacing of plants and ongoing canopy management (pruning and thinning) to allow for good air circulation, which lowers humidity and provides poor conditions for bacterial survival.

Shun the overhead irrigation system, especially during monsoon, and adopt drip or furrow irrigation to avoid the transfer of bacteria through water droplets.

B. Chemical Control:

The disease can be controlled through chemical sprays, especially when the application is done preventively or in the initial stages of infection.

Spray 0.3% Copper Oxychloride or Bordeaux mixture (1%) at 10–15 days' intervals, particularly after pruning or after heavy rain.

An admixture of Streptomycin sulfate (200 ppm) and Copper Oxychloride proved effective for inhibiting systemic as well as external bacterial development.

Where permitted and accessible, Kasugamycin (Kasumin) can also be utilized since it is effective for a wide range of bacterial pathogens.

Sprays will have to be repeated at fairly regular intervals, particularly during rainy weather, or when disease symptoms first start to manifest.

C. Biological Control

Some achievement has been seen with the use of beneficial antagonistic microbes, including *Bacillus subtilis* and *Pseudomonas fluorescens*, which can be sprayed as foliar sprays to compete or suppress the pathogen.

Keeping the rhizosphere and phyllosphere microbiome healthy by means of organic mulching, vermicompost, and natural biostimulants can minimize disease occurrence in the long run.

D. Integrated Disease Management

Taking an integrated approach that includes culture, biological, and chemical means is the most environmentally friendly method of managing the disease long-term.

Healthy planting material should be used, orchard sanitation should be practiced, pruning should be done in a timely manner, biocontrol agents should be applied, and bactericides should be used rationally.

Field observation should be regular to monitor for initial symptoms, and suspicious plant material should be cleared at once to avoid spread.

Farmworkers and orchard managers need to be instructed to identify early signs of disease and practice safe and sanitary procedures when harvesting and pruning.

Avoid all pruning and other physical work during damp or wet weather to reduce dispersal of bacteria through tools and hands.

Conclusion

Pomegranate bacterial blight is a very destructive disease that attacks all aerial parts of the tree and causes extensive economic loss due to yield reduction and poor fruit quality.

The pathogen responsible for the disease is *Xanthomonas axonopodis* pv. *punicae*, which infects rapidly in hot and humid weather and flourishes in poorly managed orchards.

Since it is not fully curable, emphasis has to be laid on early detection, prevention, and holistic management practices such as using clean planting material, field sanitation, judicious chemical usage, and biological control agents.

Implementation of these practices can totally minimize the incidence of disease and ensure a sustainable and profitable pomegranate production system.



Bacterial Blight of Pomegranate

Chapter–20

Brinjal Disease, Causal Organism, Symptoms, Etiology and Their Management

Phomopsis Blight of Brinjal (Eggplant)

(A severe fungal disease leading to blight, stem cankers, and fruit rot, which considerably lowers the yield and marketability of brinjal)

1. Causal Organism

The causal organism is the fungal pathogen *Phomopsis vexans* belonging to the class Ascomycetes, order Diaporthales, and family Diaporthaceae.

This fungus forms asexual conidia as well as sexual ascospores, facilitating rapid dissemination and survival of the disease under field conditions.

The pathogen is seed-borne as well as soil-borne, and it survives in the field for long durations under the form of infected crop residue or infested seeds.

Favorable environmental conditions, particularly warm and humid climates, promote the activity of the fungus, causing infection in different parts of the plant, including especially the leaves, stems, and fruits of brinjal.

2. Symptoms

Symptoms of **Phomopsis blight** occur on more than one part of the brinjal plant, namely leaves, stems, and fruits.

On leaves, the disease starts as tiny, round, water-soaked spots, which grow larger and turn brown to dark brown with concentric rings, leading to leaf blight later on.

As the disease advances, infected leaves can yellow, curl, and wilt prematurely, resulting in severe defoliation.

On the stems and petioles, elongated, sunken brown lesions develop, which may girdle the stems, causing dieback of shoots or even complete plant death in severe cases.

On the fruits, the disease manifests as soft, sunken, brown spots that enlarge over time, resulting in shriveling and internal rotting of the fruit.

Infected fruits often exhibit black fruiting bodies (pycnidia) of the fungus, visible on the surface, which are responsible for producing and releasing spores for further disease spread.

As a result, the fruits become deformed, dry, and unmarketable, leading to major economic losses.

3. Etiology

The pathogen ***Phomopsis vexans*** persists as pycnidia or mycelium in infected seed debris, soil, and seeds, which are the main sources of inoculum in the ensuing season.

Primary infections may result from infested seeds or diseased nursery seedlings, and with favorable conditions, quickly spread from there.

The fungus infects the host tissues through natural portals like stomata or through insect, pruning, or mechanical damage-induced wounds.

After entering, the pathogen infects intercellular spaces, interferes with nutrient and water transmission, and causes tissue necrosis, wilting, and rot.

The disease becomes secondarily spread via rain splash, irrigation water, wind movement, and infected equipment.

Ideal temperatures of 25°C to 30°C and relative humidity exceeding 80% provide suitable conditions for disease growth and numerous infection cycles within one season.

4. Management

A. Cultural Practices

Always employ disease-free certified seeds and grow seedlings under covered nursery conditions.

Do crop rotation with non-host crops such as cereals or legumes for 2–3 years to minimize the accumulation of pathogen in the soil.

Destroy all infected plant residues following harvest to reduce the source of initial infection.

Prevent excessive irrigation and maintain good field drainage to avoid moisture accumulation, which encourages fungal growth.

Provide proper spacing between plants to enhance air movement and lower humidity near the leaves.

Disinfect pruning instruments regularly to prevent mechanical spread of the pathogen.

B. Chemical Control

Treat seeds with 0.2% Carbendazim or 0.3% Thiram prior to sowing to kill seed-borne inoculum.

Spray systemic fungicides like Carbendazim (0.1%), Mancozeb (0.25%), or Chlorothalonil (0.2%) at 10–15 days' interval during onset of symptoms.

Reapply fungicides particularly after rain or irrigation, which is conducive to pathogen spread and infection.

C. Biological Control

Spray biocontrol agents such as *Trichoderma harzianum* or *Pseudomonas fluorescens* onto the nursery bed and in the field to inhibit the development of the pathogen population.

Treat with neem products containing Azadirachtin as foliar sprays to assist in reducing inoculum pressure and minimizing insect entry wounds.

D. Resistant Varieties

Although no fully resistant types exist, tolerant cultivars like Arka Nidhi and Arka Kusumakar have less severe disease incidence and can be grown in diseased areas.

Field-level resistant genotypes developed for local adaptation should be cultivated in epidemic areas.

E. Integrated Disease Management (IDM)

Use a holistic approach that integrates seed treatment, field hygiene, resistant types, biocontrol agents, and careful fungicide application.

Perform frequent field monitoring, particularly during rainy and high humidity conditions, to detect the onset of symptoms early.

Educate growers to recognize early signs of the disease and promote disposal of infected fruit and leaves.

Avoid pruning and other intercultural activities during rain to prevent pathogen propagation.

Conclusion

Phomopsis blight of brinjal due to *Phomopsis vexans* is a destructive disease that causes heavy damage to the leaves, stems, and fruits of brinjal, leading to fruit loss and poor quality.

Since the disease is seed-borne and soil-borne, reliance has to be on preventive methods, particularly with the employment of clean seeds, field hygiene, and early fungicide protection.

The most effective and sustainable approach to controlling this disease and having a productive, healthy brinjal crop is to adopt an integrated disease management (IDM) strategy that integrates cultural practices, resistant varieties, biocontrol agents, and chemical treatments.



Phomopsis Blight of Brinjal

Little Leaf Disease of Brinjal

(A severe phytoplasma-induced disease resulting in extreme stunting, decreased leaf size, poor fruiting, and severe yield losses in brinjal)

1. Causal Organism:

The Little Leaf Disease of Brinjal is caused by a highly specialized group of bacteria-like organisms called **phytoplasmas** belonging to the class Mollicutes.

These **phytoplasmas** are parasites that live inside the phloem tissues (conducting tissues) of the host plant, where they reproduce and induce systemic infection.

They do not have a cell wall, due to which they are extremely flexible and resistant to most traditional antibiotics.

As **phytoplasmas** cannot be cultured in artificial media, their detection depends on molecular methods like Polymerase Chain Reaction (PCR).

The pathogen is transmitted mainly by insect vectors, notably the leafhopper species *Hishimonus phycitis*, which gets infected while sucking sap from diseased plants and transmits it to healthy ones.

2. Symptoms:

The most prominent symptom of this disease is the formation of extremely small, narrow, and deformed leaves, which makes the plant bushy in appearance—therefore the name "little leaf".

Infected brinjal plants show highly stunted growth, with reduced internodes and excessive branching.

The leaves are usually light green to pale yellow, and the foliage appears dense and compact.

Flowers are either totally suppressed or give rise to malformed and sterile flowers that do not set fruit.

In others, when fruits are formed, they tend to be small, malformed, hard, and non-marketable.

The disease has the potential to spread quickly in densely populated vector insect areas and in ideal environmental conditions like hot temperatures and dry weather.

3. Etiology (Disease Cycle and Development):

The phloem tissues of infected host plants harbor the **phytoplasma** pathogen and is transmitted by leafhopper vectors, specifically *Hishimonus phycitis*.

The cycle of disease starts with leafhoppers feeding on a diseased plant, consuming the phytoplasma, and after a latent interval, being able to transmit it to healthy plants.

The leafhoppers, once infected, continue to be carriers for the rest of their lives and are effective vectors for disease transmission.

Phytoplasma is not transmitted through seed and moves within the plant vascular system and insect vectors.

High temperatures (25–35°C), weedy hosts, and unbroken brinjal crop growing enable the development and multiplication of both pathogen and vector populations in the field.

4. Management:

A. Cultural Practices:

Early detection and destruction of diseased plants is important to minimize disease sources within the field to control Little Leaf Disease.

Growing resistant or tolerant brinjal varieties, when available, can effectively minimize disease incidence.

Keeping the field properly sanitized by eliminating weeds and alternative hosts prevents leafhopper breeding and survival.

Yellow sticky traps can be used inside and around the field to control and minimize the population of leafhoppers effectively.

Crop rotation with non-solanaceous crops (e.g., cereals or legumes) should be done to discontinue the life cycle of the vector as well as the phytoplasma.

Farmers must also avoid overlapping the crops, as this can form a green bridge of continuous disease and survival of the vectors.

B. Chemical Control

As there is no chemical treatment to cure the phytoplasma itself, the major method of control lies in managing the leafhopper vectors.

The following insecticides are effective:

- ✓ Imidacloprid 17.8 SL at 0.3 ml per liter of water,
- ✓ Thiamethoxam 25 WG at 0.25 g per liter of water,

- ✓ Dimethoate 30 EC at 2 ml per liter water.
- ✓ Spraying must be initiated at the nursery phase and repeated at 10–15 day intervals following transplanting.
- ✓ Rotating insecticides with various modes of action is crucial to prevent vector population resistance development.

Avoiding excessive or indiscriminate use of chemicals will help protect beneficial organisms and reduce hazards to the environment.

C. Biological Control:

As a more secure and environment-friendly option, Neem-based biopesticides like Azadirachtin 1% formulations may be used to suppress leafhopper populations.

Preservation and promotion of natural enemies such as ladybird beetles, spiders, lacewings, and parasitoids also help in suppressing the vectors and ensuring ecological balance in the field.

D. Integrated Disease Management

An integrated strategy with the combination of cultural, chemical, and biological practices provides the most efficient and enduring control.

Growing disease-free seedlings under covered insect-proof nets in protected nurseries can avoid early infection.

Regular monitoring and early identification of symptomatic plants reduce the spread of the disease.

Farmers must be trained to identify early symptoms and practice safe pesticide usage.

Combining vector management, field hygiene, resistant varieties, and educational awareness can considerably reduce the disease burden.

Conclusion:

Little Leaf Disease of Brinjal is a ruinous *phytoplasma* disease caused by leafhopper vectors, resulting in extreme stunting, twisted growth, and complete crop destruction if not controlled.

Since the phytoplasma is irremovable using chemicals, vector control and premature removal of diseased plants must be the main priority.

An efficiently implemented integrated disease management plan comprising resistant varieties, biological and chemical control of vectors, and good

agricultural practices can assist farmers to control this disease and safeguard their crops.



Little Leaf of Brinjal Disease

Chapter–21

Tomato Disease, Causal Organism, Symptoms, Etiology and Their Management

Damping off Tomato Disease

(A destructive disease of tomato seedlings, especially in nurseries, resulting in death before or soon after emergence)

1. Causal Organism:

Damping-off disease of tomato is caused by a group of soil-borne fungal pathogens consisting of species within the genera ***Pythium***, ***Rhizoctonia***, and ***Fusarium***.

Of these, ***Pythium aphanidermatum***, ***Rhizoctonia solani***, and ***Fusarium spp.*** are the most widely implicated organisms causing this disease.

Pythium is of the Oomycetes class (also referred to as water molds), whereas ***Rhizoctonia*** and ***Fusarium*** are of the Basidiomycota and Ascomycota phyla respectively.

These diseases thrive in hot, wet environments, particularly in wet, poorly drained, and crowded nursery beds.

2. Symptoms:

Damping-off occurs in two different forms — pre-emergence and post-emergence damping-off.

During pre-emergence, the infected seeds decay in the soil prior to germination and develop poor or mottled seedling emergence in the nursery.

During post-emergence, the disease develops soon after seedling emergence, with water-soaked lesions at the collar area (just above the soil surface).

The lesions turn dark brown, soft, and narrowed, causing the premature collapse of the young seedlings.

The infected seedlings may appear healthy initially but suddenly topple and die within a day or two.

Under humid conditions, a cottony fungal growth may be visible near the soil line around the collar of infected seedlings.

The disease often spreads rapidly in overcrowded nurseries, under low-light conditions, and when sanitation is neglected.

3. Etiology (Disease Cycle and Development):

The damping-off pathogens persist in soil, plant residue, and on nursery equipment as resting spores like oospores (*Pythium*), sclerotia (*Rhizoctonia*), or chlamydospores (*Fusarium*).

The spores germinate when the environmental conditions are favorable — more specifically, warm temperatures (25–30°C), high humidity, soil moisture excess, and low aeration.

Once the pathogen infects a seed or seedling, the disease rapidly spreads from plant to plant through water splash, infected equipment, or direct root contact.

The disease is also intensified by factors like poor drainage, overwatering, oversowing, and reusing infested soil.

Since the fungi remain in the soil for a long time, the disease may be recurrent if management and sanitation are not upheld.

4. Management:

A. Cultural Practices:

Always plant on well-drained and sterilized nursery beds to minimize initial fungal load in the soil.

Sow seeds with proper spacing to prevent overcrowding and ensure proper air movement between seedlings.

Grow nurseries in open, well-lit places, and keep away from shaded, low-lying spots.

Make use of disease-free, certified seeds and do not make use of soil that has been previously infested.

Provide prudent irrigation – maintain moist but not waterlogged soil, and never overwater.

Check the nursery beds frequently and remove diseased seedlings instantly to avoid spreading.

B. Seed and Soil Treatment:

Treat seeds prior to sowing with fungicides such as Carbendazim 2 g/kg or Thiram 3 g/kg to destroy spores present in the seeds.

Solarize the nursery beds by covering them with transparent plastic sheeting for 3–4 weeks during the summer season to eliminate soil-borne pathogens.

Drench the nursery soil with fungicide solutions such as Captan, Mancozeb, or Metalaxyl prior to sowing to disinfect the growing medium.

C. Chemical Control:

Spray fungicide combinations like Carbendazim (0.1%) + Mancozeb (0.25%) at 7–10 days after emergence of seedlings.

Apply Metalaxyl + Mancozeb products (such as Ridomil Gold) as a soil drench in waterlogged soils.

Rotate the fungicides to avoid development of fungicide-resistant pathogen strains.

D. Biological Control:

Use *Trichoderma harzianum* or *Trichoderma viride* at 5–10 g/kg seed or 50 g/m² soil for inhibiting pathogen action.

These fungi benefitting plants colonize the root area and compete with the damping-off pathogens for space and nutrients.

Apply *Pseudomonas fluorescens* as a seed or soil treatment to prevent pathogen growth naturally.

E. Integrated Disease Management

Integrate seed treatment, nursery sanitation, biological control, and sparing use of fungicides for effective management.

Don't keep growing tomato or associated crops (such as chili or brinjal) year after year in the same nursery or field to interrupt the disease cycle.

Educate farmers and nurserymen on early symptom detection, sanitation, and timely treatment.

Employ raised seedling trays with cocopeat or sterile media for improved control over disease in protected conditions.

Conclusion:

Damping-off is the most severe early-season disease in tomato growing, leading to heavy seedling mortality in nurseries.

It is induced by a complex of soil-borne fungal root pathogens, led by Pythium, Rhizoctonia, and Fusarium, and is encouraged by hot, humid, and poorly controlled nursery conditions.

Given the lack of a satisfactory cure once seedlings are infected, prevention through seed treatment, nursery sanitation, soil sterilization, and integrated management practices is the priority.

A well-planned, proactive approach guarantees healthy seedling development and strong crop performance in the main field.



Damping of Tomato

Late Blight of Tomato Disease

(A deadly fungal-like disease attacking tomato leaves, stems, and fruits, especially under cool and wet conditions, resulting in heavy yield losses)

1. Causal Organism

Late blight of tomato is caused by an oomycete pathogen, *Phytophthora infestans*.

This pathogen falls under phylum Oomycota and class Oomycetes, and is also known as a "water mold" because it needs high humidity and moisture to develop infection.

Phytophthora infestans also causes late blight in potatoes and can spread very rapidly between tomato and potato plants.

The organism can reproduce both asexually (by sporangia and zoospores) and sexually (by oospores under favorable conditions), to assist its survival and quick spread.

2. Symptoms

The disease initiates with the development of tiny water-soaked lesions on the leaflets, especially at the margins or tips.

These lesions soon increase in size and become brown to purplish-black with a pale green or yellow halo surrounding them.

In humid conditions, a white, powdery fungal growth, made up of mycelium and spores, is visible on the lower surface of the diseased leaves.

In advanced stages of the disease, blighted leaves fall off, collapse, and die prematurely, making the plant look scorched and dead.

Stems and petioles can also exhibit black to dark brown lesions, which can cause the stem to become girdled and eventually kill branches or the whole plant.

Fruit that is infected gets dark, greasy-looking, sunken spots, usually starting at the stem end. These lesions develop into larger, soft ones and cause fruit rot.

Under extreme conditions, particularly when wet weather persists for a long time and is accompanied by cool weather, the whole tomato plant may lodge and die.

3. Etiology

The disease-causing organism survives from crop to crop as mycelium in infected plant material or in potato tubers.

Under conditions of high humidity and low temperatures (15°C to 22°C), the pathogen forms sporangia on the surface of infected plant tissues.

These sporangia may germinate directly or give rise to motile zoospores that swim in water films on leaf surfaces and cause new infections.

Rain splash, irrigation water, and wind disperse the spores, promoting long-distance transmission of the disease.

Tomato plots that are located in the vicinity of formerly infected potato plots are more susceptible to outbreaks.

It grows quickly under conditions of high relative humidity (>90%), prolonged leaf wetness, and cool temperatures, usually causing epidemics in days or less if not controlled.

Year-round growing of tomatoes or potatoes in one field with poor residue management raises the risk of carryover disease.

4. Management

A. Cultural Practices

Prevent planting tomato and potato crops near one another in order to avoid cross-infection.

Utilize certified, disease-free seeds or seedlings for transplanting.

Destroy and remove all plant debris and residues infected with the disease after harvesting.

Provide adequate drainage in the field and prevent overhead irrigation in order to minimize the moisture on the leaves.

Maintain proper spacing between plants to facilitate good circulation of air and rapid leaf drying.

Apply mulches or use plastic tunnels to prevent soil splashing and contact with infected residues.

Restrict field activity when leaves are wet to prevent mechanical spread of the pathogen.

B. Resistant Varieties

Cultivation of late blight-resistant tomato varieties, such as ‘Arka Abha’ and ‘Arka Meghali’, can reduce disease incidence.

Stay updated with local agricultural research stations or advisories for new resistant hybrids or lines.

C. Chemical Control

Prior to disease development, use protective fungicides such as Mancozeb (0.25%), Chlorothalonil, or Copper oxychloride.

During early detection of infection, move over to systemic fungicides such as Metalaxyl + Mancozeb (Ridomil Gold) or Dimethomorph, and alternate with protective fungicides.

Reapply fungicides every 7 to 10 days under cool, damp weather conditions.

To avert the onset of fungicide resistance, rotate products having disparate modes of action.

D. Biological Control

Use biocontrol agents such as *Trichoderma harzianum*, *Trichoderma viride*, or *Pseudomonas fluorescens* as seed or soil treatments.

These biocontrol products inhibit the primary inoculum and trigger plant defense.

Foliar spray with these biocontrol products can also minimize the spread of disease during crop growth.

E. Integrated Disease Management (IDM)

Follow an integrated approach incorporating cultural, biological, and chemical controls for sustainable disease management.

Optimize fungicide applications using weather-based disease forecasting systems.

Educate farmers and field workers in disease identification, the use of resistant varieties, and safe and efficient fungicide application.

Conclusion

Late blight is one of the most economically important diseases of tomato crops, having the potential to produce sudden and catastrophic losses under cool and damp conditions.

The disease is caused by *Phytophthora infestans*, a contagious oomycete that develops very quickly in wet conditions and infects both tomato and potato crops.

Since there is no treatment option once the disease has been set, successful management is based on a large extent of preventive practices.

Through resistant cultivars' combination, early application of fungicides, cultural cleanliness, and biological control, farmers can adopt a robust Integrated Disease Management (IDM) program to safeguard their tomatoes against this virulent disease.



Late Blight of Tomato

Early Blight of Tomato Disease

(A severe and debilitating foliar disease of tomato plants at every stage of development, leading to extensive loss of leaves, quality decline of fruits, and substantial reduction in yield.)

1. Causal Organism

The tomato early blight disease is largely caused by a fungus pathogen, *Alternaria solani*.

This is a fungus from the class Deuteromycetes, otherwise referred to as Fungi Imperfecti, because it lacks a recognized sexual stage of reproduction.

Alternaria solani bears characteristic dark, multicellular conidia with both transverse and longitudinal septa, hence a characteristic "beaked" shape.

The disease has the potential to last for extended periods on residues of crops, infected seeds, soil, and on alternate hosts such as potato.

It is primarily spread by wind, splash by rain, infested instruments, irrigation water, and insect vectors.

2. Symptoms

The disease usually starts on the more mature, lower leaves of the tomato plant and works upwards through the canopy over time.

The first visible signs are tiny, irregularly rounded, brown to black spots which enlarge over time and have concentric rings, creating a typical "target-like" or "bull's-eye" appearance.

As the lesions get bigger, the surrounding tissues can become yellow (a condition referred to as chlorosis), causing early defoliation.

Heavy leaf drop can expose young fruits to sunlight, and sunscald and poor fruit quality commonly result.

Stem tissue can have elongated, dark sunken lesions, especially at or just below the soil line, on young transplants or seedlings.

On fruit, the infection commonly starts at the stem end in the form of sunken, black to dark-brown lesions with concentric ring patterns.

In later stages, the fruit rots, severely degrading its value at market and post-harvest life.

3. Etiology

Alternaria solani is a polycyclic pathogen, i.e., it is capable of completing several infection cycles within one cropping season.

It overwinters on infected crop residue, in soil, seed stock, and volunteer hosts like potato and tomato.

Development and dissemination of the disease are encouraged by warm temperatures of 24°C to 29°C and relative humidity levels above 80%.

Under such conducive conditions, the pathogen forms abundant numbers of conidia, and wind, water, and mechanical dispersal spread them around.

Infection takes place through natural openings on leaves (e.g., stomata) or wounds due to mechanical injury or insect feeding.

Leaf wetness for extended periods of time by dew, rain, or overhead irrigation greatly enhances the possibility of infection.

Poor sanitation in the field, drought stress, and nutrient deficiencies (specifically potassium and nitrogen) also increase susceptibility and disease severity.

4. Management

A. Cultural Practices

Plant certified, disease-free seeds or seedlings to avoid primary introduction of the pathogen.

Implement crop rotation with non-solanaceous crops for at least 2–3 years to lower pathogen accumulation in the soil.

Destroy and remove infected plant material after harvest to remove inoculum sources.

Prevent overhead irrigation; instead, resort to drip irrigation to minimize leaf wetness and sporulation.

Provide proper plant spacing to favor air flow and rapid drying of leaves.

Fertilize well with balanced amounts of nutrients, especially with adequate potassium and nitrogen, to improve plant strength and immunity.

B. Resistant Varieties

Cultivate resistant or moderately tolerant cultivars, including 'Arka Rakshak', 'Arka Abha', and 'Arka Vikas', which exhibit decreased susceptibility to early blight under field conditions.

C. Chemical Control

Spray protective fungicides such as Mancozeb (0.25%), Chlorothalonil, or Copper oxychloride at regular frequencies, particularly during conducive weather.

Rotate systemic fungicides such as Azoxystrobin, Tebuconazole, or Difenoconazole with contact fungicides to avoid resistance development.

Initiate fungicide sprays at the earliest visible symptom and repeat every 7–10 days under conducive environmental conditions.

Always adhere to fungicide resistance management (FRM) principles by rotating fungicides with multiple modes of action.

D. Biological Control

Use *Trichoderma harzianum*, *Trichoderma viride*, or *Pseudomonas fluorescens* as a soil treatment or foliar spray on the nursery and main field to inhibit fungal growth.

Use them as seed treatments and soil amendments to minimize primary infection.

E. Integrated Disease Management (IDM)

Combine cultural, chemical, and biological methods of management to achieve effective, sustainable disease management.

Regularly inspect fields and pull out diseased leaves or plants early to help minimize pathogen spread.

Implement forecast-guided fungicide spraying with weather forecasts to maximize timing and reduce fungicide application.

Educate farmers and stakeholders on early disease detection, pesticide safety, and green solutions.

5. Conclusion

Early blight of tomato, with *Alternaria solani* as its fungal pathogen, is a serious threat to worldwide tomato production.

Its destructive nature requires an effective management strategy involving preventive cultural practices, appropriate fungicide application at the right time, and biological agents use.

Deployment of resistant varieties, agronomic care that is proper, and integrated disease management (IDM) practices ensures improved disease control and more sustainable and profitable production of tomatoes.



Early Blight of Tomato

Leaf Curl of Tomato Disease

(Severe viral disease of tomato plants, marked by distortion of leaves, dwarfing, and abortive fruit set, causing huge economic losses.)

1. Causal Organism

Tomato leaf curl disease is caused mainly by a complex of viruses called Begomoviruses, the most prevalent and widespread of which is the Tomato leaf curl virus (ToLCV).

These viruses are members of the family Geminiviridae and the genus Begomovirus. The virus particles are unique in structure, which is geminate or twin-shaped, non-enveloped, and made up of single-stranded circular DNA (ssDNA) as their genetic material.

The disease is not seed-borne, i.e., it does not spread via infected seeds.

Rather, it spreads chiefly through whiteflies, particularly *Bemisia tabaci*, which are the principal vector of the virus.

These insects pick up the virus through feeding on infected plants and spread the virus to healthy tomato plants in a persistent and circulative mode, enabling the virus to establish and multiply systemically within plant tissues.

2. Symptoms

Symptoms of leaf curl disease can occur at any plant growth stage, but infections are more damaging and severe when they happen during the seedling or early vegetative stage.

The most notable symptom is the curling of leaves upward, particularly the young apical leaves. In addition to curling, the leaves can thicken, become brittle and crinkled, and commonly turn pale yellow or light green.

As the condition advances, internodes (the spaces between leaves on the stem) become shortened, with the plant exhibiting a bushy and compact look, also referred to as rosette form.

Infected plants become stunted, exhibit less vigor, and frequently exhibit poor or no flowering.

Even when flowers do develop, they can abort prematurely, resulting in low or no fruit set.

Fruits developed on infected plants are typically few, underdeveloped, deformed, and of poor quality for marketing, impacting yield and returns.

At later stages, the disease has the potential to cause total crop loss, particularly in instances of early infection and heavy populations of whiteflies.

3. Etiology

The Tomato leaf curl virus (ToLCV) overwinters within infected plant debris, weeds, and secondary solanaceous host plants like tobacco, chili, and brinjal.

The vector whitefly is pivotal in transmitting and perpetuating the disease cycle.

Whiteflies infest infected plants and become infected with the virus, which has an incubation period in the insect's body before it can infect healthy plants.

The virus is circulative and persistent in the whitefly for a few days to weeks, enabling it to distribute the infection extensively.

Warm temperatures between 25°C to 35°C, low rainfalls, and high plant density promote quick whitefly multiplication and virus transmission.

Year-round continuous cultivation of tomato or other susceptible crops and the availability of alternate weed hosts additionally promote the epidemic development of the disease in the affected areas.

4. Management

A. Cultural Practices

Healthy and virus-free seedlings should be raised to avoid the transmission of tomato leaf curl virus, and this should be done under protected nursery conditions, ideally under insect-proof netting to exclude whiteflies.

Prevent transplanting the tomato during whitefly peak population times, particularly in the areas with history of known disease.

Periodically inspect the crop and pull out infected plants as soon as possible to stop further infection.

Destroy and remove weeds and alternate hosts growing around the crop field which can carry whiteflies or the virus.

Implement yellow sticky trap to monitor and capture whiteflies in the field and nursery.

Practice crop rotation using non-host crops like pulses, cereals, or legumes to minimize disease pressure.

Maintain good field sanitation, and don't overlap tomato crops on the same land to disrupt the whitefly-virus cycle.

B. Resistant or Tolerant Varieties

Growing resistant or tolerant varieties is perhaps one of the most effective and long-lasting management tactics.

Tomato hybrids like 'Arka Ananya', 'Arka Vikas-VIR', 'H-24', and other local adapted resistant lines exhibit a level of tolerance to leaf curl disease.

Grafting tomato scions on virus-resistant rootstocks may also be practiced in places where severe disease prevalence exists, to enhance plant resilience.

C. Vector (Whitefly) Management

Whitefly population management is essential to control the transmission of the virus.

In the early stages of plant growth, apply botanical insecticides such as neem-based products containing Azadirachtin (1%), which are eco-friendly and safe.

For severe infestations, use systemic chemical insecticides such as Imidacloprid (0.05%), Thiamethoxam, or Flonicamid, especially in nurseries and during early vegetative growth.

To prevent insecticide resistance, rotate insecticide classes and avoid repeated use of the same active ingredient.

Cover nursery beds with fine mesh nets (40–50 mesh) and apply insecticides before transplanting seedlings to the main field.

D. Biological Control

Induce the activity of natural enemies of whiteflies, including the parasitic wasp *Encarsia formosa* and predatory insects like lacewings and ladybird beetles, which suppress whitefly populations.

Use biocontrol agents such as the entomopathogenic fungi *Beauveria bassiana* and *Verticillium lecanii*, which infect and kill whiteflies naturally when used in wet environments.

E. Integrated Disease Management (IDM)

A sustainable and effective method of controlling tomato leaf curl is to incorporate all the above measures in a planned approach.

This involves use of resistant varieties, protected nursery practices, frequent vector surveillance, prompt use of biopesticides or insecticides, and uprooting of diseased plants and weeds.

Ensure optimal plant spacing to lower humidity and eliminate vector-preferring microclimates.

Enhance farmers' awareness and training on early symptom detection and proper intervention measures.

Implement forecast-based Integrated Pest and Disease Management (IPDM) practices at peak risk seasons to avoid disease outbreaks.

5. Conclusion

Tomato leaf curl disease, which is induced by Tomato leaf curl virus (ToLCV) and transmitted by whiteflies (*Bemisia tabaci*), is one of the economically most devastating viral diseases of tomato globally.

Control of this disease is largely dependent on preventive measures like the utilization of resistant varieties, protected cultivation, and control of the vector.

By embracing an integrated and holistic management system, farmers can minimize yield losses, promote crop health, and help achieve sustainable production of superior-quality tomatoes.



Tomato Leaf Curl Disease

Chapter–22

Okra Disease, Causal Organism, Symptoms, Etiology and Their Management

Yellow Vein Mosaic Disease of Okra

(A most serious and economically significant viral disease that severely lowers yield and fruit quality in okra production.)

1. Causal Organism

Yellow Vein Mosaic Disease (YVMD) of okra is caused by a virus, the Yellow Vein Mosaic Virus (YVMV).

This virus belongs to the genus **Begomovirus** and the family Geminiviridae, which are characterized by their twin (geminate) shaped virus particles.

The viral particles are enveloped-free and have circular, single-stranded DNA (ssDNA) as their genome.

The virus does not spread by seeds but very effectively by a shared insect vector, the whitefly (*Bemisia tabaci*), which picks up the virus during feeding from infected plants and transfers it to healthy plants.

Transmission is in a persistent and circulative way, i.e., once the whitefly is infected with the virus, it can transmit and carry it throughout its life.

2. Symptoms

The disease begins with slight yellowing of the veins on the younger leaves of okra plants, which is normally hard to identify at the initial stages.

As the infection continues, these veins turn yellow predominantly, developing a strong yellow network or mosaic pattern on the leaf blade.

The whole leaf can ultimately become pale yellow, deformed, and reduced in size, frequently being accompanied by curling and crumpling of the leaves.

The affected plants indicate stunted growth, with reduced internodes, weak branches, and general decrease in vigor.

Flowering is retarded or does not take place, and, when fruits are formed, they are small, deformed, rough-skinned, and unfit for the market.

In extreme cases, the disease results in high losses of yield, from 30% to as much as 90%, depending on the level of crop development at the time of inoculation.

3. Etiology (Disease Cycle and Development)

The Yellow Vein Mosaic Virus persists in various sources such as infected okra plants, wild host plants like Hibiscus species, and weedy hosts that act as reservoirs of infection.

The disease is transmitted when whiteflies feed on infected plants, become infected with the virus, and then transfer it to uninfected plants as they feed.

Whiteflies become carriers of the virus for their entire lifespan after infection and thus become extremely efficient vectors for widespread dissemination.

The whitefly population increases very quickly in the environment of high temperature (25–35°C) and low relative humidity, which is typical of tropical and subtropical regions where okra is extensively cultivated.

Okra being continuously cultivated without crop rotation, with alternate hosts, and with overlapping crop cycles favors secondary spread and epidemic outbreaks.

4. Management

A. Cultural Practices

Growers should cultivate virus-free seedlings in insect-proof net houses or protected plots to avoid infection at an early stage.

Do not plant okra during the whitefly peak seasons, especially in the hot summer months, when vector densities are high.

Ensure crop rotation with crops that are not hosts, like legumes, cereals, or oilseeds, to disrupt the disease cycle and inhibit the inoculum buildup.

Destroy and remove the infected plants, weeds as host, and any volunteer okra plants that can serve as disease carriers.

Deploy yellow sticky traps in the field to effectively monitor and control the whitefly population.

Practice good field sanitation, optimal plant spacing, and avoid repetitive cropping of susceptible varieties.

B. Resistant Varieties

Production of resistant or tolerant varieties is one of the best long-term approaches for YVMV management.

Some of the notable resistant varieties are:

Arka Anamika

Parbhani Kranti

VRO-6

Hisar Unnat

These types exhibit decreased incidence of disease and improved fruit set in field conditions.

C. Chemical Control (Vector Management)

Whitefly vector control can be managed by treating okra seeds with Imidacloprid or other neonicotinoid insecticides to guard young seedlings.

Use systemic insecticides like Imidacloprid (0.05%), Thiamethoxam, or Flonicamid at the initial crop growth stages to minimize vector counts.

Alternate insecticides with a variety of modes of action to slow or prevent development of resistance by whiteflies.

Shun excessive or incorrect use of insecticides, since it may result in population resurgence of pests, injury to beneficial insects, and environmental pollution.

D. Biological Control

Promotion of natural enemies of whiteflies is an effective long-term control technique that is eco-friendly.

Some key natural enemies are:

Encarsia formosa (parasitic wasp)

Chrysoperla carnea (green lacewing larvae)

These parasitoids and predators consume whitefly eggs and larvae, thereby decreasing their numbers.

Application of entomopathogenic fungi such as *Verticillium lecanii* and *Beauveria bassiana* as foliar sprays has also been found to decrease whitefly infestations without leaving a harmful impact on the environment.

E. Integrated Disease Management (IDM)

Combining two or more approaches of management is the best method for managing YVMV.

The major IDM strategies are:

Employment of resistant cultivars and certified virus-free seeds

Healthy nursery management,

Regulating and controlling whitefly populations,

Early removal of infected plants, and

Application of botanical insecticides (e.g., neem oil) in initial phases.

Proper plant spacing and non-overlapping plantings can limit whitefly transfer between plants and prevent reduction of infection spread.

5. Conclusion

Yellow Vein Mosaic Disease of okra is among the most devastating viral diseases of the crop, leading to significant economic losses to growers in terms of reduced yield and substandard fruit quality.

Preventive and proactive disease management involves the use of resistant varieties, vector control, and clean cultivation practices.

Farmers should implement integrated pest and disease management practices with a focus on early detection, cultural measures, biological control, and selective use of chemicals to yield healthy crops and sustainable okra production.



Okra Yellow Vein Mosaic Disease

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